# Working, but in Slums: Climate Migration and Integration in Urban Africa

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**Abstract**

How do climate-induced migrants integrate into African cities? We construct annual bilateral migration flows across eight African countries from harmonized census microdata covering 1981–2018, linking migrants' retrospective migration histories to their housing and employment conditions at destination. We exploit variation in drought exposure at origin along the same migration corridors and use instrumental-variable estimates to characterize urban integration relative to the additional migration associated with drought. We find that climate-responsive migration increases the number of migrants who work but remain in low-quality housing—the urban working poor. In our baseline estimates, 85% of the drought-associated inflow into low-quality housing is employed, compared with 61% among those in adequate housing. More broadly, for every hundred additional migrants associated with drought, approximately twenty-eight are observed in low-quality housing, amounting to roughly 110,000 people across our sample. These residential consequences are particularly pronounced in small and secondary cities, rather than large metropolitan areas. Our findings reveal an important tension in climate adaptation: migrants can participate in urban labor markets without necessarily gaining access to adequate housing, particularly in rapidly growing secondary cities where housing deprivation is already widespread.

*Keywords:* Climate Change, Migration, Low-Quality Housing, Urbanization, Labor Markets

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# 1 Introduction

African cities stand at the intersection of two major transitions: rapid urbanization and intensifying climate change and variability. By 2050, an estimated 68% of the continent's population will reside in urban areas, up from 41% in 2020 (Un-Habitat, 2020). Much of this urban growth is taking place outside the largest metropolitan areas, in smaller and emerging cities where housing and employment conditions differ substantially. Housing deprivation is more widespread in smaller cities, where agriculture also accounts for a larger share of employment, while services become increasingly important in larger urban areas. This urban transition comes with major deficits in housing: 62.7% of Sub-Saharan Africa's urban population lives in slums (Un-Habitat, 2022). Low-quality housing is characterized by inadequate access to basic services and infrastructure, exposing residents to health and environmental risks (Marx et al., 2019). At the same time, climate variability increasingly affects rural livelihoods: droughts deteriorate agricultural productivity, deplete household incomes, and push rural populations to migrate as a coping strategy (Henderson et al., 2017; Mueller et al., 2020). Yet while a growing literature documents how climate shocks affect migration decisions, much less is known about what happens to these migrants once they reach cities: where they settle, whether they find work, and how these outcomes vary across the urban hierarchy.

This paper investigates how climate-induced migration shapes urban integration in African cities, focusing on migrants' housing and employment outcomes. We study how drought-responsive migration translates into inflows of migrants observed in low-quality housing, in employment, or in both conditions simultaneously, and how these responses vary across destination cities. By examining these two dimensions jointly, we investigate whether participation in urban labor markets necessarily goes hand in hand with access to adequate housing. The first challenge is measurement: studying migrants' outcomes at destination requires knowing when and where they migrated, information that existing data for Africa seldom provide at annual frequency. We construct a novel panel of yearly bilateral migration flows across eight African countries using harmonized IPUMS census microdata. Retrospective information on migrants' administrative unit of origin and year of arrival allows us to reconstruct annual cohorts along each origin–destination corridor between 1981 and 2018. Crucially, we link these migration histories to migrants' housing conditions, measured using a multidimensional indicator based on the UN-Habitat slum definition, and employment status observed at the census. The second challenge is identification: migration flows may be correlated with characteristics of destinations that also affect migrants' housing and employment outcomes. Our empirical strategy exploits variation in drought intensity at origin across migration cohorts traveling along the same origin–destination corridors, controlling for corridor and arrival-year fixed effects and time elapsed between migration and census enumeration. We first estimate how drought affects urban migration flows, then its reduced-form effects on migrant inflows across housing and employment categories. Finally, we instrument migration flows with origin drought to express these responses per additional drought-associated migrant. These IV ratios characterize aggregate responses associated with drought-induced migration, rather than necessarily identifying the individual causal effect of migration itself.

Our first result is that drought significantly increases migration toward African cities.

Comparing migration cohorts along the same origin–destination corridors, we find that a one-standard-deviation increase in drought intensity at origin raises urban migration flows by approximately 7% relative to their mean. This response is consistent with an agricultural-push mechanism. The migration response is more than seven times larger in origins with the highest shares of harvested cropland than in those with the lowest shares. In addition, a one-standard-deviation increase in drought intensity lowers maize yields by 3.1%, without a statistically significant change in harvested area. Together, these findings suggest that agricultural losses at origin are an important driver of climate-responsive migration.

Second, drought-induced migration increases the inflow of migrants subsequently observed in low-quality urban housing. For every 100 additional migrants associated with origin drought, approximately 28 are observed living in low-quality housing at the census. This result remains positive and statistically significant when controlling for destination-specific linear trends, and is similar when using an alternative measure of cumulative drought exposure. Aggregating the baseline estimates across our eight-country sample suggests that drought-associated migration contributed approximately 110,000 additional migrants living in low-quality housing between 1981 and 2018. These estimates quantify the residential consequences of climate-responsive migration, rather than implying that climate shocks are the main driver of urban housing deprivation.

Third, we find that climate-induced migration contributes to a population of working poor in African cities: migrants who participate in urban labor markets while living in deprived housing conditions. For every 100 additional adult migrants associated with drought, approximately 68 are employed, and 22 are simultaneously employed and living in low-quality housing. Strikingly, in our baseline specification, among drought-induced migrants observed in low-quality housing, 85% are employed, compared with 61% of those in adequate housing—a difference of 24 percentage points. We further examine the types of employment associated with drought-induced migration. The estimates suggest that these migrants participate in both wage employment and self-employment, with employment responses concentrated in services and agriculture. Across our sample, drought-induced migration is associated with approximately 50,000 additional working poor. These findings suggest that migration can provide access to urban employment without necessarily ensuring adequate living conditions.

The burden of these working poor falls on small and secondary cities, and the conversion of climate-induced migrants into slums is more a question of places than of who migrates. The residential consequences of drought-induced migration vary substantially across the urban hierarchy. For every 100 additional migrants associated with origin drought, approximately 33 are observed living in low-quality housing in secondary cities of 100,000 to one million inhabitants, compared with around 10 in metropolises above one million. By contrast, the estimated employment responses do not differ significantly across city sizes. The stronger housing response in secondary cities is not explained by a disproportionately larger migration response toward these destinations; if anything, the first stage is stronger in metropolises. The distinction between labor-market and residential integration is therefore particularly pronounced outside the largest metropolitan areas. This pattern is consistent with differences in cities' capacity to accommodate migrant arrivals, although we cannot separately identify housing-market constraints from differences in prices, preferences or migrant composition.

Beyond city size, we examine how destination characteristics shape drought-induced migration and migrants' housing outcomes. Migration responses are stronger toward nearby,

fast-growing cities and destinations with established migration corridors. However, these characteristics do not systematically predict differences in the share of additional migrants observed in low-quality housing. This suggests that factors attracting climate-responsive migrants do not necessarily improve their residential outcomes.

Finally, we investigate how migrant characteristics and existing migration networks shape these responses. Drought-induced migration is stronger along shorter corridors and toward destinations with established migration networks, while new corridors disproportionately connect origins to smaller and more deprived cities. Other destination characteristics predict the volume of migration but do not systematically explain differences in housing responses per additional arrival. We also find evidence of educational selection: migration responses are weaker among adults with very low educational attainment, while less-educated migrants who do move face greater residential deprivation. Together, these results suggest that the urban consequences of climate migration reflect both the characteristics of the migrants and the destinations receiving them, with city size emerging as a particularly important dimension of residential heterogeneity. We assess the robustness of these findings across alternative specifications, measures, and samples. Our main results are robust to destination-specific linear trends and alternative measures of housing deprivation and drought exposure. We also examine potential confounding channels, including destination weather shocks, conflict, and food prices, as well as possible displacement effects on earlier migrant cohorts. Additional checks include alternative fixed-effects structures, validation of the instrument-selection procedure, alternative population denominators, and leave-one-country-out estimates. While these exercises support the stability of our main findings, we acknowledge that drought may also affect migrants' resources before departure, which limits a strictly causal interpretation of the IV estimates.

This paper contributes to the literature on climate migration by moving beyond the decision to migrate to examine how climate-responsive migrants integrate into African cities. Rather than focusing solely on whether climate migration contributes to the growth of urban slums, we investigate how migrants are accommodated across two key dimensions of urban integration: housing and employment. By examining these outcomes jointly, we show that climate-induced migration can increase urban employment while also contributing to a population of working poor—migrants who find work but remain in deprived housing conditions. We further document how these outcomes vary across the urban hierarchy, highlighting the particularly important role of small and secondary cities. Our findings therefore reveal an important tension in migration as a climate adaptation strategy: access to urban employment does not necessarily translate into adequate living conditions.

This paper makes four contributions. First, it contributes to the literature on climate migration by moving beyond migration decisions to examine migrants' housing conditions at destination. A large literature establishes that climate shocks drive migration, from rural-rural moves (Gittard, 2024; McGuirk and Nunn, 2021; Gray and Mueller, 2012) to rural-urban (Henderson et al., 2017; Cattaneo and Peri, 2016; Mueller et al., 2020) and cross-border migration (Beine and Parsons, 2016; Missirian and Schlenker, 2017), while a separate literature debates whether informal settlements are transitional or persistent poverty traps and documents the vulnerability of their residents (Marx et al., 2013, 2019). We connect these literatures by examining how drought-responsive migration translates into additional urban residents living in low-quality housing, rather than focusing solely on the decision

to leave rural areas. Second, we contribute to the literature on migrant labor-market integration by examining employment and housing outcomes jointly. Whereas existing work emphasizes barriers that keep rural-urban migrants in low-productivity activities (Gollin et al., 2014; Bryan and Morten, 2019; Alvarez, 2020; Hamory et al., 2021) and the labor-market effects of climate migration (Kleemans and Magruder, 2018; Gittard, 2024), we show that drought-induced migration increases the number of migrants who work while living in deprived housing conditions—the urban working poor. This finding highlights an important distinction between labor-market participation and residential integration: finding employment does not necessarily imply access to adequate housing. Third, we contribute to the literature on African urbanization and structural transformation (Henderson and Kriticos, 2018; McMillan et al., 2014; Jedwab et al., 2017; Gollin et al., 2016) by documenting substantial heterogeneity in the residential consequences of climate migration across the urban hierarchy. Housing responses are considerably stronger in small and secondary cities than in large metropolitan areas, highlighting the importance of looking beyond Africa's largest cities when studying how urban areas accommodate climate-responsive migration. Fourth, we contribute methodologically by constructing annual bilateral migration flows linked to migrants' subsequent housing and employment conditions, and by implementing a data-driven procedure for selecting climate instruments. Specifically, we apply post-IV LASSO to 25 candidate climate variables (Belloni et al., 2014), complemented by additional validation exercises, including a wet-shock placebo.

The remainder of the paper is organized as follows. Section 2 describes the data and the construction of the migration, weather, and housing-quality variables, including the post-IV LASSO selection of climate instruments. Section 3 presents the empirical strategy. Section 4.1 establishes the effect of drought at origin on urban migration. Section 5 reports the reduced-form effects of drought on migrant inflows across housing and employment categories, including their intersection. Section 6 then uses instrumental-variable estimates to express these responses relative to the additional migration induced by drought, characterizing housing and employment outcomes per additional migrant. Section 7 quantifies the aggregate magnitude of these responses. Section 8 examines heterogeneity across origin and destination characteristics and across migrant profiles. Section 9 provides a conceptual interpretation of the findings through two margins of urban integration. Section 10 presents robustness checks. Finally, Section 11 concludes.

# 2 Data

This section describes the construction of the weather, migration, urbanization, housing-quality and employment status variables. Appendix Table 12 reports descriptive statistics for all variables used in the analysis, and Appendix Figures 8–10 map the climate exposure of origins and the migration, urbanization, and migrant outcomes of destinations. Appendix Figure 11 maps the observed urban in-migration corridors across the eight countries, illustrating the bilateral structure of the data: each origin–destination pair contributes a yearly flow, and identification rests on within-corridor variation in the timing of droughts at origin.

## 2.1 Weather Variables

*Overview -* The empirical model selects among a large set of candidate weather variables constructed from raw climate data, using a post-IV LASSO procedure (Belloni et al., 2014). The choice of candidate variables is grounded in recent climate-econ damage function literature (Hultgren et al., 2025; Auffhammer, 2018), which identifies the most influential weather terms for agricultural production. All weather variables are computed on a common historical reference period of 1983–2023. We distinguish temperature and precipitation terms. Weather variables are defined over the growing season, denoted by a superscript $g$ (e.g. $\text{Precip}^g$), with yearly versions denoted $\text{Precip}^y$ used in robustness checks. Growing seasons are defined at the national level using the main rainy season identified from historical monthly precipitation distributions for each country. Precipitation variables are derived from CHIRPS daily data (Funk et al., 2015). Temperature variables and evapotranspiration are derived from ERA5-Land reanalysis data (Muñoz-Sabater et al., 2021).

*Precipitation variables -* We construct several candidate precipitation variables, all derived from growing-season daily precipitation aggregated to the admin unit.[1] Cumulative precipitation $\text{Precip}^g_{cum}$ is the growing-season mean; its square $(\text{Precip}^g_{cum})^2$ captures nonlinear responses. The drought dummies $\text{Drought}^g_{p10}$ and $\text{Drought}^g_{p15}$ equal 1 if $\text{Precip}^g_{cum}$ falls below the 10th and 15th percentile of its historical distribution. The number of dry days $N^g_{dry}$ counts days with daily precipitation below 1 mm. Finally, the standardized drought intensity $\text{drought_int_z}_{o,t} = \max(0, -\tilde{z}_{o,t})$, with $\tilde{z}_{o,t} = (\text{Precip}^g_{o,t} - \overline{\text{Precip}}^g_o)/\sigma^g_{\text{Precip},o}$ the seasonal precipitation anomaly, captures drought asymmetrically: it activates only for below-average rainfall and measures the number of standard deviations below the historical mean. For an extreme-precipitation analysis conducted separately, we construct two symmetric extreme-wet variables: $\text{ExtremeWet}^g_{pJ}$ is a dummy equal to 1 if $\text{Precip}^g_{cum}$ exceeds the 85th or 90th percentile of the historical distribution, and $N^g_{extreme}$ counts days with daily precipitation above 20 mm at the pixel level, spatially averaged at the admin-unit level.

*Temperature variables -* We construct three temperature candidate variables, computed at the pixel level before spatial aggregation to the admin unit.[2] The seasonal averages of daily minimum and mean 2m air temperatures, $T^g_{min}$ and $T^g_{mean}$, enter along with their squares $(T^g_{min})^2$ and $(T^g_{mean})^2$. Growing Degree Days $\text{GDD}^g$ are computed daily as $\max\left(0, \frac{1}{2}(\tilde{T}_{min,d} + \tilde{T}_{max,d}) - 8\right)$, where $\tilde{T}_{min,d}$ and $\tilde{T}_{max,d}$ are daily minimum and maximum 2m temperatures capped at 30°C, then summed over the growing season.

*Moisture variables -* Moisture-related variables include potential evapotranspiration $PET^g$ and actual evapotranspiration $ET^g$, both extracted from ERA5-Land at the daily level and

[1] Following Hsiang (2016), strongly nonlinear transformations (squared precipitation, squared temperature, GDD) are computed at the pixel level before spatial aggregation to preserve local nonlinear responses. Threshold-based drought variables ($\text{Drought}^g_{pJ}$ and drought_int_z) use admin-unit-level historical precipitation distributions, consistent with the spatial scale of our migration outcomes.

[2] Squared terms are computed after conversion to Celsius, before spatial averaging, to capture nonlinear thermal responses.

aggregated over the growing season. PET captures atmospheric moisture demand (irrigation requirements), while ET captures realized evaporation.

*LASSO selection* - The candidate set used in the post-IV LASSO selection comprises thirteen contemporaneous weather variables, all measured at the admin-unit level over the growing season - the drought dummies $\text{Drought}^g_{p10}$ and $\text{Drought}^g_{p15}$, the standardized drought intensity drought_int_z, $N^g_{dry}$, the precipitation and temperature levels and polynomials, and the degree-day and evapotranspiration measures described above - augmented with the lags and three-season means of the drought-family variables, for 25 candidates in total (Section 3.5). The procedure selects drought_int_z as the unique relevant instrument for bilateral migration. We use this single-variable selection for the main analysis to maintain clarity in interpretation. The selection is probed in several directions (Section 3.5 and Appendix Figure 14). It is stable across LASSO penalty values, from the theoretical benchmark of Belloni et al. (2014) to more permissive thresholds; among lagged candidates, the three-season mean of drought intensity dominates, consistent with the cumulative instrument used in robustness checks; and a wet-side placebo - re-running the exercise on symmetric excess-rainfall candidates - finds essentially no predictive power, confirming that the instrument's strength is specific to the dry tail of the rainfall distribution.

## 2.2 Migration data

*Sources* - Both migration and housing quality data come from IPUMS census microdata (Minnesota Population Center, 2019) for fifteen African countries over the period 1987–2019. The sample includes exhaustive population censuses from Benin (2002, 2013), Burkina Faso (1996, 2006), Cameroon (1987, 2005), Egypt (1986, 1996, 2006), Ghana (2000, 2010), Kenya (1989, 1999, 2009, 2019), Malawi (1998, 2008), Mali (1998, 2009), Mauritius (1990, 2000, 2011), Mozambique (1997, 2007, 2017), Senegal (1988, 2002, 2013), Sierra Leone (2015), South Africa (1996, 2001, 2011), Togo (2010), Uganda (2002, 2014), and Zambia (1990, 2000, 2010). These censuses provide harmonized individual-level information on demographic characteristics, migration history (place of birth, year of arrival, previous residence), and housing conditions (dwelling type, access to basic services, urban/rural classification at the enumeration area level). One contribution of this paper is the construction of *bilateral yearly migration flows* at the IPUMS administrative level for 8 countries with available year-of-arrival information, and *five-year migration brackets* for the remaining 7 countries. Table 14 gives the coverage of the origin-destination panels by country and census year for the 8 countries for which we have build the yearly bilateral migration.

*Migrant identification* - We define an individual $i$ observed at destination $d$ in census year $t_c$ as a migrant from origin $o$ in year $t$ if (i) the IPUMS variable `migyrs1` reports a years-since-arrival value $\tau$ such that $t = t_c - \tau$, and (ii) the IPUMS variable for previous administrative unit of residence (or, when unavailable, place of birth [3]) identifies origin $o$. The destination $d$ is the administrative unit where the individual resides at the census date. For all

[3]Countries for which previous administrative unit of residnce is proxied by the place of birth are Kenya and Zambia

censuses, we trace migration histories back twelve years prior to each census for the yearly migration analysis, capping the window in multi-census countries so that it does not overlap the preceding census. The individual migration indicator is:

$$\text{InMigrants}_{o,d,t} = \sum_i \mathbf{1}\{i \text{ at } d \text{ in } t_c, \text{ arrived from } o \text{ in } t\} \cdot \text{perwt}_i,$$

where $\text{perwt}_i$ is the IPUMS person-weight and the sum runs over all individuals observed at destination $d$ in census $t_c$.

*Urban Migrant* - We use the IPUMS variable `urban` to identify whether an individual resides in an urban or rural enumeration area within their destination administrative unit. This classification is finer than the admin unit level: a single administrative unit may contain both urban and rural enumeration areas. Urban migrants are individuals classified as residing in urban enumeration areas at destination (`urban` = 2 in IPUMS).

*Population denominators* - We construct two destination population measures. The *census-year population* $\text{Pop}^{\text{census}}_{d,T_c}$ is the sum of person-weights observed at destination $d$ in census year $T_c$. The *urban census-year population* $\text{Pop}^{\text{census,urban}}_{d,T_c}$ restricts to individuals classified as urban residents (`urban` = 2). For robustness, we also reconstruct annual destination populations $\text{Pop}_{d,t}$ backward from the census year via the accounting identity:

$$\text{Pop}_{d,t} = \text{Pop}_{d,T_c} - \text{InMig}_{d,[t+1,T_c]} - \text{Births}_{d,[t+1,T_c]} + \text{OutMig}_{d,[t+1,T_c]} + \text{Deaths}_{d,[t+1,T_c]},$$

where births are estimated from age or birth-year information in the census, and deaths are derived from country-specific mortality rates in the corresponding period [4]. Our main results use the urban census-year population, which is more stable; the annual reconstruction is used in robustness checks.

*Migration rates.* - The bilateral migration rate from origin $o$ to destination $d$ in year $t$ is:

$$\text{MigRate}_{o,d,t} = \frac{\text{InMigrants}_{o,d,t}}{\text{Pop}^{\text{census}}_{d,T_c}} \times 100,$$

expressed in percentage points. The urban migration rate restricts the numerator to urban-resident migrants and uses the urban census-year population as denominator:

$$\text{MigRate}^{\text{urban}}_{o,d,t} = \frac{\text{InMigrants}^{\text{urban}}_{o,d,t}}{\text{Pop}^{\text{census,urban}}_{d,T_c}} \times 100.$$

For countries with only five-year migration brackets, we construct the equivalent five-year migration rate as the sum of migrants arriving over $[t-4, t]$, normalized by the same census-year denominator. In robustness checks, we re-estimate the main results using the annual reconstructed destination population $\text{Pop}_{d,t}$ as denominator in place of the census-year population, to verify that our results are not driven by the choice of a fixed denominator.

[4] Death rates are derived fro, country-specific mortality rates in the correspondiong period, from the annual national crude death rates from the World Bank Indicator.

## 2.3 Urbanization data

*Admin-unit urbanization.* - For our heterogeneity and robustness analyses, we classify IPUMS administrative units by their urban population share at the most recent census:

$$\text{UrbanShare}_a = \frac{\sum_{i \in a, T_c} \text{perwt}_i \cdot \mathbf{1}\{\text{urban}_i = 2\}}{\sum_{i \in a, T_c} \text{perwt}_i}.$$

This admin-unit classification is distinct from the individual-level urban variable: a rural admin unit may contain some urban residents, and a city admin unit some rural enumeration areas. For the cities and rural sample restrictions, we label an admin unit as a *city* if $\text{UrbanShare}_a \geq 0.55$ and as *rural* otherwise.[5] For the urbanization heterogeneity analysis, we use three groups: rural-like ($\text{UrbanShare}_a < 0.15$), urbanizing cities ($[0.15, 0.55)$), and main cities ($\geq 0.55$).

We validate this classification against the Africapolis database (OECD/SWAC, 2024), which delineates all urban agglomerations above 10,000 inhabitants in Africa. Matching each destination to its largest agglomeration confirms that the classification orders city sizes correctly: the median largest agglomeration rises from about 47,000 inhabitants in rural-like destinations to 86,000 in urbanizing cities and 880,000 in main cities, and the share of destinations hosting a city of at least 100,000 inhabitants rises from 28% to 43% to 70%. The main-cities group contains the principal cities of each country (Nairobi, Kampala, Yaoundé, Bamako), the urbanizing-cities group classic secondary cities (Nakuru, Garoua, Fort Portal), and the rural-like group mostly small towns. Appendix 12.1.3 reports the full correspondence (Figure 13 and Table 16).

## 2.4 Housing Quality

*Individual definition of housing quality* - The household-level slum classification follows the UN-Habitat multi-criteria slum definition (UN-Habitat, 2003).[6] The classification relies on IPUMS variables covering five housing dimensions grouped in two categories. Structural quality of the dwelling captures the materials of the floor, walls, and roof. Access to basic services and living conditions captures sanitation and household crowding (persons per room).[7] In the main definition, metal-sheet roofing is classified as durable, consistent with the UN-Habitat durability criterion that durable materials protect occupants from climatic conditions; the robustness definition instead counts metal-sheet roofing as deficient, reflecting

[5] The 55% threshold is calibrated on the median urban share across our sample of African administrative units; the rolling-window estimates in Figure S4 trace the estimates continuously along the urban-share distribution, showing that no single threshold drives the results.

[6] The census-based classification captures the material dimension of housing precarity - structural quality and access to basic services - but not the legal or tenure dimension that defines informal settlements in a stricter sense (unauthorized land occupation or unregulated construction). Throughout the paper, slum refers to the UN-Habitat statistical slum definition rather than to the legal sense of informal settlement.

[7] Water access is not included. The UN-Habitat criterion counts boreholes and protected wells as improved sources, but the harmonized IPUMS water-supply variable in these censuses distinguishes types of piped supply rather than improved from unimproved sources, so it cannot reliably separate deficient from adequate access in this setting.

its use as a low-cost material in the Sub-Saharan African urban context.[8]

A household is classified as residing in a slum - or a low-quality housing - if at least one of these dimensions is deficient, following the UN-Habitat operational slum definition, under which a household lacking any one of the deprivation criteria is classified as slum (a logical OR across dimensions) (UN-Habitat, 2003). As a conservative robustness definition, we also use a stricter threshold requiring at least two deficient dimensions. The coding of each dimension - which materials, services, and crowding levels are classified as deficient, in harmonized IPUMS categories identical across countries, with dimension availability varying by census - is reported in Appendix Table 15.

From the household-level classification, two individual-level housing-quality measures are constructed: the binary indicator $\text{deprived}_i \in \{0, 1\}$ equal to 1 if individual $i$ resides in a household classified as low-quality, and the continuous count $\text{deprived_count}_i \in \{0, 1, ..., 5\}$ equal to the number of deficient housing dimensions for individual $i$'s household.

*Aggregate housing quality outcomes* - The main analysis focuses on origin-destination-year cohorts of migrants. From the individual-level housing-quality classification above, two aggregate outcomes are constructed at the corridor level. The first outcome, used in the main results to preserve the same estimation sample between first and second stages, is the *scale* outcome $S_{o,d,t}$, the low-quality housing inflow rate at destination:

$$S_{o,d,t} = \frac{\sum_i \text{perwt}_i \cdot \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t, \text{deprived}_i = 1\}}{\text{Pop}^{\text{census}}_{d,T_c}} \times 100,$$

the share of urban destination population accounted for by urban migrants in low-quality housing arriving from origin $o$ in year $t$. The second outcome, designed to isolate the propensity channel, is the *propensity* outcome $P_{o,d,t}$, the share of migrants residing in low-quality housing in each corridor:

$$P_{o,d,t} = \frac{\sum_i \text{perwt}_i \cdot \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t, \text{deprived}_i = 1\}}{\sum_i \text{perwt}_i \cdot \mathbf{1}\{i \text{ at } d \text{ in } t_c,\ \text{arrived from } o \text{ in } t\}}.$$

This second outcome is defined only on cells with at least one migrant, which yields a smaller estimation sample than $S_{o,d,t}$. The analysis is on urban-restricted versions, $S^{\text{urb}}_{o,d,t}$ and $P^{\text{urb}}_{o,d,t}$, restrict the numerator to migrants residing in urban enumeration areas at destination and use the urban census-year population as denominator.

Because $S^{\text{urb}}_{o,d,t}$ is a rate per urban resident, destinations with a very small urban population produce rates that are arithmetically valid but economically meaningless. We therefore restrict the sample to destinations with at least 2,000 urban residents, the conventional minimum size of an urban locality and the threshold underlying the Africapolis classification used below.[9] The estimation panel contains 241,540 corridor-years.

[8]The UN-Habitat definition comprises five deprivations (durable housing, sufficient living area, improved water, improved sanitation, and secure tenure); we adapt it to the dimensions the harmonized census microdata can measure reliably, and cannot capture tenure security, water access, or, in some countries, sanitation. Our measure therefore reflects the material and service dimensions of housing precarity rather than the full UN-Habitat set.

[9]The restriction is stated ex ante and applies uniformly across countries. It binds on eight destinations

*Decomposing scale and propensity.* - The scale outcome embeds two distinct mechanisms by construction. The accounting identity

$$S_{o,d,t} = \text{MigRate}_{o,d,t} \times P_{o,d,t}$$

makes explicit that a causal effect of drought-induced migration on the low-quality housing inflow can operate through two channels, that could be happening at the same time. The scale channel reflects the mechanical contribution of drought-induced migrants arriving at the baseline propensity to reside in low-quality housing. The propensity shift reflects a change in that propensity among the migrants themselves: within a given corridor, drought-induced migrants may be more or less likely to settle in low-quality housing than those who would have migrated regardless. Distinguishing the two matters for policy: whether climate change fuels slum growth through sheer demographic pressure—more arrivals at the baseline rate (the scale channel)—or through a shift in their propensity to settle in low-quality housing (the propensity shift) points to different policy implications.

A first-order decomposition of the 2SLS coefficient on the scale outcome yields:

$$\widehat{\beta}_S^{2SLS} \approx \underbrace{\overline{P}}_{\beta^{\text{scale}}} + \underbrace{\overline{\text{MigRate}} \times \widehat{\beta}_P^{2SLS}}_{\beta^{\text{shift}}},$$

where $\beta^{\text{scale}}$ is the effect that would obtain if the propensity were fixed at its sample mean, and $\beta^{\text{shift}}$ the additional contribution of a shift in that propensity. Both terms are dimensionless: $P_{o,d,t}$ is a share and the migration rate is already expressed per hundred urban residents.

Estimating the propensity shift directly requires a separate 2SLS specification with $P_{o,d,t}$ as the outcome, which raises three difficulties. First, small-sample ratio bias: $P_{o,d,t}$ is a ratio, and on corridor-years with few migrants its expectation departs from the underlying share even when both counts are unbiased. We therefore define it only on corridor-years with a positive urban migrant flow ($N = 72{,}723$).[10] Second, the shared denominator: the migrant count in the denominator of $P_{o,d,t}$ is also the numerator of the endogenous migration rate, so classical measurement error in that count would bias an OLS regression of $P$ on the migration rate downward. Instrumenting with origin drought addresses this, since the instrument moves the true flow rather than its measurement error—an additional rationale for the IV strategy. Third, interpretive ambiguity: a nonzero coefficient establishes that the migrants drought adds are housed differently from those who would have moved regardless, but not why. Origin-destination fixed effects rule out a reweighting of the flow across destinations as the source; what remains is selection on migrant characteristics—which Section 8.2 explores (and does not detect on education, age, or gender)—and differences in the conditions or timing of arrival within a corridor, which we cannot separate from it.

(3,752 corridor-years, 1.5% of the panel), five of which are census-to-census reclassification artefacts: their recorded urban population collapses by 77-99% between consecutive censuses while total population grows, reflecting a change in what the census codes as urban rather than a change in settlement.

[10] In the robustness section, results are weighted by corridor size, so that thin cells do not enter on the same footing as dense ones.

## 2.5 Other Variables

*Aridity index for heterogeneity analyses* - For heterogeneity analyses by climate baseline, we use the CGIAR-CSI Global Aridity Index (Trabucco and Zomer, 2019), computed as the long-run mean ratio of precipitation to potential evapotranspiration. This index is extracted from the CGIAR-CSI global 30-arcsec dataset and spatially averaged at the IPUMS administrative level. Lower values indicate more arid conditions; we use within-country medians to classify origins as more arid or less arid, avoiding conflation between aridity and country fixed effects.

*Built-up surface for the construction-capacity analysis* - To characterise how fast a destination was able to build before our migration flows, we use the Global Human Settlement Layer's built-up surface product, GHS-BUILT-S release R2023A (Pesaresi and Politis, 2023). The product measures, in a 100-metre grid derived from Landsat and Sentinel-2 composites, the surface in square metres covered by buildings in each cell, in twelve epochs at five-year intervals from 1975 to 2030. We aggregate the grid to IPUMS administrative units by summing built-up surface over the cells falling within each `GEOLEVEL2` polygon. For each unit $d$ and epoch $t$ we observe the total built-up surface $B_{d,t}$ in square metres and the polygon area. All 351 destinations of the estimation panel are matched, and the built-up sample is therefore the full canonical panel ($N = 241{,}540$).

Our measure of construction capacity is the log growth of built-up surface over the fifteen years ending in 1990, $\text{BuiltGrowth}_d = \log B_{d,1990} - \log B_{d,1975}$, reported in the figures as the implied proportional increase, $B_{d,1990}/B_{d,1975} - 1$: the median destination expanded its built-up surface by 99%, with a tenth-to-ninetieth percentile range of 46% to 177%. The window closes before the migration flows we study and the droughts that instrument them, so the measure is predetermined [11], and it captures a flow of construction rather than a stock, which is what a capacity interpretation requires.[12] As with aridity, destinations are split at the within-country median, since the level of built-up surface mostly reflects national development.

We emphasise that this variable is not a housing-supply elasticity in the sense of Saiz (2010): the data contain no house prices, no measure of undevelopable land and no construction-cost variation. It is a proxy for persistent urban dynamism, which mixes the capacity to build with the demand that was already being accommodated, and Section 8.1.3 reports the pattern it produces together with that caveat.

*Additional sources for the exclusion restriction and the mechanism tests* - Three further datasets enter the robustness and mechanism analysis rather than the main specification. Conflict comes from the ACLED (Armed Conflict Location and Event) Data project (Raleigh et al., 2010), whose geo-located events we aggregate to IPUMS administrative units and years; coverage begins in 1997 and reaches 70% of the estimation panel, which is why the conflict tests are run on that sub-sample against a matched no-control benchmark. Cropping

[11] For Zambia's 1990 census and the first five years of Mali's 1998 window - together 15% of the panel - the measurement period and the arrival years overlap, so the measure is predetermined for the bulk of the sample rather than for all of it.

[12] The stock analogue - the share of the unit's area built up in 1990 - has a median of only 0.19% at this spatial scale and carries no signal (Section 8.1.3).

patterns come from CROPGRIDS v1.08 (Tang et al., 2024), a global 0.05° grid of harvested area by crop for 2020, from which we compute, for each administrative unit, the share of its land area under each of six staples – maize, cassava, wheat, sorghum, rice and soybean. World prices come from the World Bank's monthly commodity price series, annualized, for a basket of nine traded crops: the four traded staples above plus cotton, coffee, tea, tobacco and cocoa, with Arabica coffee used for Kenya and Robusta elsewhere. Administrative crop statistics for the yield mechanism come from HarvestStat Africa v1.0 (Lee et al., 2025), matched to our origin districts through its areal crosswalk to IPUMS boundaries; coverage is six of the eight countries and 2,535 district-years, and Appendix Table 25 sets out its limits.

## 2.6 Descriptive Overview

*Migration patterns* - Appendix Figure 7 displays the average urban in-migration rate over time in the eight countries of the estimation sample, constructed from the yearly bilateral panel. Rates differ by an order of magnitude across countries, and within each census window they rise toward the census year - the mechanical recall and onward-migration attrition of earlier cohorts that the distance-to-census control absorbs in estimation (Section 3). The corresponding five-year-bracket rates for the full set of fifteen IPUMS countries are shown in Supplementary Appendix Figure S2.[13]

*Spatial variation in climate and migration* - Beyond temporal patterns, the data reveal substantial spatial heterogeneity within and across countries. Appendix Figure 8 maps mean growing-season precipitation and mean drought intensity by administrative unit, illustrating the diversity of agricultural climates spanned by the sample - from the arid regions of northern Mali and Kenya (below 500 mm per season) to the humid coastal zones of southern Cameroon (exceeding 2,500 mm). Figure 1 shows both the spatial and temporal variation of the drought intensity instrument.

Appendix Figures 9 and 11 map urban in-migration rates at the administrative unit level, revealing within-country variation in destination patterns, with concentrated flows toward major coastal and capital cities. This spatial granularity allows us to construct origin-destination migration matrices at a fine geographic scale and to match each origin district with its specific climate exposure.

*Slums* - Low-quality housing is widespread among the urban in-migrants we study: across the estimation panel, 41 percent of urban in-migrants in the average corridor reside in housing classified as low-quality under the main UN-Habitat definition, and 36 percent once corridors are weighted by their migrant flow (Appendix Table 12), a magnitude consistent with the prevalence UN-Habitat reports for Sub-Saharan Africa's urban population (Un-Habitat, 2022). Appendix Figure 10 maps this share by destination, showing variation both across and within countries. Appendix Figure 12 reports the index dimension by dimension, for urban residents and for in-migrants separately, and Supplementary Appendix Figures S17 and

[13]A potential concern with retrospective migration data is recall bias: individuals may report their year of arrival with diminishing accuracy as the gap between migration year $t$ and census year $t_c$ widens. All specifications include a linear control for the survey-to-arrival gap $|t - t_c|$, absorbing systematic differences in measurement quality across migration cohorts.

S20 show which material and facility categories fall on the deprived side of each rule. Cross-country levels partly reflect census vintage - Malawi and Mali, the highest in the sample, are measured in 2008 and 2009, while Zambia's are measured in its 1990 and 2000 censuses, the earliest we observe - and are absorbed by the corridor fixed effects in estimation.

# 3 Empirical Strategy and Identification

## 3.1 Empirical Model

This paper examines how drought shocks at origin reshape migration flows to African cities and the urban integration of migrant cohorts, as reflected in their housing and employment outcomes. We first estimate the migration response to origin drought, then the corresponding responses of migrant inflows into different housing and employment categories. We subsequently scale these outcome responses by the migration response to provide an economically interpretable measure of their relative magnitude.

Our unit of observation is an origin–destination–arrival-year cohort, indexed by origin $o$, urban destination $d$, and reported migration year $t$. We estimate the migration equation, the first stage:

$$\text{MigRate}_{o,d,t} = \alpha_1 Z_{o,t} + \gamma_1 |t - t_c| + \delta_{o,d} + \lambda_t + \varepsilon_{o,d,t}, \tag{1}$$

where $\text{MigRate}_{o,d,t}$ is the count of migrants in cohort $(o, d, t)$, normalized by the destination's urban population and expressed per 100 residents. The climatic shock $Z_{o,t}$ is growing-season drought intensity at origin selected by the Post-IV lasso procedure explained in Section 3.5, defined as

$$Z_{o,t} = \max\{0, -\widetilde{z}_{o,t}\},$$

where $\widetilde{z}_{o,t}$ is the standardized precipitation anomaly (Section 2.1). Origin–destination fixed effects $\delta_{o,d}$ absorb time-invariant corridor characteristics; arrival-year fixed effects $\lambda_t$ absorb common annual shocks. The control $|t - t_c|$, where $t_c$ is the census year, accounts flexibly only to the extent of its linear specification for differences in the length of the retrospective observation window. In particular, earlier cohorts may be less completely observed because of onward or return migration, and housing and employment are measured at enumeration rather than at arrival. Standard errors are clustered by administrative unit of origin, the level at which $Z_{o,t}$ varies.

We next estimate a reduced-form equation for each outcome $Y$:

$$Y_{o,d,t} = \beta^Y Z_{o,t} + \gamma^Y |t - t_c| + \delta_{o,d} + \lambda_t + \nu^Y_{o,d,t}. \tag{2}$$

The main outcomes are the migrant inflow observed in low-quality housing, the inflow of employed adult migrants, and the joint inflow of employed adults migrants observed in low-quality housing as defined in Section 2. For example, the housing inflow is

$$S_{o,d,t} = \frac{\sum_{i \in \text{Migrants}(o,d,t)} \text{perwt}_i \; \mathbf{1}\{\text{deprived}_i = 1\}}{\text{Pop}^{\text{census}}_{d,t_c}} \times 100.$$

We denote the employment and joint employment–housing inflows by $E_{o,d,t}$ and $W_{o,d,t}$, respectively. The housing measure covers all migrants, whereas employment is defined for adults.

Accordingly, any ratio involving adult outcomes uses the corresponding adult migration response, with identical observations, controls, and population normalizations. Coefficients $\alpha_1$ and $\beta^Y$ summarize, respectively, the migration and outcome responses associated with the same change in drought intensity.

## 3.2 Identifying Assumption

Our identifying assumption is that, conditional on corridor and arrival-year fixed effects and the distance-to-census control, the residual variation in origin drought is orthogonal to unobserved determinants of the corresponding migrant inflows. Identification therefore exploits changes in the timing and intensity of drought within an origin–destination corridor, relative to the common arrival-year component. Under this assumption, the reduced-form coefficient $\beta^Y$ captures the total response of the corresponding migrant inflow to origin drought. This response encompasses changes in the number of migrants, their selection and resources, and their subsequent housing and labor-market circumstances.

Origin drought may be correlated with contemporaneous weather and economic shocks at destination, particularly for geographically close corridors. Corridor fixed effects absorb permanent differences in destination conditions, but not destination-specific shocks that vary over time. We therefore examine controls for destination weather, alternative fixed-effect structures including destination-by-arrival-year effects, and placebo and timing specifications. We also investigate potential confounding factors, such as conflict and agricultural commodity prices, which may be correlated with drought at origin and independently affect migrants' housing outcomes at destination. The results and their implications are reported in the robustness analysis (Section 10.1).

## 3.3 Wald Ratios and 2SLS Estimation

To express the outcome response relative to the migration response associated with origin drought, we define the Wald ratio

$$\theta^Y = \frac{\beta^Y}{\alpha_1}. \tag{3}$$

Provided both responses are estimated on the same sample, with the same controls and compatible normalizations, $100\theta^Y$ expresses the increase in outcome-$Y$ inflows per 100 additional migrant arrivals associated with drought. This is a ratio of aggregate responses, rather than a share computed from identified individual migrants. In particular, it can incorporate both additional arrivals and changes in the housing and employment circumstances of people who would have migrated regardless of the shock.

We compute the ratio using conventional just-identified two-stage least squares, instrumenting the migration rate with origin drought. The associated second-stage specification is

$$Y_{o,d,t} = \theta^Y \widehat{\text{MigRate}}_{o,d,t} + \gamma_2^Y |t - t_c| + \delta_{o,d} + \lambda_t + u_{o,d,t}^Y, \tag{4}$$

where the equation denotes the second stage of the full 2SLS procedure, including the exogenous regressors in both stages. With one instrument for one endogenous migration rate, the 2SLS coefficient equals the Wald ratio in Equation (3). We use cluster-robust inference

and report Anderson–Rubin confidence sets, which remain valid asymptotically under weak first-stage identification. The first-stage $F$-statistic is therefore informative about the precision of the ratio, while the reduced-form outcome response can be estimated and tested directly.

The main interpretation is the response of migrant outcomes per additional drought-associated arrival. Interpreting the same numerical 2SLS coefficient as the causal effect of migration itself would additionally require an exclusion restriction ruling out other pathways from drought to migrant outcomes. Our analysis of drought responses includes those pathways as part of the phenomenon of interest. Since our question is what becomes of the migrants drought brings to cities — not the structural effect of migration per se — the total drought response is precisely the object of interest.

## 3.4 Composition of Observed Migrant Cohorts

We also examine whether drought is associated with changes in the housing composition of observed migrant cohorts. The propensity $P_{o,d,t}$ is the share of migrants in cohort $(o, d, t)$ observed in low-quality housing at the census date. It is defined only for corridor-years with positive migration flows. Because drought can itself alter whether a corridor-year has positive migration, the propensity estimates characterize a conditional, potentially selected sample. We therefore report them as complementary evidence on the composition of observed migrant cohorts, alongside the unconditional inflow responses.

## 3.5 Choice of Climatic Shock: Post-IV LASSO

Our measure of drought intensity is selected from a set of candidate growing-season climate indicators using a data-driven procedure based on the post-IV LASSO approach of Belloni et al. (2014). The procedure applies LASSO to both the migration and outcome equations after partialling out the fixed effects and the distance-to-census control and examines the union of the selected climatic predictors. The candidate set and penalty choices are described in Appendix 12.2.1.

The climatic exposure measure is selected using a data-driven procedure. We assess the sensitivity of our findings to alternative drought measures, including the binary precipitation indicator used in an earlier specification, and discuss the implications of variable selection for statistical inference in Appendix X.

## 3.6 Scope of the Estimates

The census records migrants at their place of residence and employment status at enumeration, potentially years after the reported migration event. The estimated responses therefore concern the inflows of migrant cohorts still observed at destination, together with their census-measured housing and employment outcomes. They characterize corridor-level migration responses and their distribution across observed migrant categories and urban destinations.

# 4 First-Stage : Drought and Urban Migration

## 4.1 First Stage Results

Table 1: First Stage: Drought Intensity and Urban Migration

| | *DV: Urban migration rate* | | | | | |
|---|---|---|---|---|---|---|
| | *Contemporaneous drought* | | | *3-season mean drought (lagged)* | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Main spec. | | | Main spec. | | |
| Drought intensity | 3.618*** | 2.076*** | 1.592** | 9.468*** | 3.928** | 2.360 |
| | (0.664) | (0.604) | (0.732) | (1.638) | (1.745) | (2.130) |
| First-stage $F$ | 29.65 | 11.83 | 4.74 | 33.42 | 5.07 | 1.23 |
| Effect of 1-SD drought (% mean) | +6.8% | +3.9% | +3.0% | +10.5% | +4.3% | +2.6% |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Destination linear trend | No | Yes | No | No | Yes | No |
| Destination × Year FE | No | No | Yes | No | No | Yes |
| Dist. to census control | Yes | Yes | No[a] | Yes | Yes | No[a] |
| $R^2$ | 0.486 | 0.494 | 0.512 | 0.499 | 0.507 | 0.524 |
| Within $R^2$ | 0.010 | 0.007 | 0.000 | 0.011 | 0.007 | 0.000 |
| Mean migration rate | 26.820 | 26.820 | 26.820 | 26.804 | 26.804 | 26.804 |
| Observations | 241,540 | 241,540 | 241,540 | 235,920 | 235,920 | 235,920 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: the urban migration rate. Sample: urban migrants, all destinations, 8 countries, 12-year window, columns (4)–(6) on the sample with complete lag coverage. Specification: columns (1)–(3) use contemporaneous drought intensity at origin, $\max(0, -\tilde{z})$, and (4)–(6) its three-season mean over $\{t, t-1, t-2\}$; within each, the paper's main specification, then the same with a linear time trend specific to each destination, then origin-destination pair with destination × year fixed effects. [a]The distance-to-census control is mechanically absorbed under the last. First-stage $F$ is the cluster-robust Wald statistic on the instrument. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean migration rate × 100, on each column's own sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 1 reports the first stage under alternative sets of fixed effects, for the sample of urban migrations to all destinations. The table shows the strength and the stability of the instrument for in-migration towards urban areas. Column (1) is the paper's main specification: origin-destination pair and arrival-year fixed effects with the distance-to-census control. Column (2) adds a linear time trend specific to each destination, and column (3) replaces the arrival-year fixed effects with destination-by-year fixed effects. Columns (4) to (6) repeat the same three specifications with the cumulative instrument, the three-season mean of drought intensity, on the lag-restricted sample.

Destination-specific trends leave the coefficient positive and significant at the 1% level, at

2.076 against 3.618 in column (1). They absorb a smooth path of destination growth while leaving 96% of the identifying variation the main specification uses, and the standard error stays of the same order, 0.604 against 0.664: what moves between columns (1) and (2) is the coefficient itself, and the first-stage $F$ of 11.8 follows from it. Under destination-by-year fixed effects (column (3)) the coefficient remains positive and significant at the 5% level, at 1.592, and the first-stage $F$ drops below conventional thresholds. These fixed effects absorb about a third of the identifying variation the main specification uses, rainfall shocks being spatially correlated across origins and destinations within a country. The cumulative instrument follows the same ordering in columns (4) to (6): the main specification gives the strongest first stage of the table, $F = 33.4$, destination trends give $F = 5.1$, and destination-by-year fixed effects leave the coefficient indistinguishable from zero.

Quantitatively, the main specification in column (1) implies that a one standard deviation increase in the dryness of the growing season at origin raises in-migration to urban areas by 6.8% of the mean migration rate. Let's consider a destination of 10,000 urban inhabitants. At the mean migration rate, this corridor receives 2.68 migrants in a normal year. A one standard deviation drier growing season raises the migration rate by $3.618 \times$ SD(drought) per 100,000 urban residents, which scaled by the destination population adds about 0.18 migrants — bringing the inflow from 2.68 to 2.86, a 6.8% increase.

This magnitude is in line with existing estimates of drought-induced migration in sub-Saharan Africa (Henderson et al., 2017; Mueller et al., 2020; Gittard, 2024; Hoffmann et al., 2020). Relative to this literature, we focus on in-migration and identify the first stage within origin-destination corridors at annual frequency: a corridor receives more migrants precisely in the years its origin is hit by drought, net of fixed corridor characteristics and common year shocks.

Table 17 in Appendix 12.3 assesses the robustness of the first stage along two dimensions: the sample and the drought exposure window, reporting the paper's main specification throughout. Table 18 in the same appendix varies the controls, adding the growing-season temperature at origin and at destination.

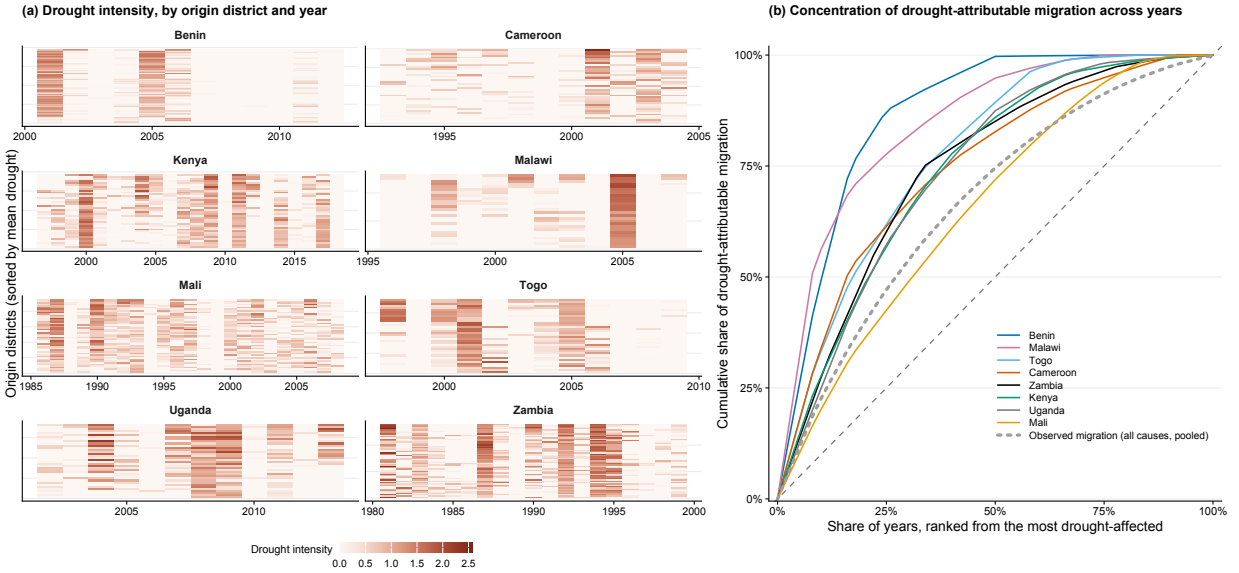
## 4.2 Synchronous Shocks

A destination draws on several origins, and whether those origins are hit together changes how many people arrive. Panel (a) of Figure 1 plots drought intensity for each administrative unit and each year: some years are national droughts in which nearly every origin is hit at once, others show dry and wet districts side by side. We quantify that structure with three destination-level measures built on predetermined weights: the temporal concentration of the destination's induced inflow across the years of its window, the variance of the shock it receives relative to the independence benchmark, and the average pairwise correlation of drought across its origins.[14]

Drought reaches a destination through many origins at once. The median destination draws on origins whose droughts correlate at 0.67, and the variance of the shock it receives

[14]The weights are the origin shares $w_{od}$ of each destination's inflow in the three years preceding its recall window, census years $-13$ to $-15$, so that no variation from the estimation window enters the classifier. This is the same pre-period migration table as Section 8.1.4, covering 5,146 corridors and 344 destinations and carrying 95.6% of the urban flow. Appendix Table 19 gives the three definitions.

Figure 1: The Spatio-Temporal Structure of Drought Shocks and of Drought-Attributable Migration




*Notes:* Panel (a) plots drought intensity $\text{drought}_{ot} = \max(0, -\text{rain_z}_{ot})$ for every origin district, rows sorted within country by mean drought, and every year of the country's span, with free scales because the windows differ in length. It is drawn from the rainfall data and not from the estimation sample, so it also covers years the migration windows leave out; white cells are genuine zeros and not missing data, the instrument being censored at zero, which is 33 to 70% of origin-years depending on the country. Panel (b) ranks, within each destination–census cycle, the years from the most to the least drought-affected and plots the cumulative share of the cycle total against the rank fraction, averaged by country, with the uniform diagonal as reference. The solid series is the first-stage prediction summed over a destination's origins with equal weights; the faint line uses pre-period flow weights, the classifier of Table 20, and correlates 0.906 with it. Since a Lorenz curve normalises by the within-group total, $\hat{\beta}$ cancels exactly: the curve is a transformation of the instrument and not evidence that the first stage drives its shape. The dashed grey line is observed migration from all causes, discussed in the text.

is 3.4 times what independence would imply (Appendix Table 19). A permutation test fixes the magnitude without any tuning parameter: shuffling years independently within each origin destroys the synchrony and nothing else, and the standard deviation of a country's mean drought falls from 0.292 to 0.081 ($p < 0.001$), so synchrony alone accounts for 92% of the year-to-year variance of the average shock. The induced inflow is compressed in time to match: Panel (b) of Figure 1 gives a median of 5.0 effective years out of twelve, with the most affected quarter of years carrying 70% of it.[15] Supplementary Appendix Figure S1 reports the same facts country by country.

Shock structure amplifies the first stage. Appendix Table 20 interacts the instrument

[15]Panel (b) also plots total observed migration, whose curve sits closer to the diagonal, 8.1 effective years against 5.0. The first stage implies that observed flows are roughly 93% baseline migration and 7% drought-induced, and a concentration curve on that mixture is dominated by the baseline: injecting the estimated induced component into the observed series moves its concentration from 9.56 to 9.54 effective years, 0.5% of the distance to the predicted curve. Total flows are mechanically incapable of revealing shock synchrony at this effect size, and the observed curve is a descriptive contrast.

with the three measures. Where the induced inflow is most concentrated in time, the first stage is 53% stronger; the variance ratio gives +23%; the average pairwise correlation gives +10%, not distinguishable from zero ($p = 0.54$), and it is the noisiest of the three and covers the fewest destinations. The first two survive controlling for squared drought and for drought interacted with city size, so neither a non-linearity in drought nor the size of the destination is doing the work. The amplification operates on volume alone: the same interactions leave the composition margin untouched (+0.13 pp per standard deviation for temporal concentration, $p = 0.760$; −0.03 pp for the variance ratio, $p = 0.908$), with a minimum detectable effect between 92% and 162% of the main propensity effect.

These results matter for climate projections. Projections for the Sahel and East Africa imply droughts that are more spatially extensive, hence more correlated across the origins feeding any given city (IPCC, 2021). For the same average intensity, a more spatially correlated shock regime delivers larger and more concentrated migration surges toward urban destinations, so projections built on mean intensity alone will understate the pressure on receiving cities.

# 5 Reduced Form : Drought and Migrants' Urban Outcomes

This section examines how drought shocks at origin affect the housing and employment conditions of migrant cohorts subsequently observed in African cities. Beyond the number of migrants arriving in urban areas, the central question is whether these additional inflows are associated with residential deprivation, labor-market participation, or both. We first examine housing and employment responses separately, then investigate their intersection to assess whether migrants observed in low-quality housing also participate in urban labor markets. Finally, we explore the types of employment associated with these responses.

We estimate Equation (2) with origin–destination and arrival-year fixed effects, controlling for time elapsed between migration and census enumeration, and cluster standard errors at the origin administrative-unit level. Outcomes are expressed as migrant inflows in each category per 100 destination residents. The estimates therefore capture aggregate responses in the number of migrants subsequently observed in different housing and employment conditions, rather than individual causal effects on these outcomes. Section 6 relates these responses to the corresponding increase in overall migration.

## 5.1 Low-Quality Housing

Table 2 shows that drought intensity at origin increases the inflow of migrants observed in low-quality housing at destination. Columns (1)-(3) use the contemporaneous drought instrument and columns (4)-(6) its cumulative (three-season mean) counterpart on the lag-restricted sample; within each group, the first column is our main specification and the remaining columns progressively saturate the fixed effects, up to destination-by-year effects that absorb all destination-year shocks.

Column (1) of Table 2 shows that a one-unit increase in drought intensity at origin significantly raises the inflow of migrants observed in low-quality housing by 1.029 per 100,000

urban residents at destination. For comparison, the average origin-destination corridor receives 11.96 migrants in low-quality housing per 100,000 destination residents in a given year. A one-standard-deviation increase in origin drought intensity therefore raises this inflow by approximately 0.52 migrants per 100,000 residents, equivalent to a 4.3% increase relative to its mean.[16] Using the cumulative three-season drought measure (column 4), a one-standard-deviation increase is associated with a 6.6% increase in low-quality-housing inflows.

The estimated housing response remains positive across specifications, although its magnitude and statistical precision are sensitive to the inclusion of more demanding fixed effects. The estimated response remains positive and significant at the 5% level when destination-specific linear trends are added to the corridor fixed effects (column 2). Adding destination-by-year fixed effects attenuates the coefficient and renders it statistically insignificant. In particular, destination-by-year fixed effects absorb all shocks common to migrants arriving in the same city in a given year, leaving identification to variation in drought exposure across origins within each destination-year. The alternative three-season drought measure also yields a positive and statistically significant response (column 4). We further investigate these sensitivities in Sections 8.1.1 and 10.1.

Drought intensity at origin is also associated with changes in the residential composition of observed migrant cohorts. Appendix Table 21 examines two dimensions of housing deprivation: the extensive margin, measured by the share of migrants residing in low-quality housing, and the intensive margin, measured by the average number of deficient housing dimensions. Among corridor-years with positive migration flows, a one-standard-deviation increase in contemporaneous drought intensity is associated with a 0.37 p.p. increase in the share of migrants observed in low-quality housing, relative to a sample mean of 40.6%, corresponding to approximately 41.0%. The cumulative three-season drought measure is associated with an increase of 0.58 p.p. Turning to the intensive margin, contemporaneous drought has no statistically significant association with the number of housing deprivations. In contrast, a one-unit increase in cumulative drought intensity is associated with 0.032 additional deficient housing dimensions per migrant, approximately 4.8% of the outcome mean of 0.662. Because these estimates are conditional on positive migration flows, which may themselves respond to drought, we interpret them as complementary evidence on the residential composition of observed migrant cohorts.

Appendix Figure 17 illustrates the relationship between origin drought intensity and both migration and low-quality housing inflows, after residualizing the fixed effects and controls. Both relationships appear approximately linear across the observed range of drought intensity.

## 5.2 Employment

Table 3 examines how drought intensity at origin affects migrant inflows across three labor-force statuses: employed, unemployed, and inactive. The analysis focuses on migrants aged 15 or older at arrival whose employment status is observed at census enumeration.

[16]The drought indicator is defined as $Z_{ot} = \max(0, -\widetilde{z}_{ot})$, so its empirical standard deviation differs from one despite the standardization of the underlying climate anomaly. We calculate the percentage change as $100 \times \widehat{\beta}^{S} \times \mathrm{SD}(Z)/\overline{S}$, where $\mathrm{SD}(Z) = 0.502$ and $\overline{S} = 11.96$.

Table 2: Reduced Form: Drought Intensity and Low-Quality Housing Inflows

| | *Contemporaneous drought* | | | *3-season mean drought (lagged)* | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Main spec. | | | Main spec. | | |
| Drought intensity | 1.029*** | 0.825** | 0.546 | 2.659*** | 2.307*** | 1.496 |
| | (0.350) | (0.334) | (0.437) | (0.735) | (0.865) | (1.012) |
| Effect of 1-SD drought (% mean) | +4.3% | +3.5% | +2.3% | +6.6% | +5.8% | +3.7% |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Destination linear trend | No | Yes | No | No | Yes | No |
| Destination × Year FE | No | No | Yes | No | No | Yes |
| Dist. to census control | Yes | Yes | No[a] | Yes | Yes | No[a] |
| $R^2$ | 0.370 | 0.374 | 0.393 | 0.383 | 0.388 | 0.407 |
| Within $R^2$ | 0.003 | 0.003 | 0.000 | 0.003 | 0.003 | 0.000 |
| Mean outcome | 11.964 | 11.964 | 11.964 | 11.892 | 11.892 | 11.892 |
| Observations | 241,540 | 241,540 | 241,540 | 235,920 | 235,920 | 235,920 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: the low-quality-housing inflow rate per 100,000 urban residents. Sample: urban migrants, all destinations, 8 countries, 12-year window, columns (4)–(6) lag-restricted. Specification: reduced form, the outcome regressed directly on drought at origin with the migration rate left uninstrumented. Columns (1)–(3) use the contemporaneous instrument and (4)–(6) its cumulative three-season counterpart; within each, the specifications are those of Table 1 in the same order. [a]The distance-to-census control is mechanically absorbed under destination × year fixed effects. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean outcome × 100, on each column's own sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Drought at origin significantly increases migrant inflows across all three employment categories. In our main specification, a one-standard-deviation increase in drought intensity raises employed migrant inflows by 7.4% relative to their mean, compared with 12.1% for unemployed migrants and 4.0% for inactive migrants. Evaluated at their respective sample means, these responses correspond to increases from 9.72 to approximately 10.44 employed migrants, from 1.14 to 1.28 unemployed migrants, and from 5.37 to 5.59 inactive migrants per 100,000 urban residents.

Although unemployed migrant inflows increase more in proportional terms, their baseline level is considerably smaller. As a result, employed migrants account for the largest share of the total reduced-form response among adults with observed labor-force status (67.2%), compared with 12.9% for unemployed migrants and 19.9% for inactive migrants. The distribution of the response is broadly similar to that of usual migrant arrivals, suggesting that drought-associated migration is accompanied by substantial labor-market participation rather than being concentrated in unemployment or inactivity.

The positive employment response persists when controlling for destination-by-year fixed effects, which absorb shocks common to all migrant corridors arriving in the same destination

in a given year. This result suggests that the employment response is not solely driven by contemporaneous destination-wide shocks. We next examine the types of employment in which these migrant cohorts are observed.

Table 3: Reduced Form: Drought Intensity and the Labour-Force Status of the Adult Inflow: Scale

| | *Contemporaneous drought* | | | | *3-season mean (lagged)* | | | |
|---|---|---|---|---|---|---|---|---|
| | *All, status known* | Employed | Unemployed | Inactive | *All, status known* | Employed | Unemployed | Inactive |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Drought intensity | 2.128*** | 1.429*** | 0.275*** | 0.423** | 5.826*** | 3.028*** | 0.962*** | 1.837*** |
| | (0.392) | (0.254) | (0.051) | (0.171) | (1.004) | (0.597) | (0.137) | (0.420) |
| Effect of 1-SD drought (% mean) | +6.6% | +7.4% | +12.1% | +4.0% | +10.6% | +9.2% | +24.6% | +10.2% |
| Share of the response (%) | 100.0 | 67.2 | 12.9 | 19.9 | 100.0 | 52.0 | 16.5 | 31.5 |
| Fieller 95% CI | – | [56.8, 80.2] | [9.1, 18.4] | [5.7, 29.8] | – | [42.9, 60.5] | [12.9, 22.0] | [23.4, 38.3] |
| Share of average arrivals (%) | 100.0 | 64.2 | 10.8 | 24.9 | 100.0 | 64.2 | 10.9 | 24.9 |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean outcome | 16.231 | 9.721 | 1.139 | 5.371 | 16.294 | 9.772 | 1.162 | 5.360 |
| Observations | 241,540 | 241,540 | 241,540 | 241,540 | 235,920 | 235,920 | 235,920 | 235,920 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: in-migrants of each labour-force status per 100,000 urban residents at destination, unwinsorized. Sample: urban in-migrants aged 15 or over at arrival whose labour-force status is known, as in Table 30; the three categories are exhaustive within it and sum to column (1), respectively column (5). Columns (5)–(8) use the three-season mean of drought intensity on the lag-restricted sample. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: each inflow is regressed directly on origin drought and the migration rate is not instrumented. 'Share of the response' divides each coefficient by the exhaustive total of its group: the migration response cancels in that ratio, so the share needs no first stage, and its Fieller interval inverts the ratio test on the joint cluster-robust covariance of the reduced-form coefficients. 'Share of average arrivals' is the migrant-weighted share of the same category among all in-migrants of the sample. Response shares are ratios of reduced-form coefficients and are distinct from a regression of an observed share on the instrument. Table 22 reports the same reduced forms on all ages. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## 5.3 What Kind of Work?

We next investigate the types of employment observed among migrant cohorts responding to origin drought. Information on class of worker and sector of activity is available for different subsets of country-censuses. We therefore estimate the reduced-form responses separately on each corresponding sample, restricting employment outcomes to migrants aged 15 or older at arrival.

Table 4 shows that the employment response is divided between wage work (59.5%) and self-employment (40.7%), with no detectable increase in unpaid work. Turning to sectors of activity, Table 5 shows that services account for 66.7% of the estimated employment response, followed by agriculture (25.0%) and industry (8.3%). The industrial response is not

statistically distinguishable from zero, while an occupation-based measure of agricultural employment yields a similar positive reduced-form response. Notably, agriculture accounts for a substantially larger share of the drought-associated employment response than of average migrant employment inflows in the same sample (a difference of 17.2 percentage points).

These shares are calculated as ratios of category-specific reduced-form coefficients to the total employed reduced-form coefficient within a common sample, rather than as individual transition probabilities. Overall, the results suggest that drought-responsive migration is accompanied primarily by employment in services and agriculture, with limited evidence of an additional industrial inflow.

Table 4: Reduced Form: Drought Intensity and the Class of Worker of the Inflow

| | *All* | Wage/salary | Self-employed | Unpaid/other |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Drought intensity | 1.063*** | 0.633*** | 0.433*** | -0.003 |
| | (0.248) | (0.169) | (0.100) | (0.043) |
| Effect of 1-SD drought (% mean) | +6.1% | +7.0% | +6.3% | −0.2% |
| Share of the response (%) | 100.0 | 59.5 | 40.7 | -0.3 |
| Fieller 95% CI | – | [44.0, 74.0] | [28.7, 58.1] | [-11.7, 6.8] |
| Share of average arrivals (%) | 100.0 | 59.3 | 33.4 | 7.4 |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |
| Mean outcome | 8.772 | 4.531 | 3.440 | 0.802 |
| Observations | 224,668 | 224,668 | 224,668 | 224,668 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: the inflow of employed in-migrants aged 15 or over at arrival, by class of worker, per 100,000 urban residents; column (1) sums the remaining columns. Sample: IPUMS `CLASSWK`, carried by 10 of the 11 urban census samples. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: each inflow is regressed directly on origin drought and the migration rate is not instrumented. Response shares are ratios of reduced-form coefficients, with Fieller 95% intervals. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## 5.4 Working Migrants in Low-Quality Housing

We next examine housing and employment jointly to assess whether the additional migrants observed in low-quality housing following origin drought are also excluded from urban labor markets. This distinction matters because residential deprivation may reflect limited access to both jobs and housing, or persist despite employment.

Table 6 reports reduced-form responses across four mutually exclusive employment and housing categories. Drought shocks significantly increase the inflow of migrants who are

Table 5: Reduced Form: Drought Intensity and the Sector of the Inflow

| | *All* | Agriculture | Industry | Services |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Drought intensity | 0.670*** | 0.168** | 0.055 | 0.447*** |
| | (0.206) | (0.069) | (0.057) | (0.147) |
| Effect of 1-SD drought (% mean) | +4.1% | +6.4% | +2.1% | +4.0% |
| Share of the response (%) | 100.0 | 25.0 | 8.3 | 66.7 |
| Fieller 95% CI | – | [7.2, 47.7] | [-14.7, 22.6] | [43.5, 93.8] |
| Share of average arrivals (%) | 100.0 | 7.8 | 15.9 | 76.3 |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |
| Mean outcome | 8.146 | 1.313 | 1.293 | 5.541 |
| Observations | 209,548 | 209,548 | 209,548 | 209,548 |
| OD pairs | 15,586 | 15,586 | 15,586 | 15,586 |

*Notes:* Outcome: the inflow of employed in-migrants aged 15 or over at arrival, by sector, per 100,000 urban residents; column (1) sums the remaining columns. Sample: IPUMS `INDGEN`, carried by 9 of the 11 urban census samples, three-way. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: each inflow is regressed directly on origin drought and the migration rate is not instrumented. Response shares are ratios of reduced-form coefficients, with Fieller 95% intervals. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

both employed and living in low-quality housing, while the estimated response among non-employed migrants in low-quality housing is not statistically distinguishable from zero. Among the additional adult migrant inflows into low-quality housing associated with drought, the estimated employment share is 85%, compared with 61% among those observed in adequate housing—a difference of 24 percentage points, statistically significant at the 10% level. These shares are ratios of aggregate reduced-form responses rather than individual employment probabilities.

These findings show that the increase in migrants observed in deprived housing is not confined to those without work. Instead, a substantial part of the housing response consists of migrants who are employed at the time of the census. This highlights why labor-market participation alone is an incomplete measure of urban integration and why housing conditions are an important dimension of the urban consequences of climate-induced migration.

Drought-associated migration is predominantly absorbed into employment and adequate housing. Yet a substantial minority of additional arrivals is observed in low-quality housing, including migrants who are employed. These findings highlight the coexistence of labor-market participation and residential deprivation among drought-responsive migrant cohorts.

Table 6: Reduced Form: Drought Intensity across Employment and Housing Cells: The Working Poor

| | *In low-quality housing* | | *Not in low-quality housing* | |
|---|---|---|---|---|
| | Employed | Not employed | Employed | Not employed |
| | (1) | (2) | (3) | (4) |
| Drought intensity | 0.465*** | 0.080 | 0.962*** | 0.614*** |
| | (0.120) | (0.097) | (0.184) | (0.135) |
| Effect of 1-SD drought (% mean) | +5.7% | +1.4% | +8.5% | +8.6% |
| Share of the response (%) | 21.9 | 3.8 | 45.3 | 28.9 |
| Fieller 95% CI | [13.5, 30.6] | [-6.9, 11.0] | [34.8, 58.5] | [20.7, 37.9] |
| Share of average arrivals (%) | 21.4 | 12.5 | 42.8 | 23.3 |
| Employed share of housing group (%) | 85.3 | | 61.0 | |
| Fieller 95% CI | [64.0, 148.2] | | [50.3, 71.9] | |
| baseline inflow (%) | 63.1 | | 64.8 | |
| Gap between housing groups (pp) | +24.3* ($p = 0.071$) | | | |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |
| Mean outcome | 4.084 | 2.918 | 5.647 | 3.601 |
| Observations | 241,540 | 241,540 | 241,540 | 241,540 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: in-migrants per 100,000 urban residents in the four employment × low-quality housing cells; column (1) is the working-poor cell. Sample: in-migrants aged 15 or over at arrival whose status and housing are both observed, 71.3% of urban migrants, within whom the four cells are exhaustive. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: each cell is regressed directly on origin drought and the migration rate is not instrumented. Shares are ratios of reduced-form coefficients, with Fieller 95% intervals. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

# 6 Urban Integration of Drought-Induced Migrants

The reduced-form results establish how droughts at origin affect migrant inflows across housing and employment outcomes. We now relate these responses to the corresponding increase in urban migration to characterize the additional arrivals associated with drought. Following Section 3.3, we estimate Wald ratios that divide each outcome's reduced-form coefficient by the corresponding migration first-stage coefficient, using consistent samples, controls, and population definitions. These ratios express outcome responses per 100 additional migrants and can be interpreted as IV estimates under the exclusion restriction. However, drought may also affect migrants' resources before departure, making it difficult to isolate the effect of migration itself from the consequences of the origin shock. We therefore interpret the ratios primarily as characterizing the housing and employment responses associated with

drought-induced migration, rather than as individual-level causal effects of migration. We first examine housing outcomes, then labor-market integration, and finally their intersection. We use weak-instrument-robust Anderson–Rubin confidence intervals to account for uncertainty in the migration response.

## 6.1 Low-Quality Housing

Table 7: Drought-Induced Migration and Low-Quality Housing Inflows

| | *DV: Low-quality housing inflow* | | | |
|---|---|---|---|---|
| | *Main specification* | | *+ Destination linear trend* | |
| | OLS | 2SLS | OLS | 2SLS |
| | (1) | (2) | (3) | (4) |
| Urban migration rate | 0.469*** | 0.285*** | 0.471*** | 0.398*** |
| | (0.025) | (0.063) | (0.025) | (0.093) |
| First-stage $F$ | | 29.65 | | 11.83 |
| AR 95% CI | | [0.132, 0.397] | | [0.149, 0.591] |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Destination linear trend | No | No | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |
| $R^2$ | 0.697 | 0.647 | 0.700 | 0.692 |
| Within $R^2$ | 0.521 | 0.441 | 0.522 | 0.509 |
| Mean outcome | | 11.964 | | |
| Mean migration rate | | 26.820 | | |
| SD(drought intensity) | | 0.502 | | |
| Observations | | 241,540 | | |
| OD pairs | | 16,846 | | |

*Notes:* Outcome: the inflow of migrants in low-quality housing per 100,000 urban residents at destination; the endogenous regressor is the urban migration rate, in the same units. Sample: urban migrants, all destinations, 8 countries, 12-year window, one estimation sample across the four columns. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Columns (3)–(4) add a linear time trend specific to each destination. Migration is instrumented by contemporaneous drought intensity at origin. Its first stage is 3.618 (0.664) and 2.076 (0.604) across the two groups (Table 1). AR 95% CI is the cluster-robust Anderson–Rubin interval, valid under a weak first stage. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7 reports the OLS and 2SLS estimates for urban low-quality housing inflows. Columns (1)–(2) present the baseline specification, while columns (3)–(4) additionally include destination-specific linear trends. All specifications include origin–destination pair and arrival-year fixed effects and the distance-to-census control. The IV estimates use contemporaneous drought intensity at origin as an instrument for urban migration.

The main scale result shows that drought-induced migration increases the inflow into low-quality housing at destination. The main result is the 2SLS coefficient in column (2): a 1 pp increase in the drought-induced urban migration rate raises the low-quality housing inflow rate at destination by 0.285 pp, under the IV identification assumptions. Since both rates are expressed relative to the same destination urban population, the coefficient reads directly as a ratio of flows: for every additional hundred migrants associated with origin drought, about 28 are observed living in low-quality housing at the census (see Appendix Table 26 for more details on the calculation). The estimate increases to 0.398 when destination-specific linear trends are included and remains statistically significant. The result is also similar under the cumulative drought instrument, which yields a baseline estimate of 0.281.

The OLS estimates (columns (1) and (3)) give the within-corridor correlation between migration and low-quality housing inflows, without a causal interpretation: corridors with more migration may differ systematically in ways correlated with housing quality. They are substantially larger than their 2SLS counterparts—0.469 against 0.285 in the baseline specification, and 0.471 against 0.398 with destination-specific trends. This difference may reflect endogeneity in migration flows or differences in the variation captured by OLS and IV. Neither comparison answers whether the migrants that drought adds are individually more likely to end up in low-quality housing than those who would have moved anyway, a question about the composition of the inflow examined in the reduced-form analysis.

Table 27 in Appendix 12.4 restricts the sample first to destinations that are main cities, and then to rural-to-main-city corridors. The effect is robust across both restrictions and is strongest along the rural-to-urban margin, consistent with the agricultural-push mechanism at the core of the paper.

## 6.2 Labor-Market Integration

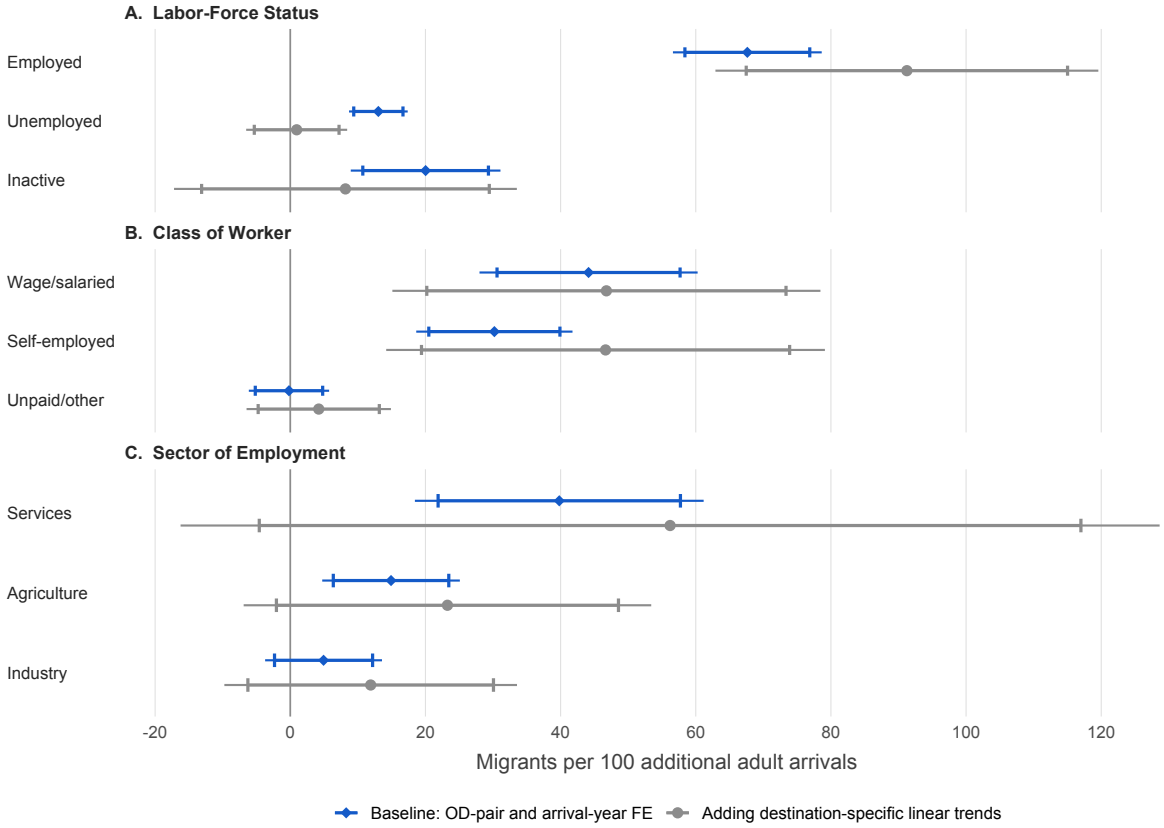
The housing results show that drought-induced migration increases the inflow of migrants living in low-quality housing. But does this residential deprivation also translate into poor labor-market integration? We find that a substantial part of the additional adult migrant inflow is observed in urban employment, including wage and self-employment, with suggestive evidence of employment in services and agriculture.

Figure 2, Panel A, presents the IV estimates for labor-force status. The analysis focuses on migrants aged 15 or older at arrival, with employment status observed at the census. In the baseline specification, the 2SLS coefficient for employed migrants is 0.676 (95% Anderson–Rubin confidence interval [0.57, 0.81]). Since the employed inflow and the adult migration rate share the same destination-population denominator, the coefficient reads directly as a ratio of flows: for every 100 additional adult migrants associated with origin drought, approximately 68 are observed in employment at the census, compared with 13 unemployed and 20 inactive. The employment response remains positive and statistically significant with destination-specific linear trends, with an estimated ratio of 0.912.[17]

Panels B and C examine the types of employment. In the class-of-worker sample, approximately 74 migrants are observed in employment per 100 additional adult arrivals, including

[17]The corresponding Anderson–Rubin confidence interval is considerably wider ([0.70, 1.52]), reflecting a weaker first stage ($F = 10.40$). The unemployment and inactivity responses are no longer statistically significant under this specification.

Figure 2: Employment Outcomes per 100 Additional Drought-Induced Adult Migrants




*Notes:* Markers are 2SLS estimates of employment inflows per 100 additional adult migrants, blue diamonds for the baseline specification, origin–destination pair and arrival-year fixed effects with the distance-to-census control, and grey circles for the addition of destination-specific linear trends; the horizontal bar is a conventional 95% confidence interval and the short ticks mark the 90% interval. Migration is instrumented by contemporaneous drought at origin. The population is in-migrants aged 15 or over at arrival and employment is observed at the census. Standard errors clustered at the origin district. *Source:* Authors' computations using IPUMS census and CHIRPS climate data.

44 in wage or salaried jobs and 30 in self-employment. In the sector sample, approximately 40 work in services, 15 in agriculture, and 5 in industry, with the latter response not statistically distinguishable from zero. These results suggest that the employment response extends beyond agricultural activities to wage employment, self-employment, and services. However, the sectoral estimates are less precisely identified: the first-stage $F$-statistic falls from 8.4 in the baseline to 1.5 with destination-specific trends, yielding unbounded Anderson–Rubin confidence intervals.[18]

The OLS estimates imply a different distribution of the adult inflow, with approximately 53 employed, 7 unemployed, and 40 inactive migrants per 100 additional arrivals, compared with 68, 13, and 20 in IV. This difference does not establish that drought-induced migrants are individually more likely to work than other migrants, and the reduced-form changes in employment composition are sensitive to destination-specific trends (Appendix Table 24).

[18]For class of worker, the first-stage $F$-statistic falls from 13.9 to 4.8 when destination-specific trends are included. Detailed IV estimates and weak-instrument-robust confidence intervals are reported in Appendix Table 31.

Nevertheless, the increase in employed migrant inflows highlights the importance of examining labor-market and residential integration jointly: employment does not necessarily imply access to adequate housing, a distinction we examine next.

## 6.3 Working Poor

Table 8: Drought-Induced Migration across Employment and Housing Cells: The Working Poor

| | *In low-quality housing* | | *Not in low-quality housing* | |
|---|---|---|---|---|
| | Employed 2SLS (1) | Not employed 2SLS (2) | Employed 2SLS (3) | Not employed 2SLS (4) |
| Urban migration rate | 0.22015*** | 0.03794 | 0.45505*** | 0.29052*** |
| | (0.04108) | (0.04177) | (0.05618) | (0.04122) |
| First-stage $F$ | 28.84 | 28.84 | 28.84 | 28.84 |
| AR 95% CI | [0.136, 0.307] | [-0.069, 0.110] | [0.351, 0.587] | [0.209, 0.381] |
| Mean outcome | 0.004084 | 0.002918 | 0.005647 | 0.003601 |
| Employed share of group (%) | 85.3 | | 61.0 | |
| baseline inflow (%) | 63.1 | | 64.8 | |
| Observations | 241,540 | 241,540 | 241,540 | 241,540 |

*Notes:* Outcomes: in-migrants per 100 urban residents in the four employment × low-quality housing cells, main definition, measured on the same individuals; column (1) is the working-poor cell. Sample: migrants with both labour-force status and housing observed, 71.3% of urban migrants, all destinations, 8 countries, 12-year window, destinations of at least 2,000 urban residents; no outcome is winsorized. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. The estimates describe the corridor-level composition of the marginal inflow and not an individual-level causal link between employment and housing. Column (2) is imprecise, its Anderson–Rubin interval including zero. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8 cross-tabulates the two dimensions on the same individuals. Of every hundred additional adult migrants associated with origin drought, 45 are observed employed and adequately housed, 22 employed and in low-quality housing, 29 not employed and adequately housed, and 4 in neither: deprived housing receives about a quarter of the inflow, not most of it. Read by housing, the table says what neither margin could alone. Among the additional adults observed in low-quality housing, 85% are employed; among those in adequate housing, 61%. This difference is suggestive and not robust to destination-specific trends. By contrast, the working-poor response remains positive and statistically significant with destination-specific trends, rising from 22 to approximately 33 per hundred additional adult arrivals. Climate migration thus adds working people to deprived housing.

The detailed employment estimates are reported in Appendix Tables 30 and 31.

# 7 The Aggregate Picture

The corridor-level estimates can be aggregated to gauge the magnitude of drought-associated migration and its urban integration outcomes. Over 1981–2018, across eight African countries, the censuses record approximately 11.6 million urban in-migrants in the corridors we observe. Applying the first-stage estimates to realized drought exposure yields approximately 388,000 additional migrants, or 3.3% of observed urban in-migration. Of these, approximately 110,000 are estimated to be living in low-quality housing at the census, based on the 2SLS estimate of 0.285 (Table **??**).[19]

The corresponding estimates for migrants aged 15 or older at arrival imply approximately 226,000 additional adult migrants, of whom around 153,000 are observed in employment at the census. Importantly, applying the joint housing and employment estimates from Table 8 yields approximately 50,000 additional working poor—adult migrants simultaneously employed and living in low-quality housing. These figures illustrate how drought-associated migration can contribute to urban employment while also increasing residential deprivation.

These estimates are illustrative aggregates over the observed census corridors and migration windows, rather than estimates of the total effect of drought across Africa. They exclude migration flows not captured by the census design and do not account for potential equilibrium responses at destination.

# 8 Heterogeneity

## 8.1 Origin and Destination Characteristics

A destination can matter in two ways: by receiving more climate migration, or by housing its migrants differently. The first is a property of the first stage, the second of the second stage, and in our data they do not move together. This section tests destination heterogeneity along seven characteristics – city size, corridor distance, baseline slum shares (which could capture housing saturation), baseline share of labor force in industry, past built-up expansion, baseline migration share, and the pre-period corridor network. We also look at origin characteristics such as the share of area under harvested cropland. The organising fact is an asymmetry: these characteristics shift the volume of climate migration substantially, but its conversion into bad housing and employment is driven by the size of the city. The residential burden – the working poor, migrants who find work but not decent housing – concentrates in secondary and small cities.

### 8.1.1 Destination City Size

We examine whether the effect varies with the type of destination, classifying destinations by the population of the largest Africapolis agglomeration located in or within 10 km of

[19] We multiply each corridor-year's drought intensity by the estimated first-stage coefficient and destination urban population, and aggregate across the observed corridors and years. This yields approximately 387,652 additional migrants, of whom $387{,}652 \times 0.285 \approx 110{,}306$ are estimated to be observed in low-quality housing. The calculation assumes zero drought intensity in the counterfactual.

the destination unit (OECD/SWAC, 2024): small towns (< 100k), secondary cities (100k–1M) and metropolises (≥ 1M). We use this size-based classification as our main cut for three reasons: it is external to the census data that produce our outcomes, it does not require choosing a threshold on a continuous and smoothly distributed urban share, and the agglomeration boundaries are those of the urban-settlement literature rather than of administrative units. Section S1.6 reports the same exercise on the census urban share as a complementary check. We estimate a single specification in which the migration rate is fully interacted with the destination-type indicators:

$$S_{o,d,t} = \sum_{k=1}^{3} \beta_k \big(\text{MigRate}_{o,d,t} \times \mathbf{1}\{d \in k\}\big) + \sum_{k=1}^{3} \delta_k \big(\text{dist}_{o,d,t} \times \mathbf{1}\{d \in k\}\big) + \alpha_{od} + \mu_{t \times k} + \varepsilon_{o,d,t}, \tag{5}$$

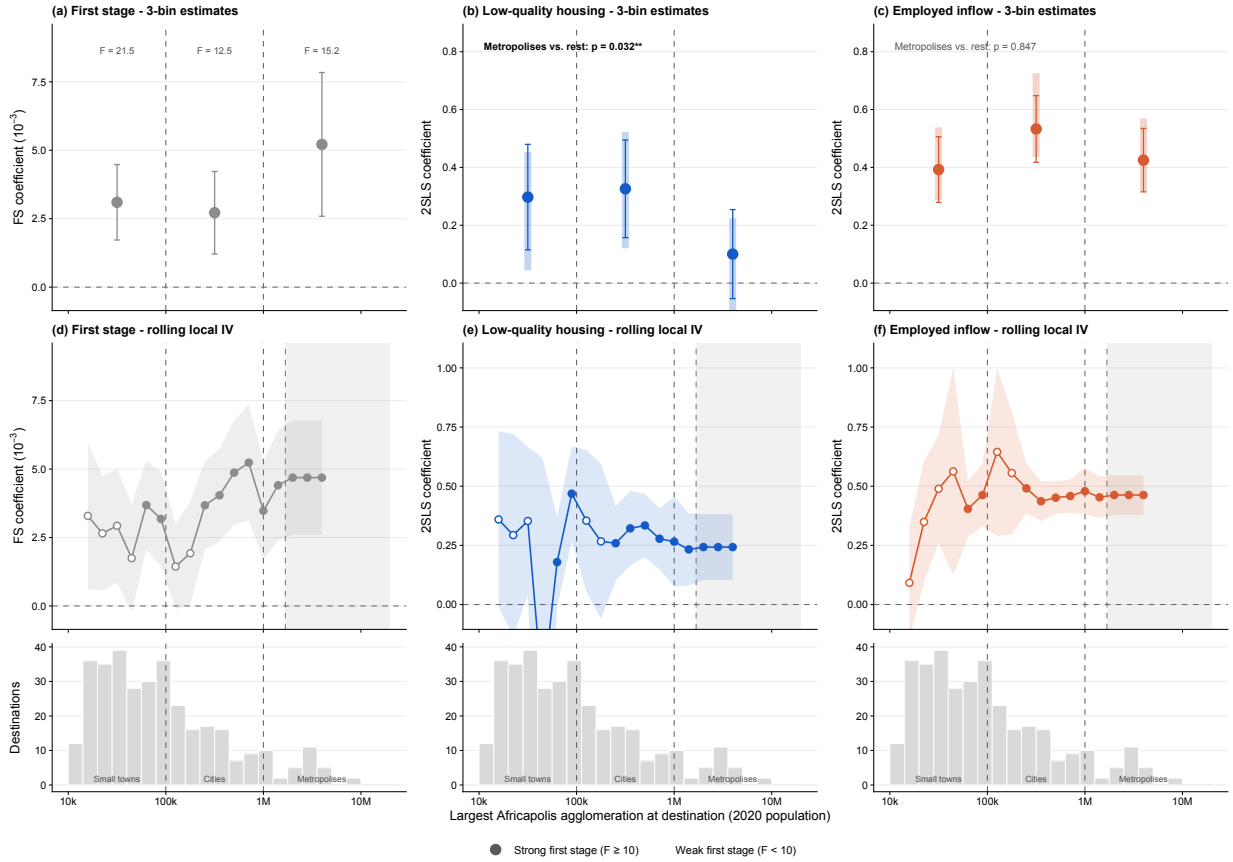
where each interacted migration term is instrumented by the corresponding interaction of drought intensity at origin, $Z_{o,t} \times \mathbf{1}\{d \in k\}$ (three endogenous variables, three instruments: just-identified), $\alpha_{od}$ are corridor fixed effects, and $\mu_{t \times k}$ arrival-year × bin fixed effects. No uninteracted migration term is included, so each $\beta_k$ measures the absolute effect within destinations of type $k$ rather than a difference from a reference category. The system is numerically equivalent to estimating the main specification separately by destination type, while the pooling delivers the covariance across the $\beta_k$ used to test for differences.[20] Figure 3 reports the binned estimates with the first stage by group (panels (a)–(c)), alongside a continuous rolling-window counterpart that re-estimates the main specification along the logarithm of destination city size (panels (d)–(f), indicative only) and the distribution of destinations.

*The housing burden falls on secondary and small cities, not metropolises.* - Of every 100 marginal migrants, about 33 enter low-quality housing in a secondary city of 100k–1M (0.326) and about 30 in a small town below 100k (0.297); the two are statistically indistinguishable ($p = 0.809$), so the effect is uniform across the non-primate tier. In metropolises above one million the coefficient falls to 0.100, with an Anderson–Rubin interval that includes zero. This attenuation at the top is the one difference the data establish: the metropolis coefficient lies significantly below the other zones pooled ($p = 0.032$) and below secondary cities pairwise ($p = 0.035$), and the rolling estimates in panel (e) trace the same profile continuously along city size.

*Employment absorption, by contrast, does not vary with city size.* - Drought migrants find work in destinations of every size – 0.393 in small towns, 0.533 in secondary cities, 0.425 in metropolises, each well away from zero – and metropolises do not differ from the rest ($p = 0.847$). The two margins therefore separate: urban labor markets absorb drought migrants everywhere, while the housing burden concentrates in secondary and small cities.

[20] Saturation by bin is required for identification: with pooled year fixed effects or a continuous interaction, the interacted instruments do not separately identify the interacted endogenous variables (Sanderson-Windmeijer conditional $F$ near zero).

Figure 3: Heterogeneity by Destination City Size (Africapolis): Low-Quality Housing and Employment




*Notes:* Markers are plotted at zone midpoints, whiskers are 95% Wald intervals and shaded bars Anderson–Rubin 95% intervals; hollow points flag a Sanderson–Windmeijer conditional $F$ below 10. Destinations are classified by the 2020 population of the largest Africapolis (OECD/SWAC, 2024) agglomeration in or within 10 km of the unit: small towns below 100k, secondary cities from 100k to 1M, metropolises above, $N = 240{,}340$ corridor-years. The classification is independent of the census urban share. Panels (a)–(c) come from a three-bin saturated interacted IV, migration × bin instrumented by drought × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control. On housing, secondary cities exceed metropolises and metropolises differ from the other two zones pooled; no employment contrast is significant. Panels (d)–(f) are a rolling-window local IV along log city size, indicative only, the shaded zone marking where the window no longer slides. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census, CHIRPS climate, and Africapolis 2024 data.

*The gradient is a second-stage result, not a first-stage artifact.* - Drought does not send disproportionately more migrants to smaller destinations; if anything the first stage is strongest in metropolises (0.00521 against 0.00272 in secondary cities, $p = 0.095$), precisely where the housing coefficient is smallest. A causal forest confirms that drought's effect on migration varies with corridor distance and origin rurality, not destination size (the destination urban share enters the best linear projection at $p = 0.123$). Because the volume of arrivals is not tilted toward smaller destinations, the size gradient in the housing effect reflects where migrants sort once they arrive, not where drought sends them. The two forces offset in the reduced form, which is flat across city sizes ($p = 0.328$); the contrast is identified by the 2SLS.

*The inflow geography is consistent with the size result.* - The share of each destination's in-migration attributed to drought declines with destination size, averaging 7.4% among destinations receiving fewer than 50,000 migrants over the panel against 3.3% among those above 200,000, and reaching 10 to 14% in secondary destinations such as Bushenyi, Kisii and Sikasso against 1.2% in Mombasa (Appendix Figure 21). Classifying destinations by census urban share yields the same direction – the coefficient declines as destinations urbanize – though no contrast along that gradient reaches significance; Section S1.6 reports that exercise.

### 8.1.2 Origin Characteristics

*At origin, the rural world responds uniformly.* - The counterpart of city size at origin is rurality, which the agricultural-push mechanism would predict to concentrate the effect. It does not: within the rural world the response is flat. Splitting the 315 rural origin districts (88% of the panel) at their median urban share, the first stage is statistically indistinguishable across halves ($p = 0.618$), as are both reduced forms and the 2SLS conversions. The direction is monotone – less rural origins respond slightly more in every split we construct – but never separable from zero, and it vanishes within country. Drought displaces the rural world fairly uniformly, which justifies treating it as a single block; the only discontinuity is at the urban threshold, where the 36 majority-urban origins respond markedly more in levels.[21]

*Cropland, not rurality.* - The census urban share classifies origins by where people live; the agricultural-push mechanism needs to be established on where do they come from and whether this affects their outcomes at destination. For every origin district we compute the share of its area under harvested cropland from CROPGRIDS v1.08 (Tang et al., 2024), summing the harvested area of six staples – maize, cassava, wheat, sorghum, rice and soybean – over the district's 0.05° grid cells and dividing by its land area. [22] Along this classifier the first stage shows that climate-induces migrants are coming from agricultural areas. Split into terciles [23] it rises 0.99, 1.56 and $7.42 \times 10^{-3}$, a contrast of $+6.43 \times 10^{-3}$ ($p = 0.0006$) and a ratio of roughly seven and a half. The drought-induced migration this paper studies originates almost entirely in farmland.

The two measures are therefore not substitutes: the census urban share captures settlement, which drought does not discriminate on, while cropland share captures exposure to rainfall, which it does.

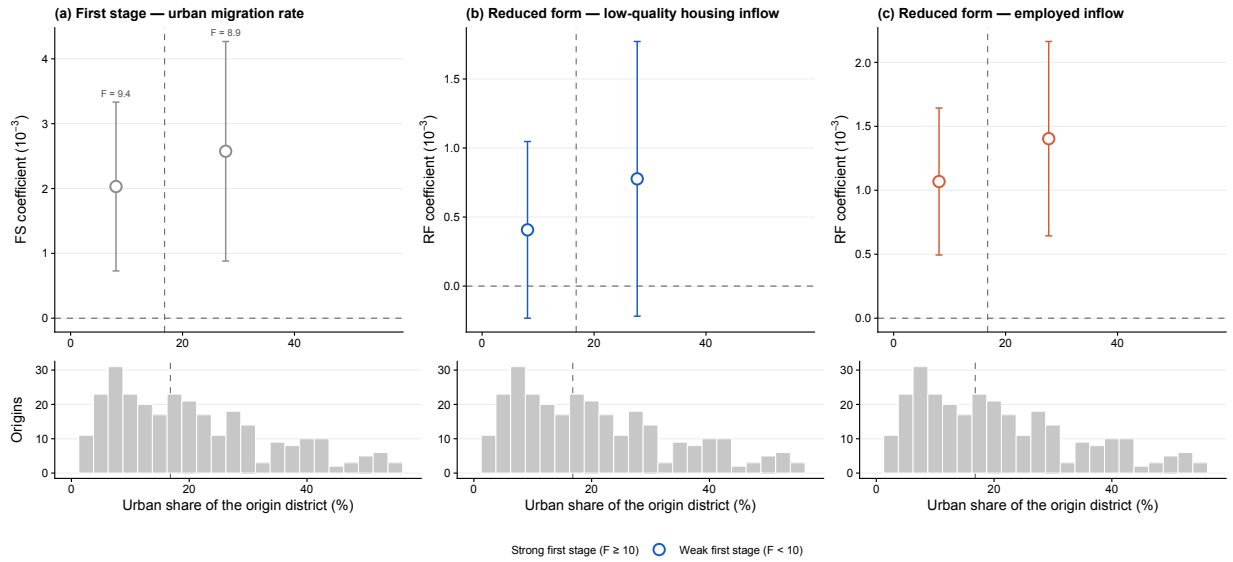
*The agricultural-push channel is corroborated on crop yields.* - Drought should destroy the harvest before it drives migration, and crop statistics let us check this directly. Matching HarvestStat Africa (Lee et al., 2025) to our origin districts (2,535 district-years, 214 districts, six countries), a one-standard-deviation increase in drought lowers maize yield per hectare by 3.1% ($p = 0.0004$) and production by 3.0% ($p = 0.081$), while harvested area does not

[21] Splitting the rural world into terciles leaves conditional first stages between 3.7 and $10.8 \times 10^{-3}$, so these nulls reflect limited power as much as an absence of effect.

[22] The measure is available for 350 of our 354 origins, covering 99.5% of the panel, and spreads them widely.

[23] The terciles cut at 4.3% and 18.7% of district area, with median cropland shares of 1.2%, 9.7% and 40.3%

Figure 4: Heterogeneity Within the Rural World: Origin Districts Below the Urban Threshold



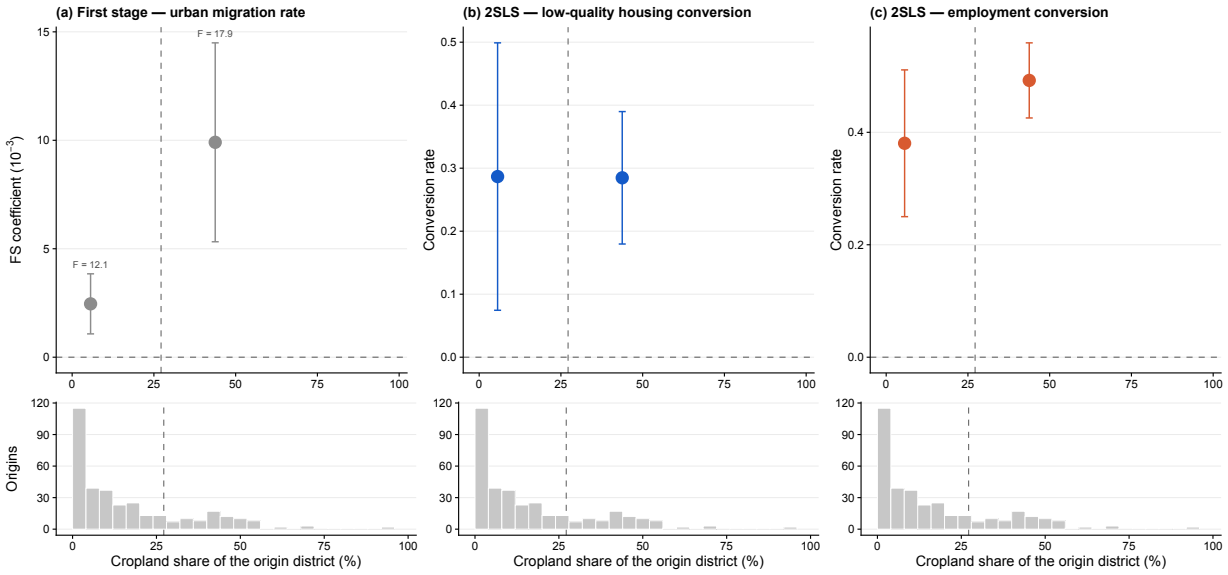

*Notes:* Whiskers are 95% Wald intervals and hollow points flag a conditional first stage below ten, which both halves are. Origin districts whose census urban share is below 0.55, 315 origins, 14,739 corridors and $N = 212{,}481$ corridor-years, are split at the pooled median of that share, 16.8%, marked by the dashed line; the classifier is time-invariant, so origin–destination fixed effects stay nested within bins. Panel (a) is the first stage in each half and panels (b)–(c) the two reduced forms, from a saturated interacted specification: instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control. No contrast is significant, on the first stage or on either reduced form. Standard errors clustered at the origin district.

move (+0.1%). Drought destroys the harvest on the land that is farmed. Together with the cropland gradient above, this is the agricultural-push channel in two pieces – drought destroys the harvest at origin, and migration concentrates where that harvest is the livelihood. We characterise the channel this way rather than through a formal mediation of migration on yields, which the data (within-district yield reliability of 0.30, 69% panel coverage, effect driven by Kenya) cannot identify.

### 8.1.3 Other Destination Dimensions

The city-size contrast of Section 8.1.1 is the paper's pre-specified primary hypothesis. This section asks, of a secondary family of destination characteristics, whether each changes the number of drought-induced migrants a city receives and what becomes of them. The answer is uniform: every characteristic moves the volume, none moves the conversion. Figure 6 plots the contrasts between bins, each characteristic being split at its within-country median. It shows where climate-induced migrants go: toward destinations that are close ($p = 0.0003$), had been building fast ($p = 0.0002$), were already migrant-intensive ($p = 0.016$) and have an industrial labour market ($p = 0.049$), but not toward those whose housing stock is already saturated ($p = 0.008$). Saturation here is the share of the destination's *resident* urban population — not of its migrants — living in low-quality housing at the last usable

Figure 5: The Drought Response by Cropland Share of the Origin District



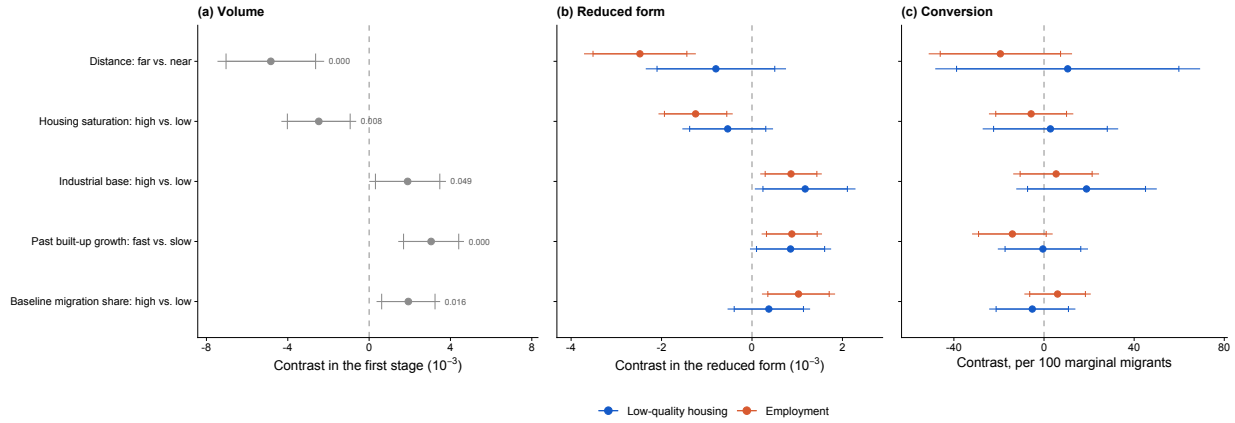

*Notes:* Whiskers are 95% Wald intervals. Cropland share is the harvested area of six staple crops over the origin district's 0.05° cells divided by its land area, from CROPGRIDS v1.08 (Tang et al., 2024), crop absence being coded −1 in the source rasters and recoded to zero before the zonal sums. It is available for 350 of 354 origins, 99.5% of the panel, and split at the upper quartile, 27.2%, marked by the dashed line: 262 origins and 180,334 corridor-years below, 88 and 60,006 above. The classifier is time-invariant, so origin–destination fixed effects stay nested within bins. Panel (a) is the first stage and panels (b)–(c) the two 2SLS conversions, from a saturated interacted specification, migration × bin instrumented by drought × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control. Both conditional first stages exceed ten, which is why the quartile split is used rather than terciles. The first-stage contrast is significant while neither conversion contrast is. Standard errors clustered at the origin district.

census, so the classifier is a property of the place, measured outside the flow this paper studies. It acts as a ceiling rather than a gradient: the first stage collapses to zero beyond roughly 75%, so destinations whose stock is already overwhelmingly informal stop receiving drought migrants altogether. What attracts is the composition of local employment — the industrial, formally employing part visible from a distance, not the overall employment rate ($p = 0.157$) — and proximity: drought migration is overwhelmingly short-distance, sixfold stronger below the within-country median distance than above. A Romano–Wolf correction over the six characteristics leaves two volume contrasts significant at 5% and one at 10%, and none of the twelve conversion contrasts (Appendix Table S3).

Past built-up expansion, built from the GHSL data described in Section 2.5, rules out the natural supply-side reading: a supply constraint predicts the effect should concentrate where building was slow, yet the first stage is two and a half times larger where past expansion was fast ($p = 0.0002$). A binding constraint would deter arrivals, not attract them, so we read built-up growth as a proxy for urban dynamism rather than supply elasticity in the sense of Saiz (2010).

Across all five characteristics the 2SLS conversion of arrivals into low-quality housing is statistically indistinguishable (Appendix Figures S11–S16; $p = 0.853$, 0.236, 0.726, 0.962 and 0.594), as is employment. Housing saturation, industrial composition, distance, past built-up

Figure 6: Destination Characteristics: the Volume of Climate Migration Varies, its Conversion into Low-Quality Housing Does Not




*Notes:* Each contrast spans its 95% Wald interval, with ticks at the bounds of the 90% interval, so a contrast is significant at 10% where the ticks fall on the same side of zero and at 5% where the whole line does. Each row is one destination characteristic from a two-bin saturated interacted IV: instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control. Splits are at the within-country median, except city size, where the pre-specified contrast is metropolises of at least 1M in Africapolis against the other two zones pooled. Panel (a) reports the contrast in the first stage, panel (b) in the two reduced forms and panel (c) in the two 2SLS conversions, blue for low-quality housing and orange for the employed inflow. Of the twelve conversion contrasts only one differs from zero, metropolises housing drought-induced migrants better than the rest, while every other characteristic moves the volume and leaves the conversion unchanged. Samples, top to bottom: $N = 240{,}340$; 157,556; 230,676; 215,556; 241,540; 241,540. Supplementary Appendix Table S6 reports the coefficients bin by bin. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census, CHIRPS climate, Africapolis and GHSL data.

expansion and baseline migrant intensity thus determine how many drought migrants a city receives; city size determines what becomes of them. Where climate migration lands and how badly it is housed are separate questions, and only the second concerns absorption.

### 8.1.4 Where the Pressure Lands

Climate Opens New Corridors to Secondary Cities. The heterogeneity results establish one half of the answer. Secondary and small cities convert drought-induced migrants into low-quality housing at roughly three times the metropolitan rate (0.325 and 0.296 against 0.097), and city size is the only destination characteristic that moves the conversion at all: distance, housing saturation, industrial composition, past built-up expansion and baseline migrant intensity each shift the volume a city receives, none shifts what becomes of the arrivals. But a destination effect alone cannot explain why the burden falls on that tier, since it says nothing about how many migrants reach it. This section supplies the other half: the geography of the flows. Drought opens migration corridors that did not exist, and those new corridors lead to smaller, more deprived cities. New migration corridors disproportionately connect drought-affected origins to smaller, more deprived cities. Although these corridors account for a minority of drought-induced migration, they contribute to the residential burden associated with climate shocks.

Drought does not only intensify established flows; it opens new routes, which carry a small share of the inflow but a disproportionate share of its housing burden. For every corridor we measure a pre-period network: the urban migrants who travelled it in census years −13

Table 9: Climate Opens New Corridors, and They Lead to Smaller, Worse-Housed Cities

*Panel A. Extensive versus intensive margin (causal)*

| | No pre-period network (extensive) | Pre-period network (intensive) |
|---|---|---|
| First stage: urban migration rate | +0.00174*** | +0.00441*** |
| | (0.00051) | (0.00131) |
| First-stage $F$ | 11.84 | 11.36 |
| Reduced form: low-quality housing | +0.00047* | +0.00102 |
| | (0.00028) | (0.00066) |
| Reduced form: employed inflow | +0.00056*** | +0.00245*** |
| | (0.00018) | (0.00059) |
| Share of drought-induced migration | 13.2% | 86.8% |
| Share of the low-quality-housing effect | 15.2% | 84.8% |
| Share of the employment effect | 8.1% | 91.9% |
| Corridor-years | 168,396 | 73,144 |
| Corridors | 12,034 | 4,812 |

*Panel B. What the new corridors are (descriptive, flow-weighted)*

| | Existing | New | Difference | 95% CI |
|---|---|---|---|---|
| *Destination* | | | | |
| Metropolises ($\geq$ 1M), % | 65.6 | 8.9 | -56.7*** | [-60.9, -52.5] |
| Small towns ($<$ 100k), % | 11.0 | 55.4 | +44.4*** | [+40.8, +48.0] |
| Urban share of destination, % | 69.9 | 25.0 | -44.8*** | [-48.0, -41.6] |
| Low-quality housing at destination, % | 45.7 | 58.9 | +13.3*** | [+10.3, +16.3] |
| Built-up growth 1975–1990, % | 69.6 | 101.5 | +31.8*** | [+26.1, +37.5] |
| *Corridor and origin* | | | | |
| Distance, km | 200 | 336 | +136*** | [+111, +161] |
| Urban share of origin, % | 34.8 | 29.4 | -5.5* | [-11.7, +0.8] |
| Mean drought intensity at origin | 0.365 | 0.375 | +0.010 | [-0.007, +0.026] |

*Notes:* A corridor has no pre-period network if no urban migrant travelled it in the three years preceding its 12-year recall window, census years $-13$ to $-15$, measured from the same IPUMS retrospective coding as the estimation panel but strictly outside it. Panel A reports the first stage by group. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. The causal object is the first stage on all corridors without a pre-period network: estimating on the new corridors alone would select on the outcome, since being new requires a positive flow. Shares of the aggregate effect weight each group's coefficient by its realised drought exposure and destination population. Panel B compares the 5,855 corridors that opened during the window with the 4,812 that already existed, excluding the 6,179 that never carried any flow; means are weighted by corridor flow and differences come from a regression on a new-corridor indicator. Panel B is descriptive: it characterises where the new corridors lead and not a causal effect of novelty. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

to −15, outside the estimation window and therefore predetermined with respect to every drought realisation we exploit.[24] Of the 16,846 corridors, 4,812 carried migrants before the window and together move 95.6% of the urban flow; 5,855 carried none then and some during it, moving the remaining 4.4%. Panel A of Table 9 estimates the main specification separately on the two groups. On corridors with no prior network, drought raises the urban migration rate by 0.00174 ($p = 0.0006$, $F = 11.84$), against 0.00441 on established ones. In levels the established response is two and a half times larger, as one expects of corridors an order of magnitude bigger to begin with; relative to their own baseline the ranking reverses, a one-standard-deviation drought raising migration by 7.7% of the mean on corridors without a prior network against 3.6% on those with one. The extensive margin is thus the more climate-sensitive of the two, which is why it carries 15.2% of the low-quality-housing effect for 13.2% of the volume.

New corridors reach smaller, poorer, more distant destinations than established ones, yet drought-induced migrants arriving along them are housed at the same rate as everyone else — the extra deprivation they face is the local rate, not a penalty for arriving by a new route. Panel B characterises the new corridors against the established ones, weighting by flow. They lead elsewhere: 8.9% of the new flow reaches metropolises of at least a million against 65.6% of the established flow, while small towns below 100,000 take 55.4% against 11.0%. Their destinations are markedly worse housed — 58.9% of the resident urban population in low-quality housing against 45.7%, a gap of 13.3 points — lie further away (336 km against 200), and differ in the formality of their labour markets rather than in employment as such: wage and salaried work accounts for 14.1% of the urban population against 23.4%, agriculture for 11.3% against 5.0%. They are not declining places, having built faster over 1975–1990 (101.5% against 69.6%). Panel C then asks whether a migrant arriving along one of these routes fares differently, estimating both conversions jointly on the same panel so that their difference is testable. On housing, nothing: 27.1 of every 100 drought-induced migrants enter low-quality housing on new corridors against 23.0 on established ones ($p = 0.766$), even though the stock at those destinations is 13.3 points more deprived. The extra deprivation migrants face there is the local rate, not a penalty attached to arriving by a new route. On employment there is a gap — 0.321 against 0.555 ($p = 0.026$) — consistent with destinations offering half as much wage work, though we read it as suggestive: it is a step rather than a gradient, and having no prior network is strongly correlated with being a small corridor.

## 8.2 Heterogeneity by Migrant: Gender, Age and Education

Appendix Table 37 profiles the drought-induced migrants themselves, comparing arrivals in high- versus no-drought years with the complier characterization of Abadie (2003). Both readings agree: drought-induced migrants are observationally similar to other migrants - the same gender balance and comparable or slightly higher education - but younger, with an over-representation of children under fifteen (42% versus 29%), consistent with whole-household displacement rather than the selective migration of young workers.

[24] Older recall is noisier, but since the variable only stratifies the sample, measurement error attenuates contrasts without biasing coefficients. Corridors with no prior network are kept in the estimation sample whether or not they later carry flow: restricting to those that do would condition on the outcome.

The effect itself is equally unselective on gender and age. Appendix Figures 19 and 20 re-estimate the main specification with subgroup-specific inflows and migration rates: drought displaces women and men, children and adults, proportionally alike (panel (a): relative first stages of +6 to +7% of each group's mean migration rate), and the housing and employment responses are correspondingly uniform across both dimensions.

Education is the exception, and it cuts in opposite directions on the two stages. Drought does not displace the least-educated adults: their migration response is indistinguishable from zero and significantly below that of the secondary-educated (−8.6 percentage points of relative response, $p = 0.001$) - a trapped-population pattern. Among the adults it does displace, the migration response tilts if anything toward the more educated (+8.8% against +5.6%, $p = 0.064$), yet the housing effect runs the other way - twice as strong for the primary-educated ($p = 0.036$, both groups well identified with conditional first stages above 14) - while employment absorption is education-blind ($p = 0.787$). Education thus eases displacement but shields from low-quality housing: the least educated stay behind, the secondary-educated escape bad housing, and the lower-educated among the displaced bear the residential burden. We verify that this selection operates along a single, additive dimension: a fully saturated model crossing gender, age at arrival, and education into twelve profile cells rejects equality of the first-stage responses (joint $p = 0.024$) but finds no interaction beyond the additive margins (joint $p = 0.59$) - drought selects migrants along a monotone educational gradient, from near-zero mobility among the least educated to the strongest response among the secondary-educated, without any distinct combined profile. The separate margins of Figure 19 are therefore a sufficient statistic for who drought moves (Supplementary Appendix Table S11). Attainment is measured at census and may post-date arrival, so the two displacement-side gradients remain suggestive. Full estimates are in Supplementary Appendix Tables S7–S10. Together with the profile evidence, the pattern points to barriers rather than composition: similar people, absorbed uniformly into work, sorted selectively into housing.

# 9 Discussion: Interpreting Urban Integration

The results highlight a distinction between labor-market and residential integration. Drought-induced migration increases both employment and low-quality housing inflows, with a substantial number of migrants observed in both conditions simultaneously. Moreover, housing responses vary considerably across destination city sizes, while employment responses are more similar across the urban hierarchy. These findings raise the question of why migrants can participate in urban labor markets without necessarily accessing adequate housing.

We use a simple two-margin Roy framework to illustrate one possible explanation (Appendix Figure 23). Migrants sort between high- and low-integration states along two dimensions: employment and housing. In each dimension, their choices depend on their relative returns to the two states and on potential barriers to accessing the higher-integration state. A barrier to adequate housing may therefore prevent migrants from accessing better residential conditions even when they participate in urban employment. Because these barriers need not be identical across the two dimensions, employment and residential integration can diverge.

This framework offers an interpretation of the stronger low-quality housing response observed in small and secondary cities. Differences in residential access across destinations could contribute to this pattern, alongside variation in housing prices, preferences, and migrant composition. Distinguishing these channels would require additional information on local housing markets. Our results therefore point to the importance of residential conditions in understanding climate-related urban integration, particularly outside the largest metropolitan areas.

# 10 Robustness

This section presents a series of robustness checks and alternative specifications designed to assess the stability of the main results and the validity of the identification strategy. Several of these checks are reported in the appendix - the fixed-effects saturation in Table 2, the definition sweep, and the wet-side placebo of Figure 14 - and the remaining estimates are in progress.

*Alternative specifications.* - We consider four alternative specifications. First, we estimate a manually specified two-instrument design combining $\text{Drought}^g * p10$ and $N^g * dry$ to examine whether the results are robust to multi-instrument identification despite a weaker joint first stage. Second, we replace drought intensity with extreme precipitation measures based on high-percentile rainfall shocks to test whether migration responses are symmetric across the rainfall distribution or primarily driven by adverse drought events. Third, following Hultgren et al. (2025), we perform an out-of-sample validation to evaluate the stability of the post-LASSO instrument selection across held-out countries. Fourth, we replace the census-based destination population denominator with the annually reconstructed population series, $\text{Pop}_{d,t}$, to assess the sensitivity of the estimated migration rates to the choice of normalization.

*Alternative samples.* - To assess the external validity of the main findings, we extend the analysis to all fifteen countries in the dataset by exploiting five-year migration intervals rather than restricting attention to the eight countries with annual migration information. This exercise evaluates whether the estimated scale effect generalizes beyond the baseline sample.

*Sensitivity analyses.* - We conduct two additional sensitivity analyses. First, we replace the binary low-quality housing indicator with a continuous measure of housing deprivation defined as the number of deficient UN-Habitat dimensions (ranging from 0 to 5), allowing us to test whether drought-induced migration affects the entire distribution of housing quality rather than only the extensive margin of low-quality housing residence. Second, we perform a leave-one-country-out exercise to verify that the estimated effects are not driven by any single country in the sample.

*Leave-one-country-out.* - Because the eight countries differ in size, census vintage and housing levels, we re-estimate the main specification eight times, each time dropping one

country (Appendix Figure 22). The estimate stays positive in every subsample and ranges from 0.197 to 0.340 against 0.285 on the full sample, with each subsample estimate inside the full-sample confidence interval. Precision falls where the first stage weakens: dropping Benin or Kenya lowers the first-stage $F$ to about 13 or 15, and in those two subsamples the Anderson–Rubin interval includes zero, a loss of power rather than a change in the estimate.

*Level of clustering.* - All estimates in the paper cluster standard errors at the origin district, which allows arbitrary correlation across the destinations and years a given origin feeds. Appendix Table S4 reports the four main results under three alternatives. Because the clustering level affects only the variance, **the coefficients are identical across columns**; what changes is precision. The two scale results survive at $p < 0.0001$ under every level. The housing propensity is the margin that moves: its $p$-value goes from 0.043 in the main specification to 0.058 when clustering at the origin $\times$ year cell and to 0.092 under two-way clustering, which is consistent with it being the paper's least robust result and reinforces the reading of Section 12.4.5 that the aggregate effect is a scale phenomenon. The employment propensity holds at the 10% level throughout. The demanding check is two-way clustering on origin and year, since it additionally permits arbitrary correlation across all origins within a year – precisely the spatial synchrony of shocks documented in Section 4.2, where the weighted pairwise correlation of drought across a destination's origins has a median of 0.67. All four results survive it. The first-stage $F$ falls to 8.08 in that column, but this reflects the year dimension supplying only 38 clusters rather than any weakening of the instrument: the first-stage coefficient is unchanged, and the $F$ is 29.65 and 24.66 under the two one-way levels on the same sample. We retain origin-level clustering as the main specification.

## 10.1 Exclusion Restriction

nterpreting the 2SLS estimates as causal effects of migration requires that origin drought affect destination outcomes only through migration flows. Because drought may also affect migrants' resources before departure, we primarily interpret these estimates as housing and employment responses relative to drought-induced migration. The natural threat is geographic. Origins lie close to the destinations they feed, so origin drought may be correlated with shocks *at destination* – its own rainfall, conflict, agricultural prices – that could move destination housing and employment directly, bypassing migration altogether. This subsection tests that possibility on three channels, in each case by adding the destination shock as a control and by asking whether it moves the outcome on its own.

*How much of the threat survives the fixed effects.* - Less than one might fear. The raw correlation between origin drought and destination drought is 0.55 – not the 0.95 that the shift-share exposure index displays, because that index averages over a whole portfolio of nearby origins and is therefore far more collinear with local weather than any single origin is. Residualising both on corridor and arrival-year fixed effects brings the correlation down to 0.35, so only about 12% of the instrument's variance is even potentially contaminated, and the instrument retains 81% of its raw dispersion. The design absorbs most of the problem before any control is added, because it compares years within a corridor rather than

destinations with each other.

*Controlling for destination shocks.* - Table 10 reports the two scale results with and without each destination control. Because the three controls have different coverage, each forms a two-column group estimated on *its own* subsample, with the no-control column reported alongside: the comparison within a group is a pure control effect, and differences across groups are differences of coverage. This matters most for conflict, where ACLED's 1997 start restricts the panel to 70% of corridor-years, so the relevant benchmark is the 0.256 estimated on that subsample rather than the 0.285 of the full panel.

No control moves either result. The housing coefficient goes from 0.285 to 0.308 when destination drought is added, from 0.256 to 0.255 under conflict, and from 0.278 to 0.266 under crop prices; the employed inflow is unchanged in all three. The first-stage $F$ is reported for both columns of each group and is essentially unaffected – 18.99 against 19.38 for conflict, 29.38 against 27.83 for prices – which is what one expects of controls measured at the destination: they do not touch the origin-to-migration relationship. Destination drought is the one that costs something, 29.65 to 23.73, for the spatial-correlation reason above. Entering all three jointly is a more demanding exercise and is reported separately in Supplementary Appendix Table S2; the housing result survives there too, but with the instrument visibly weakened.

*Placebos.* - The control test asks whether the destination shock changes the estimate; the placebo asks the prior question of whether it moves the outcome at all. Table 11 reports reduced forms in which no migration rate appears. Neither destination drought nor destination conflict has any detectable effect on the low-quality-housing inflow ($p = 0.50$ and $p = 0.98$), while origin drought retains its effect in the same regressions. There is simply no destination-weather or destination-conflict channel into housing to worry about.

The one placebo that bites is crop-price exposure, and it bites in the wrong direction to be a threat: the coefficient is *negative* and precisely estimated ($-0.0104$, $p < 0.001$). A favourable world price in a cropping destination *reduces* the inflow into low-quality housing, which is what one would expect if agricultural income retains people in the rural part of the unit rather than pushing them into its urban part. A channel of that sign cannot manufacture the positive housing effect we estimate; if anything it works against it.

*The effect is specific to the arrival cohort.* - A direct destination shock would affect everyone living there, not only that year's arrivals. It does not. Adding leads and lags of origin drought to the housing reduced form, the contemporaneous term is $+0.00109$ ($p = 0.025$) while drought three years *ahead* is $-0.00004$ ($p = 0.935$): a future shock at origin tells us nothing about the housing of the cohort arriving today. The small negative lag ($-0.00062$, $p = 0.096$) is consistent with modest intertemporal substitution – a past drought has already displaced some of the people who would otherwise have moved – rather than with a destination-wide shock.

*What this leaves unresolved.* - Two things, stated plainly. First, these tests cover the three destination-level confounders we can measure; they cannot speak to one we have not thought

of. Second, the propensity margin is more fragile than the scale margin here as elsewhere: Supplementary Appendix Table S1 shows that within each control group it barely moves, but that the no-control estimates differ substantially *across* groups – from 1.216 on the full panel to 1.831 on the ACLED subsample – so its apparent sensitivity is largely a matter of coverage, on a sample where the first stage is in any case weaker. The scale results, which carry the paper's conclusions, are not threatened by destination-level confounding through any of the three channels we can test. A shift-share design on destination outcomes would be the natural way to go further; Section 4.2 explains in a footnote why the spatial synchrony of the shocks and the absence of an earlier census for five of our eight countries make it infeasible here.

Table 10: Exclusion Restriction: Destination Shocks and the Scale Results

| | Destination drought | | Destination conflict | | Destination crop prices | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | No control | With control | No control | With control | No control | With control |
| *Panel A. Low-quality housing inflow per 100 urban residents* | | | | | | |
| Urban migration rate | 0.28455*** | 0.30748*** | 0.25556*** | 0.25529*** | 0.27763*** | 0.26575*** |
| | (0.06323) | (0.07332) | (0.05460) | (0.05391) | (0.06453) | (0.06707) |
| First-stage $F$ | 29.65 | 23.73 | 18.99 | 19.38 | 29.38 | 27.83 |
| AR 95% CI | [0.13, 0.40] | [0.12, 0.44] | [0.12, 0.36] | [0.12, 0.36] | [0.12, 0.39] | [0.10, 0.38] |
| Observations | 241,540 | 241,540 | 169,920 | 169,920 | 240,340 | 240,340 |
| *Panel B. Employed inflow per 100 urban residents* | | | | | | |
| Urban migration rate | 0.42760*** | 0.40278*** | 0.53594*** | 0.53583*** | 0.42959*** | 0.43142*** |
| | (0.04330) | (0.04918) | (0.05234) | (0.05190) | (0.04440) | (0.04605) |
| First-stage $F$ | 29.65 | 23.73 | 18.99 | 19.38 | 29.38 | 27.83 |
| AR 95% CI | [0.35, 0.53] | [0.32, 0.53] | [0.44, 0.68] | [0.44, 0.68] | [0.35, 0.54] | [0.35, 0.54] |
| Observations | 241,540 | 241,540 | 169,920 | 169,920 | 240,340 | 240,340 |

*Notes:* Outcomes: the paper's two scale results, the low-quality-housing and employed in-migrant inflows. Sample: one subsample per control group, the odd column of each pair reporting that subsample without the control, so a within-group comparison is a pure control effect while differences across groups are differences of coverage. Destination drought applies the instrument's construction to the destination unit (100% of the panel); destination conflict is any battle, violence against civilians or remote-violence event plus $\log(1 + \text{fatalities})$, from ACLED matched by point-in-polygon, 70% of the panel from 1997; crop prices are $\sum_c \text{share}_{cd} \log(P_{ct})$ and its first difference, from CROPGRIDS shares for nine traded crops and World Bank Pink Sheet prices, 99.5% of the panel. Specification: 2SLS of the urban migration rate instrumented by contemporaneous drought intensity at origin, origin–destination-pair and arrival-year fixed effects, distance-to-census control, with cluster-robust Anderson–Rubin 95% intervals; the instrument is origin drought in every column and every control is measured at the destination. No control moves either result, and the first-stage $F$, reported for both columns of each group, is essentially unaffected; destination drought is the one that costs something, 29.65 to 23.73, rainfall being spatially correlated between an origin and its destinations. Supplementary Appendix Table S2 enters the three controls jointly. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: Placebo: Do Destination Shocks Move Destination Outcomes Directly?

| | (1) Low-quality housing inflow | (2) Employed inflow | (3) Share in low-quality housing |
|---|---|---|---|
| *Panel A. One shock at a time, each on its own maximal coverage* | | | |
| Destination drought | +0.00020 | +0.00084*** | +0.00428 |
| | (0.00029) | (0.00022) | (0.00366) |
| Destination conflict (any violent) | +0.00001 | +0.00060* | +0.00597 |
| | (0.00022) | (0.00032) | (0.00487) |
| Destination log(1 + fatalities) | +0.00019 | +0.00044** | +0.00078 |
| | (0.00015) | (0.00020) | (0.00225) |
| Destination crop-price exposure | -0.01038*** | -0.00772*** | -0.01534 |
| | (0.00195) | (0.00160) | (0.03940) |
| Destination crop-price shock | -0.00079 | +0.00331 | +0.05995 |
| | (0.00278) | (0.00295) | (0.06026) |
| Observations | 241,540 | 241,540 | 72,723 |
| *Panel B. All shocks jointly, common subsample* | | | |
| Destination drought | +0.00099*** | +0.00131*** | -0.00179 |
| | (0.00028) | (0.00027) | (0.00361) |
| Destination conflict (any violent) | -0.00011 | +0.00045 | +0.00680 |
| | (0.00022) | (0.00032) | (0.00491) |
| Destination log(1 + fatalities) | +0.00023 | +0.00048** | +0.00061 |
| | (0.00015) | (0.00020) | (0.00226) |
| Destination crop-price exposure | -0.00776*** | -0.00744*** | -0.02104 |
| | (0.00189) | (0.00159) | (0.03859) |
| Destination crop-price shock | -0.00241 | +0.00430 | +0.02952 |
| | (0.00280) | (0.00322) | (0.06308) |
| Origin drought (the instrument) | +0.00041 | +0.00137*** | +0.01041** |
| | (0.00028) | (0.00033) | (0.00415) |
| Observations | 168,820 | 168,820 | 52,593 |

*Notes:* Outcomes: destination outcomes regressed on destination shocks and origin drought jointly. No migration rate enters these regressions, so they ask whether a destination shock moves destination outcomes through any channel. Sample: Panel A estimates each shock on its own maximal coverage, destination drought on 100% of the panel, conflict on 70% and crop prices on 99.5%; Panel B puts all of them in one regression on the subsample where all are observed. Column (1) is the outcome the paper's main result is built on. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Neither destination drought nor destination conflict has a direct effect on the low-quality-housing inflow, while origin drought retains its effect throughout. Crop-price exposure does enter column (1), with the sign opposite to the paper's effect: a positive world price shock in a cropping destination lowers the inflow, consistent with farm income retaining people, so it cannot generate the estimated effect through a direct channel. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

*Exclusion restriction.* - The identifying assumption may be violated if drought at origin affects destination outcomes through channels other than migration, such as food price transmission, remittance flows, or government infrastructure responses. Origin–destination pair fixed effects absorb time-invariant correlations between origin characteristics and destination outcomes, while arrival-year fixed effects capture aggregate shocks common to all migration corridors.

*Crowding-out.* - The scale outcome counts every migrant of the arrival cohort living in low-quality housing, not only those whom drought displaced. A positive coefficient is therefore consistent with two mechanisms: drought-induced migrants settling in low-quality housing themselves, or drought-induced migrants crowding other arrivals out of decent housing into it. The aggregate margins cannot separate them, because on the sample where housing is observed the decent-housing response is the exact arithmetic complement of the low-quality one.

Two checks bound the concern. First, drought-induced migration does not degrade the housing of migrants who arrived on the same corridor in *earlier* years, who cannot be compliers of the current shock: the weak-instrument-robust interval on that outcome excludes any effect above 0.138 per marginal migrant, and a placebo on their total number is flat (Appendix Table 39). Crowding-out, if it operates, does not spill onto previously settled migrants. Second, the estimated effect (0.285) lies *below* the benchmark that no crowding-out and average sorting would imply — the baseline low-quality-housing share among migrants, 0.359 — whereas crowding-out would push it above.

Neither check identifies crowding-out *within* the arrival cohort, which is the channel that would bias the estimate. A direct test on the one migrant group whose own inflow does not respond to drought — adults with less than primary education, and therefore almost entirely always-takers — is inconclusive: their low-quality-housing propensity rises with drought-induced migration by 1.25, of the same order as the aggregate propensity effect, but the Anderson–Rubin interval, $[-0.21, +3.53]$, contains both zero and substantial crowding-out (Appendix Table 40). We therefore read the coefficient as a total effect on the corridor — each hundred additional drought-induced migrants is associated with twenty-eight additional migrants living in low-quality housing — rather than as a claim about which individuals occupy which dwellings.

*Comparison with other migration drivers.* - Two additional exercises will compare drought-induced migration with migration generated by alternative shocks. The first instruments migration using conflict events at origin (ACLED data); the second uses exogenous variation in mine openings at the admin-unit level. These comparisons sharpen the interpretation of the LATE estimated in the main analysis by contrasting push driven (drought, conflict) and pull-driven (mining) migration mechanisms and their respective urban absorption patterns.

# 11 Conclusion

This paper examines how climate-induced migration shapes urban integration in Sub-Saharan Africa, focusing on migrants' housing and employment outcomes. Using harmonized census

microdata from eight African countries, we reconstruct annual bilateral migration flows over 1981–2018 and link migrants' retrospective migration histories to their subsequent living conditions. Exploiting variation in drought exposure at origin along the same migration corridors, we estimate the housing and employment responses associated with drought-induced migration and express these responses relative to the additional migration generated by drought.

Our findings reveal an important tension between labor-market participation and residential integration. For every hundred additional migrants associated with drought, approximately twenty-eight are observed living in low-quality housing, corresponding to roughly 110,000 additional residents across our sample. Yet many of these migrants participate in urban labor markets. In our baseline estimates, 85% of the drought-associated inflow into low-quality housing is employed, compared with 61% among those in adequate housing, although this difference is not robust to destination-specific trends. More importantly, drought increases the number of migrants who are simultaneously employed and living in deprived housing conditions—the urban working poor—a finding that remains robust across specifications. These housing responses are particularly pronounced in small and secondary cities, where approximately 30–33 of every hundred additional migrants are observed in low-quality housing, compared with around 10 in metropolises. The migration response is also substantially stronger in origins with greater shares of cultivated land, consistent with an agricultural-push mechanism.

These findings have implications for both climate adaptation and urban policy. Migration can provide access to urban employment without necessarily ensuring adequate housing, and these residential challenges are particularly pronounced outside the largest metropolitan areas. Urban policies focused primarily on large cities may therefore overlook smaller and secondary cities receiving climate-responsive migrants, where housing deprivation is already widespread. Strengthening access to adequate housing and basic urban services in these destinations could improve living conditions for climate migrants alongside other urban residents.

More broadly, our results suggest that evaluating migration as a climate adaptation strategy requires looking beyond migration decisions and employment outcomes to consider the conditions under which migrants live at destination. Further research combining migration histories with housing prices, incomes, and local housing-market institutions could help identify the constraints underlying residential deprivation and evaluate the welfare effects of policies that improve access to adequate housing.

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# 12 Appendix

## 12.1 Descriptive Statistics

Table 12 reports descriptive statistics for all variables used in the analysis: the corridor-year estimation panel (Panel A), the migrant propensity outcomes (Panel B), and destination characteristics (Panel C).

Table 12: Descriptive Statistics

| Variable | Mean | SD | P25 | Median | P75 | $N$ |
|---|---|---|---|---|---|---|
| *Panel A: Corridor-year panel (main estimation sample)* | | | | | | |
| Urban migration rate (per 100 urban residents) | 0.0268 | 0.1003 | 0.0000 | 0.0000 | 0.0099 | 241,540 |
| Low-quality housing inflow rate (per 100) | 0.0120 | 0.0589 | 0.0000 | 0.0000 | 0.0000 | 241,540 |
| Employed inflow rate (per 100) | 0.0106 | 0.0405 | 0.0000 | 0.0000 | 0.0000 | 241,540 |
| Unemployed inflow rate (per 100) | 0.0013 | 0.0095 | 0.0000 | 0.0000 | 0.0000 | 241,540 |
| Inactive inflow rate (per 100) | 0.0118 | 0.0522 | 0.0000 | 0.0000 | 0.0000 | 241,540 |
| Drought intensity at origin | 0.350 | 0.502 | 0.000 | 0.000 | 0.625 | 241,540 |
| Cumulative 3-season drought at origin | 0.366 | 0.297 | 0.106 | 0.347 | 0.551 | 235,920 |
| Distance to census $\lvert t - t_c \rvert$ (years) | 6.08 | 3.29 | 3.00 | 6.00 | 9.00 | 241,540 |
| Urban population at destination ('000) | 123.8 | 315.6 | 16.0 | 41.4 | 95.9 | 241,540 |
| *Panel B: Migrant propensity outcomes (corridors with positive denominators)* | | | | | | |
| Share of migrants in low-quality housing | 0.406 | 0.388 | 0.000 | 0.333 | 0.750 | 72,723 |
| Mean number of deprived dimensions (0–5) | 0.662 | 0.818 | 0.000 | 0.429 | 1.000 | 72,723 |
| Share of migrants employed | 0.505 | 0.362 | 0.200 | 0.500 | 0.842 | 71,503 |
| Share of migrants unemployed | 0.058 | 0.155 | 0.000 | 0.000 | 0.036 | 71,503 |
| Share of migrants inactive | 0.437 | 0.358 | 0.000 | 0.429 | 0.667 | 71,503 |
| *Panel C: Destination characteristics (one row per destination, most recent census)* | | | | | | |
| Urban population share of the destination | 0.256 | 0.241 | 0.085 | 0.188 | 0.351 | 351 |

*Notes:* Variables used in the analysis. Panel A is the corridor-year estimation panel: 8 countries with Egypt excluded, Mali 1998 included, 12-year capped windows excluding the census year, destinations of at least 2,000 urban residents. The panel is the full origin–destination rectangle and keeps pairs that never record an urban migrant, 37% of the 16,846 pairs over the window and 70% of corridor-years, which is what puts the lower quartile of the rate variables at zero; those corridor-years are kept because dropping them would condition the sample on the endogenous regressor. Rates are counts per 100 urban residents at destination and no outcome is winsorized. Drought intensity is the standardized negative seasonal rainfall anomaly at origin, $\max(0, -\tilde{z})$, zero by construction in wetter-than-average seasons; the cumulative version averages the current and two preceding seasons, where lags exist. The inactive inflow counts migrants out of the labour force with determined status. Panel B reports shares among migrants, with migrants as the housing denominator and migrants of known labour-force status as the employment one, on corridor-years with a positive denominator. Panel C is the census urban population share, one observation per destination at its most recent census.

Table 13 breaks the two outcome shares down by migrant education and by destination city size. It is purely descriptive, and its role is to give the baseline against which the causal

estimates are read. Three features of that baseline matter. First, the housing gradient in education is steep: 49.5% of urban in-migrants with less than primary schooling live in low-quality housing, against 35.0% of the primary-educated and 29.1% of those with secondary schooling or more. Second, the same gradient is present on the employment margin and is of the same magnitude – in log-odds, +0.87 on housing and +0.87 on employment between the top and bottom education groups – so education is not a dimension along which the two margins separate. Third, the low-quality-housing share falls with city size, from 44.1% in small towns to 37.1% in secondary cities and 33.8% in metropolises, while the employment share is nearly flat across the three (49.2, 48.6 and 53.1%). It is the destination, not the education of the migrant, that separates the residential from the occupational margin, and it does so in the raw data before any instrument is applied.

We stress what this table does not establish. A difference in shares across education levels mixes any access friction with the composition of skills, since education shifts both, and no observational comparison can separate them. The paper's causal statements rest instead on the instrumented estimates, which compare drought-induced migrants with the infra-marginal migrants arriving on the same corridor – a comparison that holds skills fixed because, as Section 8.2 shows, drought-induced migrants are not selected on education. The purpose of the table is to show that the gradients above are features of the migrant population as a whole, so that the finding that drought-induced migrants settle at their destination's local rate means the climate shock adds arrivals to this landscape rather than altering it.

Table 14 documents the panel census by census. For each of the eight countries whose censuses record a year of arrival, it reports the range of arrival years that census can describe, the number of origins, destinations and corridors it contributes, and the urban in-migrants it observes. The eleven census samples together yield 241,540 corridor-years, 16,846 corridors and 11.6 million urban in-migrants, of whom 36% live in low-quality housing and 51% are employed. Both figures are shares of the urban in-migrants themselves, not of the destination's resident population, and they vary widely across census samples – from 15% to 55% on housing and from 31% to 65% on employment – which is why the estimating equations absorb destination and arrival-year effects throughout. Coverage is uneven by construction: Benin 2013 alone supplies 28% of the corridor-years, since its 77 origins and 75 destinations generate 5,700 corridors over a full twelve-year window, while Malawi 2008 supplies 3%. Arrival windows are shorter than twelve years in three cases because a preceding census in the same country already describes the earlier years – Kenya 2019, Mali 2009 and Zambia 2000 – and in one further case, Zambia 1990, because the CHIRPS rainfall record begins in 1981 and earlier arrival years cannot be matched to a shock.

### 12.1.1 Descriptive Maps

### 12.1.2 Housing-Quality Definition

Figure 12 shows how the index behaves in the data. For each country it reports the share of households counted deprived on each dimension, separately for all urban residents and for the urban in-migrants on whom the paper's outcome is built. Three features are worth noting. First, in-migrants are less deprived than the resident urban population in every country and on almost every dimension, by 6 to 19 percentage points on the summary measure: recent

Table 13: Housing and Employment of Urban In-Migrants, by Education and Destination City Size

| | Share in low-quality housing (%) | Share employed (%) | Migrants |
|---|---|---|---|
| *Panel A. By education (migrants aged 25 and over)* | | | |
| Less than primary | 49.5 | 60.3 | 900,280 |
| Primary completed | 35.0 | 69.2 | 2,380,194 |
| Secondary or more | 29.1 | 78.4 | 2,647,353 |
| *All aged 25 and over* | 34.6 | 72.0 | 5,927,827 |
| *Panel B. By destination city size (all migrants)* | | | |
| Small town (<100k) | 44.1 | 49.2 | 1,500,424 |
| Secondary city (100k–1M) | 37.1 | 48.6 | 2,754,880 |
| Metropolis (≥1M) | 33.8 | 53.1 | 7,285,082 |
| *Panel C. Education × destination city size (aged 25 and over)* | | | |
| *Small town (<100k)* | | | |
| Less than primary | 62.9 | 62.1 | 158,170 |
| Primary completed | 42.7 | 72.9 | 268,554 |
| Secondary or more | 31.1 | 81.0 | 289,399 |
| *Secondary city (100k–1M)* | | | |
| Less than primary | 53.3 | 60.2 | 237,723 |
| Primary completed | 37.2 | 70.4 | 520,449 |
| Secondary or more | 29.0 | 79.9 | 535,977 |
| *Metropolis (≥1M)* | | | |
| Less than primary | 43.4 | 59.8 | 500,238 |
| Primary completed | 32.9 | 68.2 | 1,581,991 |
| Secondary or more | 28.9 | 77.6 | 1,807,966 |
| All urban in-migrants | 35.9 | 51.5 | 11,603,276 |

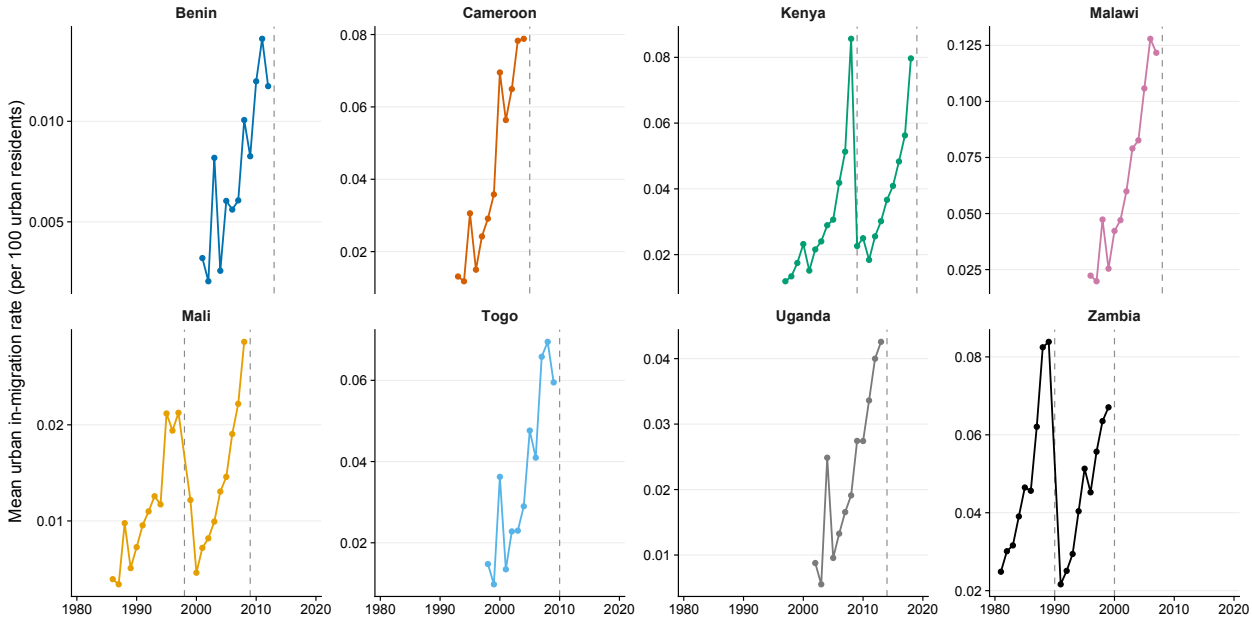
*Notes:* Outcomes: observed shares among urban in-migrants in the estimation panel, pooled over corridor-years and weighted by migrant counts, so each cell is an individual-level share and not an average of corridor shares. Low-quality housing follows the paper's main definition with migrants whose housing is observed as the denominator, and employment is the share employed among those whose labour-force status is observed, so the two columns differ slightly in denominator and the 'Migrants' column reports the housing one. Sample: education is observed only from age 25, so Panels A and C cover that group, 5,927,827 migrants or 51% of the panel, while Panel B and the final row cover all migrants; that age restriction, and not the education split, is why employment is higher in Panels A and C, and the subtotal row makes the two comparable. City size is the Africapolis class, available for 99% of the panel. The table is descriptive and carries no causal content: a difference in shares across education levels mixes any access friction with the composition of skills, which observational data cannot separate. It supplies the baseline against which the instrumented estimates of Tables **??** and 28 are read, those comparing drought-induced migrants with the infra-marginal arrivals on the same corridor.

Table 14: Coverage of the Estimation Panel, by Country and Census

| Census | Arrival window | Years | Origins | Destinations | Corridors | Corridor-years | Urban in-migrants | Share of urban in-migrants (%) | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | In low-quality housing | Employed |
| *Benin* | | | | | | | | | |
| 2013 | 2001–2012 | 12 | 77 | 75 | 5,700 | 68,400 | 470,870 | 38 | 46 |
| *Cameroon* | | | | | | | | | |
| 2005 | 1993–2004 | 12 | 39 | 39 | 1,482 | 17,784 | 1,914,190 | 19 | 31 |
| *Kenya* | | | | | | | | | |
| 2009 | 1997–2008 | 12 | 38 | 38 | 1,406 | 16,872 | 2,595,370 | 33 | 64 |
| 2019 | 2009–2018 | 10 | 38 | 38 | 1,406 | 14,060 | 2,954,250 | 49 | 55 |
| *Malawi* | | | | | | | | | |
| 2008 | 1996–2007 | 12 | 26 | 26 | 650 | 7,800 | 385,220 | 54 | 46 |
| *Mali* | | | | | | | | | |
| 1998 | 1986–1997 | 12 | 47 | 47 | 2,162 | 25,714 | 190,090 | 55 | 43 |
| 2009 | 1999–2008 | 10 | 47 | 25 | 1,150 | 11,500 | 310,060 | 44 | 48 |
| *Togo* | | | | | | | | | |
| 2010 | 1998–2009 | 12 | 36 | 35 | 1,225 | 14,700 | 447,140 | 36 | 54 |
| *Uganda* | | | | | | | | | |
| 2014 | 2002–2013 | 12 | 36 | 36 | 1,260 | 15,120 | 1,140,030 | 15 | 65 |
| *Zambia* | | | | | | | | | |
| 1990 | 1981–1989 | 9 | 53 | 54 | 2,810 | 25,290 | 552,210 | 46 | 40 |
| 2000 | 1991–1999 | 9 | 55 | 50 | 2,700 | 24,300 | 641,590 | 39 | 38 |
| All | 1981–2018 | 38 | 354 | 351 | 16,846 | 241,540 | 11,601,020 | 36 | 51 |

*Notes:* Rows are the census samples of the eight countries that record a year of arrival, so that bilateral yearly flows can be constructed; censuses listed in Section 2 but absent here lack that question. Origins and destinations are IPUMS units, GEOLEVEL2 except in Malawi and Uganda, which enter at GEOLEVEL1. The arrival window is twelve years, excludes the census year, and is truncated where a preceding census in the same country already describes the earlier years (Kenya 2019, Mali 2009, Zambia 2000) or where it would reach before 1981, when the CHIRPS record begins (Zambia 1990). The last two columns are shares of the urban in-migrants themselves and not of the destination's resident population, the share in low-quality housing among those whose housing is observed and the share employed among those whose labour-force status is observed, so the two denominators differ slightly from each other and from the 'Urban in-migrants' column, which reports the housing one. The panel holds 241,540 corridor-years of a constructed table of 300,431. One sample is constructed and cannot enter, Zambia 2010, whose harmonized urban classification is missing on every record.

Figure 7: Average Urban In-Migration Rates Over Time, by Country



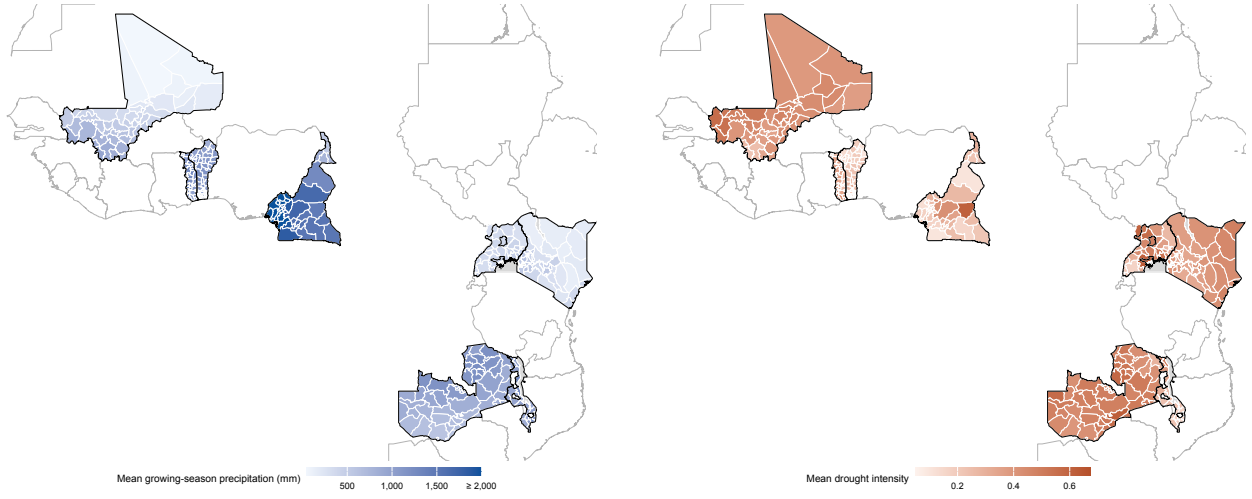

*Notes:* Mean urban in-migration rate, in-migrant arrivals per 100 urban residents at destination, across corridor-year observations of the yearly estimation panel, by calendar year and country; dashed vertical lines mark census years, each census contributing a capped retrospective window of yearly migration cohorts. Within each window, rates rise mechanically toward the census year, reflecting recall and the onward-migration attrition of earlier cohorts; the distance-to-census control absorbs this gradient in estimation. Y-axis scales differ across panels.

*Source:* Authors' computations from IPUMS census microdata.

Figure 8: Descriptive Maps: Climate at Origin

(a) Mean growing-season precipitation

(b) Mean drought intensity



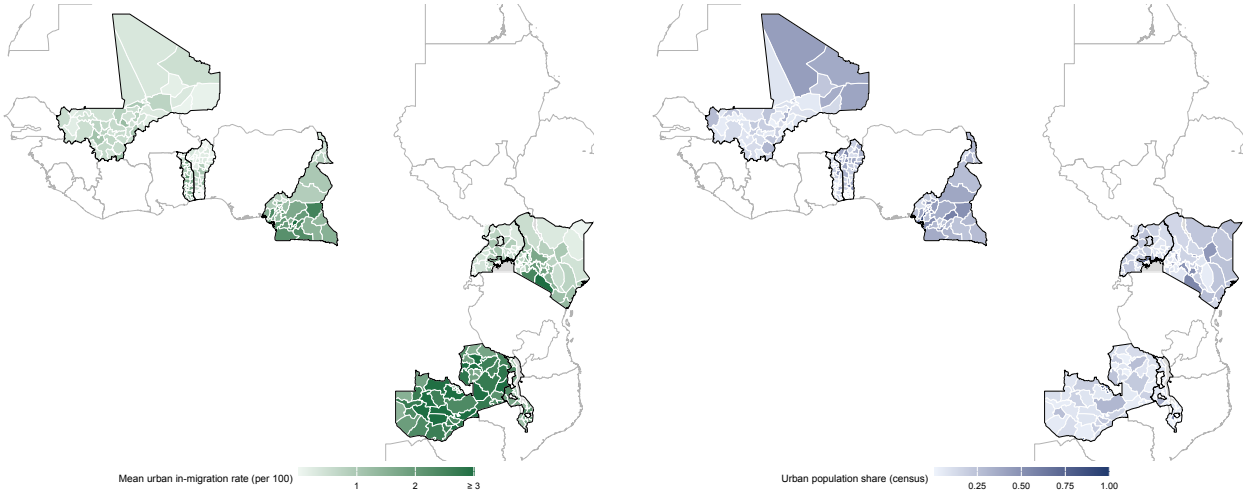

*Notes:* The eight countries of the estimation sample are outlined in black, with their administrative units, second level and first level for Malawi and Uganda; other African countries covered by IPUMS are outlined in grey. Panel (a) is the mean cumulative precipitation over the growing season in millimetres by origin unit, the $\text{Precip}^{g}_{cum}$ measure underlying the instrument, and panel (b) the mean of the standardized drought intensity $\max(0, -\tilde{z})$ over the panel years. The colour scale of panel (a) is capped at 2,000 mm: fourteen of the 354 origin units, all in southern Cameroon, lie above that level and are drawn in the darkest shade, which keeps the gradient readable across the other seven countries.

*Source:* Authors' computations from CHIRPS data and IPUMS harmonized shapefiles.

Figure 9: Descriptive Maps: Migration and Urbanization at Destination

(a) Mean urban in-migration rate

(b) Urban population share



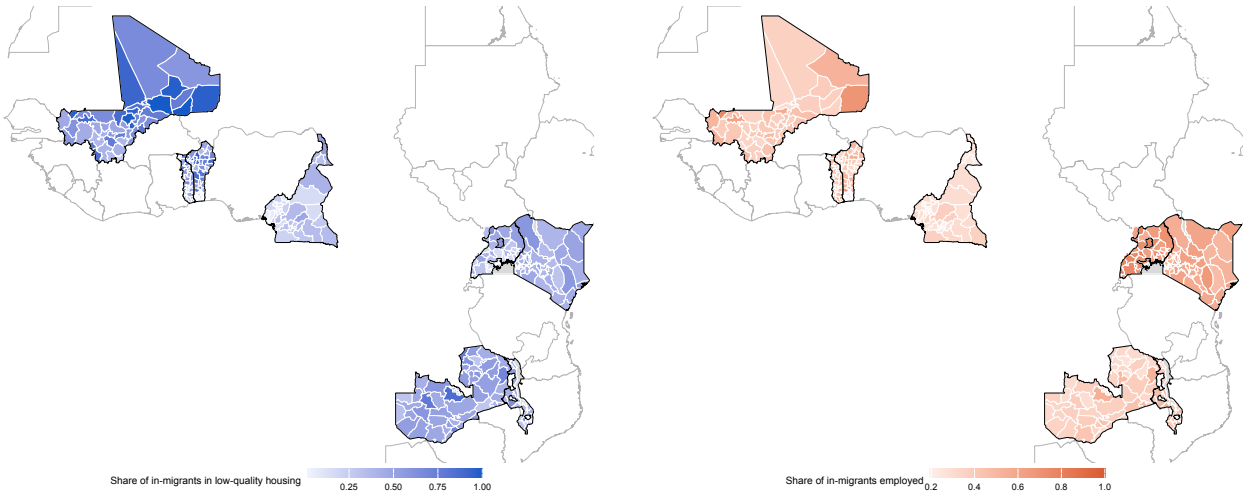

*Notes:* Panel (a) is the mean yearly urban in-migrant arrivals per 100 urban residents by destination unit, aggregated over all origins, and panel (b) the urban population share of the destination, urban over total census population at the most recent census. The colour scale of panel (a) is capped at 3 arrivals per 100 residents: sixteen of the 351 destination units, mostly Zambian, lie above that level and are drawn in the darkest shade. Grey units are not observed in the estimation panel.

*Source:* Authors' computations from IPUMS census microdata.

Figure 10: Descriptive Maps: Migrant Outcomes at Destination

(a) Share of in-migrants in low-quality housing

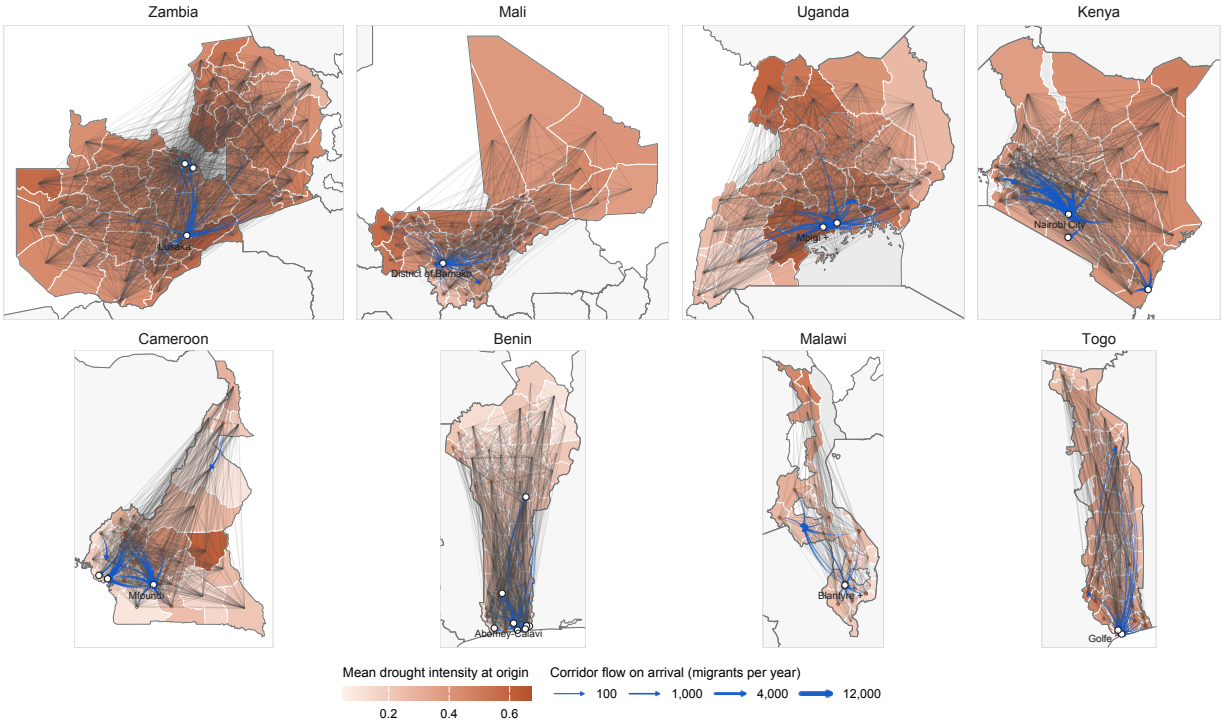
(b) Share of in-migrants employed




*Notes:* The propensity outcomes of the paper, aggregated by destination over the panel: among urban in-migrants, the share residing in low-quality housing in panel (a), main UN-Habitat definition with migrants as denominator, and the share employed in panel (b), with migrants of known labour-force status as denominator. Grey units are those where no in-migrants are observed.

*Source:* Authors' computations from IPUMS census microdata.

Figure 11: Observed Urban In-Migration Corridors, by Country




*Notes:* Arcs run from origin, an open dot, to destination, an arrowhead, with width proportional to the square root of the mean annual flow; totals are not comparable across countries, Kenya, Mali and Zambia contributing two censuses each. The faint web shows all 10,399 corridors with a positive urban flow and the highlighted arcs the 189 largest, the top 25 per country, carrying 44% of urban in-migration. One panel per country, 241,540 corridor-years. Units are shaded by mean drought intensity at origin, on the scale of Figure 8(b), and grey units are never observed as an origin. White circles mark main-city destinations, a census urban share of at least 0.55. Flows count urban in-migrants, the numerator of the endogenous variable. Centroids are interior points of administrative polygons and not city locations.

*Source:* Authors' computations from IPUMS census microdata and IPUMS harmonized shapefiles.

arrivals are not, in the raw data, the worst-housed group in the city. This is not in tension with the paper's estimates, which compare the marginal drought-induced migrant with the average migrant arriving on the same corridor, holding the corridor fixed, and never compare migrants with residents. If anything it works against a mechanical reading of the results, since the unconditional correlation between migration and housing deprivation runs the other way. Second, crowding is the one dimension on which the two populations coincide: migrants occupy dwellings of comparable or better materials but at similar densities, which locates the pressure of in-migration on living space rather than on the built structure. Third, the dimensions contribute very unequally. Floors, walls and crowding do most of the work; roofs fire rarely once sheet metal is treated as adequate; and sanitation fires only where latrines are absent rather than unimproved. Water is absent from the index altogether, for the reason given in Section 2: these censuses distinguish only types of piped supply, while UN-Habitat counts boreholes and protected dug wells as improved sources, so the variable cannot separate deficient from adequate access. The index should therefore be read as a measure of housing durability and living space, not as the full four-pillar UN-Habitat definition, and its levels are not comparable to published slum headcounts. The category composition behind each dimension is reported in Supplementary Appendix Figures S17–S20.

Table 15: Coding of the Housing-Quality Dimensions

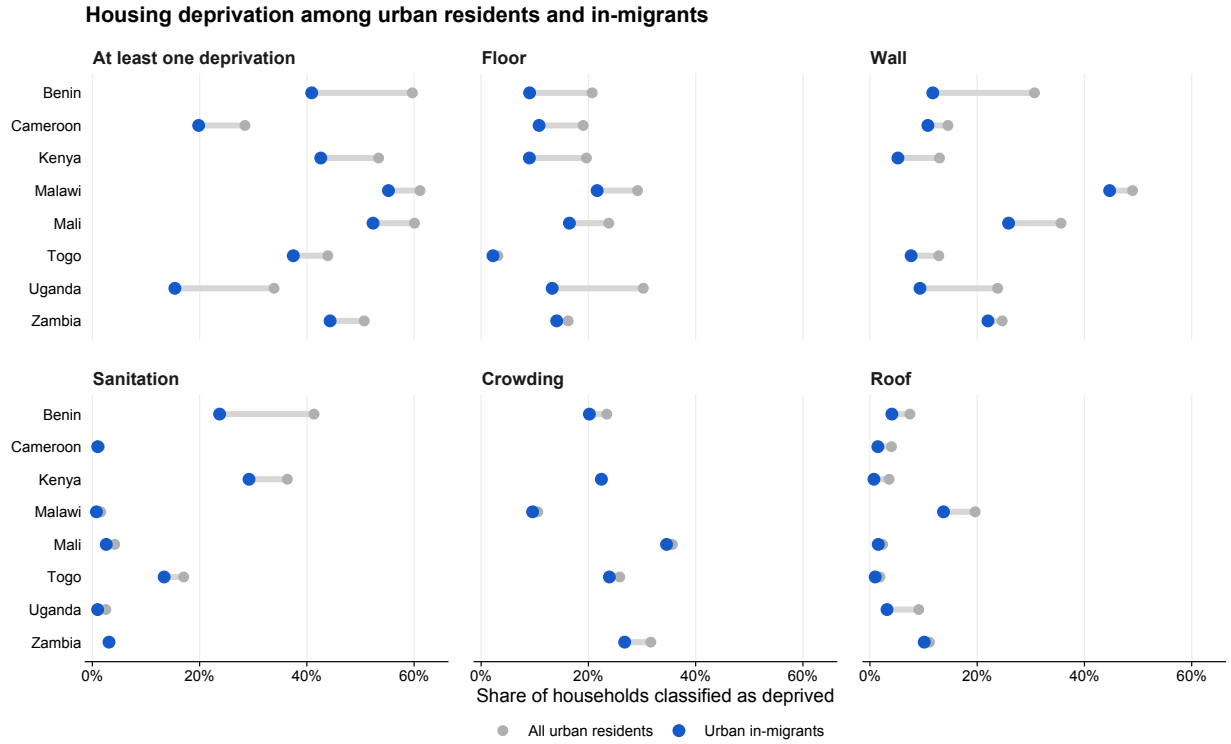
| Dimension | Classified as deficient |
|---|---|
| *Panel A: Structural quality of the dwelling* | |
| Floor | Unfinished natural floor – earth, sand, or dung – or salvaged wood (codes 100–120, 235) |
| Walls | No walls, or rudimentary, organic, or salvaged materials – cane, palm, logs, grass, scrap (codes 100–299, 405, 407) – and non-permanent earth materials: mud, adobe, clay, unburnt brick, pole and mud (codes 513, 515, 527, 533, 535, 539, 540, 541). Cement-stabilised earth, burnt or stabilised brick, and wood are treated as permanent |
| Roof | Wood, thatch, or palm; mud or earth; cardboard, plastic, or salvaged materials; other (codes 40–53, 60–61, 70–73, 90) |
| *Panel B: Access to basic services and living conditions* | |
| Sanitation | No toilet facility (code 10) |
| Crowding | More than three persons per habitable room |

*Notes:* Coding of the five housing-quality dimensions, in harmonized IPUMS categories (codes in parentheses). The coding is identical across countries; the availability of each dimension varies by census (Egypt records no floor, wall, or roof variable and is excluded from the housing analysis; sanitation is unavailable in Kenya 2009, and crowding in Cameroon 2005 and Uganda 2014, whose censuses record only rooms used for sleeping). Water access is not part of the index: the UN-Habitat criterion counts boreholes and protected wells as improved sources, but the harmonized IPUMS water-supply variable in these censuses distinguishes types of piped supply rather than improved from unimproved sources, so it cannot reliably separate deficient from adequate access in this setting. Sanitation follows the UN-Habitat standard, under which a pit latrine with a slab counts as improved, so only the complete absence of a facility is coded deficient. The crowding denominator is the habitable room; for Zambia, where the census records living rooms and bedrooms separately, it is their sum. Sheet-metal roofing (codes 30–39) is classified as durable in the main definition and as deficient in the robustness definition. A household resides in low-quality housing if at least one available dimension is deficient (main definition); the robustness definition requires at least two. The continuous deprivation count sums the deficient dimensions (0–5).

### 12.1.3 Urbanization Data

Table 16 and Figure 13 validate the census-based urbanization classification against an external measure of city size, the population of the largest Africapolis agglomeration at destination. The two measures line up where it matters: across the three bins, the median agglomeration size rises from 47,000 to 86,000 to 880,000, the share of destinations hosting a city of at least 100,000 inhabitants rises from 28 to 43 to 70 percent, and the main-cities bin contains the region's actual metropolises (Nairobi, Kampala, Yaoundé, Bamako). The destination-level rank correlation is positive but moderate (Spearman 0.33), as expected: the census share captures the urban composition of the destination's population, while Africapolis measures the physical size of the nearest agglomeration, and coarser administrative units in Malawi and Uganda mechanically dilute urban shares (panel (b)). Appendix Figure S3 reports the country-specific cumulative distributions of both measures.

Figure 12: Housing Deprivation among Urban Residents and Urban In-Migrants




*Notes:* Grey points are all urban residents, blue points urban in-migrants, and the segment joins the two for the same country and dimension; the outcome is the share of households counted deprived on each dimension of the housing index. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window; both populations are restricted to urban areas and person-weighted, and sheet-metal roofs count as adequate. The first panel applies the paper's threshold, at least one deprived dimension. Water deprivation is virtually absent in all samples and is omitted. Crowding cannot be constructed for Cameroon 2005 and Uganda 2014, whose censuses record only rooms used for sleeping, so those two countries are absent from the crowding panel and mechanically lower in the summary panel.

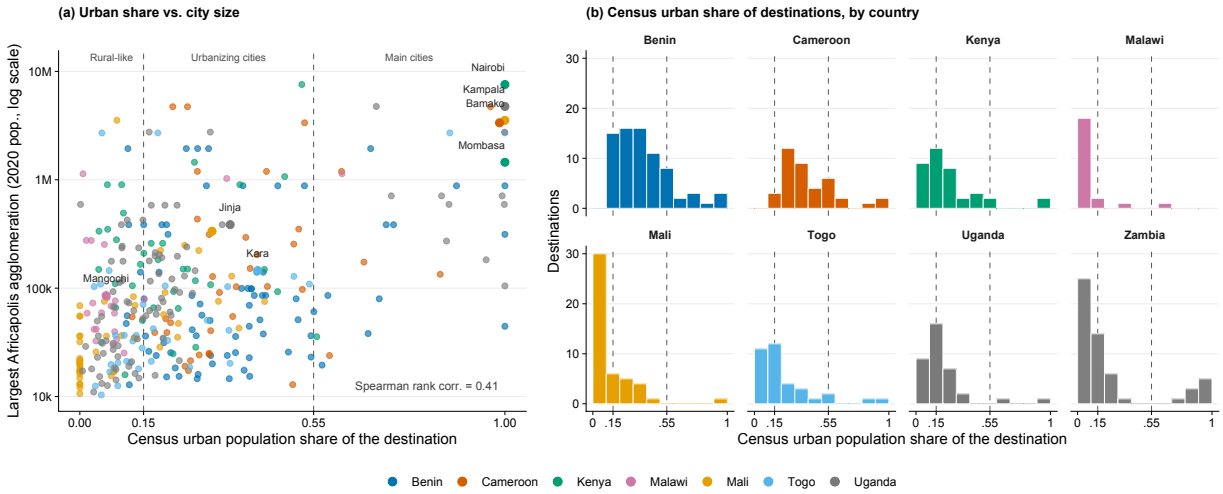
*Source:* Authors' computations from IPUMS census microdata.

Table 16: Validation of the Urbanization Classification: Census Urban Share versus Africapolis City Size

| | Destinations | Corridor-years | Largest agglomeration | | Example cities |
|---|---|---|---|---|---|
| Bin (urban share) | | | Median ('000) | $\geq$ 100k (%) | |
| Rural-like ($< 0.15$) | 142 | 91,019 | 36 | 23% | Naivasha, Arua, Molo, Kitui |
| Urbanizing cities ($[0.15, 0.55)$) | 169 | 122,821 | 90 | 43% | Nakuru, Lilongwe, Kisumu, Fort Portal |
| Main cities ($\geq 0.55$) | 36 | 27,700 | 652 | 78% | Nairobi, Kampala, Yaounde, Bamako |

*Notes:* Correspondence between the census urbanization bins and Africapolis city sizes, on the corridor-years of the estimation sample, $N = 241{,}540$, all eight countries. Bins are the census urban population share of the destination. Agglomeration populations and names are the largest Africapolis (2024) agglomeration in or within 10 km of the destination unit, 2020 population, from the recovered boundary file including Nairobi and 15 further Kenya units. Example cities take the bin of their own unit, the highest urban share among the units they map to. Four rows carrying conurbation-cluster totals, Kisumu, Embu, Mbale and Kikima, are corrected by each agglomeration's pre-merge decadal growth.

Figure 13: Census Urban Share versus Africapolis City Size at Destination




*Notes:* Panel (a) plots one observation per destination, coloured by country: the horizontal axis is the destination's urban population share in the most recent census and the vertical axis the 2020 population of the largest agglomeration in the Africapolis 2024 database located within, or within 10 km of, the destination unit, on a logarithmic scale. Panel (b) shows the distribution of census urban shares across destinations by country, the dashed vertical lines marking the classification thresholds of 0.15 and 0.55. Destinations without a matched Africapolis agglomeration of at least 10,000 inhabitants are excluded, and four Africapolis records reporting conurbation-cluster totals are corrected as described in the notes to Table 16. The Spearman rank correlation between the two measures is 0.33. Appendix Figure S3 reports their country-specific cumulative distributions.

*Source:* Authors' computations using IPUMS census data and the Africapolis 2024 database.

## 12.2 Identification Strategy

### 12.2.1 Post-IV Lasso Selection

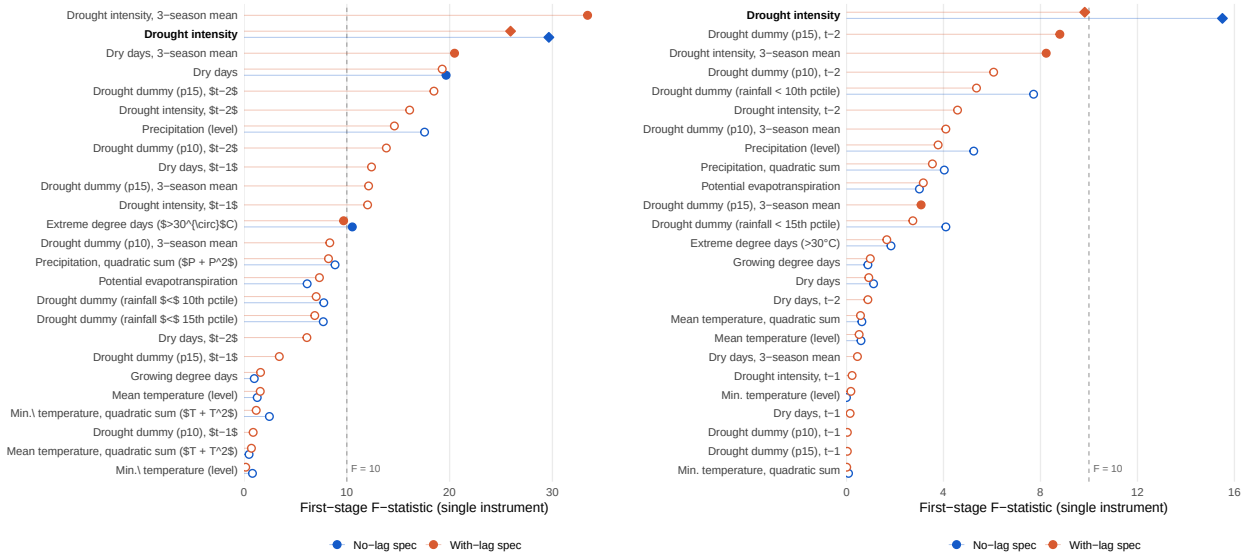
Figure 14 reports a data-driven validation of the instrument choice. We build a pool of candidate instruments - drought intensity, drought dummies at alternative thresholds, precipitation and temperature polynomials, degree-day and evapotranspiration measures, and their lags and 3-season means - and apply the post-IV-LASSO selection of Belloni et al. (2014), alongside the single-candidate first-stage $F$. We split the exercise between drought shocks (panels (a)-(b)) and their extreme-wet counterparts (panels (c)-(d)).

The drought-intensity instrument is the strongest single candidate on both samples ($F = 23.9$ on the scale sample, $F = 13.0$ on the propensity sample) and is selected by the LASSO in both; adding lagged candidates, its 3-season mean dominates ($F = 27.4$), consistent with the cumulative instrument used throughout. At the default penalty the LASSO retains a small set of dry-side candidates rather than a single winner (drought intensity, dry days, and extreme degree days on the scale sample); we nevertheless use the single strongest candidate, both for transparency and to keep the model just-identified, so that Anderson-Rubin inference remains exact and the complier population is not mixed across instruments. he robustness section reports the main results under the LASSO-selected instrument sets, showing that our findings are not tied to this choice.

By contrast, Figure 15 shows the results for the excess-rainfall intensity - the exact positive mirror of the drought instrument - has essentially no predictive power ($F = 4.0$ and

$F = 0.03$), and the flood dummies are similarly weak. The instrument is thus data-selected, and its power comes specifically from the dry tail of the rainfall distribution.

Figure 14: Post-IV-LASSO Instrument Selection: Drought Candidates




(a) Scale sample

(b) Propensity sample

*Notes:* Each point is the first-stage $F$-statistic of a single candidate instrument for the urban migration rate under the main specification, origin–destination and year fixed effects with the distance-to-census control; filled markers denote the candidates selected by post-IV-LASSO. The panels compare candidate sets without and with lagged instruments, for the scale and the propensity samples. Standard errors clustered at the origin district.

*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

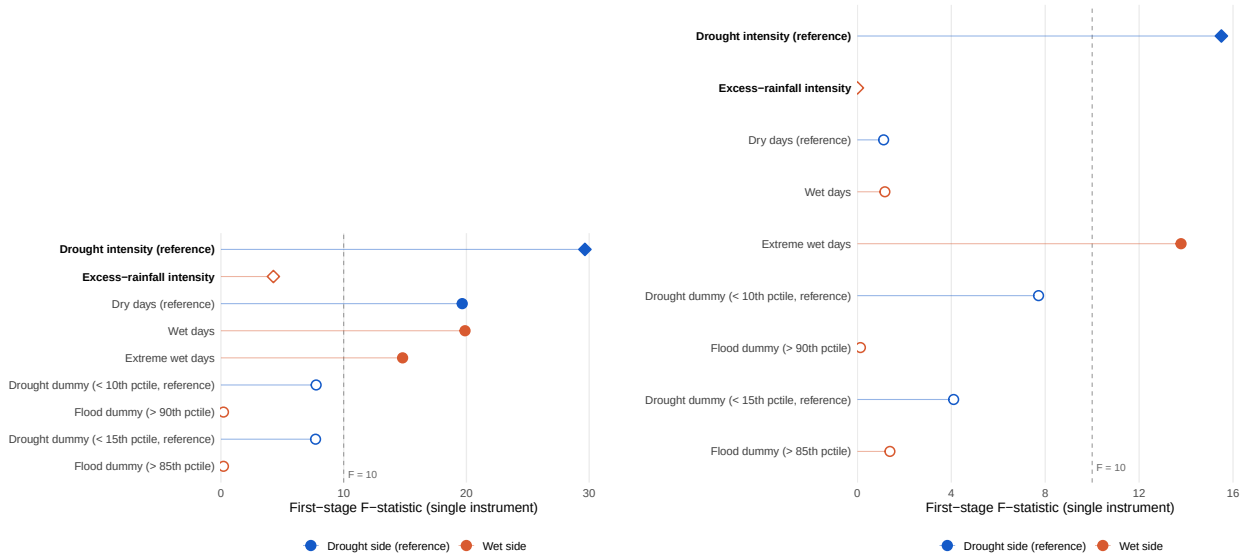
## 12.3 First Stage and Reduced Form

### 12.3.1 First Stage

Table 17 assesses the robustness of the first stage to sample restrictions and to the drought exposure window, reporting the paper's main specification for each sample and instrument; specification robustness itself is established on the full sample in Tables 1 and 18. Columns (1)-(2) cover the main sample of urban migrations to all destinations, columns (3)-(4) restrict destinations to main cities,
footnoteMain cities are defined as destinations with an urban population share weakly above the threshold, $\text{share}_d^{\text{urb}} \geq 0.55$, measured at the country's most recent census wave and held fixed across waves. Rural origins are defined as the complement: locations with $\text{share}_o^{\text{urb}} < 0.55$ or with no measured urban share. Robustness checks vary this threshold and use alternative definitions and datasets. and columns (5)-(6) further restrict to moves from rural origins to main cities. Within each pair, the first column uses the contemporaneous drought instrument and the second its cumulative (three-season mean) counterpart on the lag-restricted sample.

The contemporaneous instrument is a strong and significant predictor of migration in every sample, including moves from rural origins to major cities. On the full sample, the

Figure 15: Placebo Instrument Selection: Wet-Side Mirror Candidates




(a) Scale sample

(b) Propensity sample

*Notes:* Each point is the single-instrument first-stage $F$-statistic under the main specification, with each drought candidate replaced by its wet-side mirror, excess-rainfall intensity being the positive mirror of drought intensity. The weaker first stages indicate that instrument strength is specific to the dry tail of the rainfall distribution. Standard errors clustered at the origin district.

*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

cumulative instrument delivers an even stronger first stage ($F = 27.4$ against 23.9), but its strength collapses once the sample is restricted to city destinations ($F$ below conventional thresholds in columns (4) and (6)). This asymmetry, together with exclusion-restriction considerations-a multi-year accumulation of drought is more likely to affect destination outcomes through channels other than the timing of migration, such as persistent income losses at origin-leads us to retain the contemporaneous instrument as the paper's main one. The cumulative instrument is nonetheless reported alongside the main results throughout, and delivers consistent conclusions wherever its first stage is strong.

Table 18 varies the controls and the fixed-effect structure on the full sample, reporting the contemporaneous instrument throughout. Column (1) is the paper's main specification, reproduced from column (1) of Table 1. Comparing columns (1) and (2) shows that this control is not driving the first stage: the raw specification in column (2), without any control, already delivers a positive and highly significant coefficient, and adding the distance to the census mainly improves precision. Column (3) confirms that the instrument captures rainfall deficits rather than heat, and column (4) adds temperature at destination to absorb a potential climatic pull factor; column (5) includes all controls jointly. In all three cases the coefficient is essentially unchanged and the first stage remains strong. Column (6) is the destination-by-year specification of column (3) of Table 1.

Table 17: First Stage: Restricted Samples and Lagged Instrument (Main Specification)

| | *All destinations* | | *Main cities* | | *Rural → main cities* | |
|---|---|---|---|---|---|---|
| | Contemp. | Lagged | Contemp. | Lagged | Contemp. | Lagged |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A. Scale sample: all corridor-years* | | | | | | |
| Drought intensity | 0.00362*** | 0.00947*** | 0.00312*** | 0.00386 | 0.00250*** | 0.00369* |
| | (0.00066) | (0.00164) | (0.00092) | (0.00317) | (0.00065) | (0.00196) |
| First-stage $F$ | 29.65 | 33.42 | 11.61 | 1.48 | 14.54 | 3.55 |
| Effect of 1-SD drought (% mean) | +6.8% | +10.5% | +4.8% | +3.4% | +5.1% | +4.3% |
| Mean migration rate | 0.02682 | 0.02680 | 0.03317 | 0.03350 | 0.02487 | 0.02509 |
| Observations | 241,540 | 235,920 | 27,700 | 26,764 | 23,968 | 23,176 |
| OD pairs | 16,846 | 16,846 | 1,948 | 1,948 | 1,678 | 1,678 |
| *Panel B. Propensity sample: corridor-years with a positive urban flow* | | | | | | |
| Drought intensity | 0.00601*** | 0.01007*** | 0.00317** | 0.00429 | 0.00256*** | 0.00473** |
| | (0.00153) | (0.00351) | (0.00129) | (0.00430) | (0.00081) | (0.00239) |
| First-stage $F$ | 15.50 | 8.24 | 6.07 | 1.00 | 9.90 | 3.93 |
| Effect of 1-SD drought (% mean) | +3.4% | +3.3% | +3.1% | +2.4% | +3.2% | +3.4% |
| Mean migration rate | 0.08908 | 0.08881 | 0.05319 | 0.05393 | 0.04086 | 0.04134 |
| Observations | 72,723 | 71,207 | 17,273 | 16,624 | 14,587 | 14,066 |
| OD pairs | 10,537 | 10,510 | 1,633 | 1,632 | 1,395 | 1,394 |

*Notes:* Outcome: the urban migration rate. Sample: 8 countries, 12-year window; main cities are destinations with a census urban share of at least 0.55 and the rural columns additionally restrict origins below that threshold. Panel A is the sample on which the scale results are estimated, Panel B the corridor-years where the share of migrants in low-quality housing is defined, which is the sample of the propensity estimates. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. 'Contemp.' is $\max(0, -\tilde{z})$ at origin and 'Lagged' its three-season mean over $\{t, t-1, t-2\}$ on the lag-restricted sample. The first-stage coefficient is larger in Panel B, and so is the mean migration rate, which is why the same shock is a smaller percentage of the mean there. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean migration rate × 100. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

### 12.3.2 Reduced Form

*Residential Status* - This section reports the reduced-form counterparts of the paper's main estimates: each outcome is regressed directly on the drought instrument at origin, under the main specification, so that no result depends on the strength of the first stage. Table 21 reports the reduced form for the propensity margin. Columns (1)-(4) use the contemporaneous instrument and columns (5)-(8) the cumulative instrument on the lag-restricted sample; within each instrument, the first pair of columns reports the share of in-migrants living in low-quality housing under the main definition (dummy) and the second pair the mean number of deprived housing dimensions among the same migrants (continuous), each under the paper's main specification and with a linear trend specific to each destination. The significance pattern of its baseline columns mirrors the Anderson-Rubin intervals of Table 28, as

Table 18: First Stage: Controls and Fixed-Effect Structure

| | *DV: Urban migration rate* | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Main spec. | | | | | |
| Drought intensity | 3.618*** | 2.437*** | 3.956*** | 3.892*** | 4.003*** | 1.592** |
| | (0.664) | (0.683) | (0.765) | (0.715) | (0.772) | (0.732) |
| First-stage $F$ | 29.65 | 12.72 | 26.77 | 29.66 | 26.91 | 4.74 |
| Effect of 1-SD drought (% mean) | +6.8% | +4.6% | +7.4% | +7.3% | +7.5% | +3.0% |
| Origin FE | No | No | No | No | No | No |
| Destination FE | No | No | No | No | No | No |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Country × Year FE | No | No | No | No | No | No |
| Destination × Year FE | No | No | No | No | No | Yes |
| Dist. to census control | Yes | No | Yes | Yes | Yes | No[a] |
| Destination controls | No | No | No | Yes | Yes | No |
| Origin controls | No | No | Yes | No | Yes | No |
| $R^2$ | 0.486 | 0.481 | 0.486 | 0.486 | 0.486 | 0.512 |
| Within $R^2$ | 0.010 | 0.000 | 0.010 | 0.010 | 0.010 | 0.000 |
| Mean migration rate | 26.820 | 26.820 | 26.820 | 26.820 | 26.820 | 26.820 |
| Observations | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: the urban migration rate. Sample: urban migrants, all destinations, 8 countries, 12-year window. Specification: column (1) is the paper's main one, reproduced from Table 1, and the remaining columns vary the controls and the fixed-effect structure as indicated; origin and destination controls are the growing-season mean temperature of each, and [a] marks where `dist_census` is mechanically absorbed. The instrument is contemporaneous drought intensity at origin. First-stage $F$ is the cluster-robust Wald statistic. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean migration rate × 100, on each column's own sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 19: How Synchronous Are the Shocks Reaching a Destination?

| | Median | 25th pct. | 75th pct. | Destinations |
|---|---|---|---|---|
| Pairwise correlation of drought across a destination's origins | 0.674 | 0.552 | 0.796 | 318 |
| Variance ratio of induced inflow (= 1 if shocks independent) | 3.38 | 2.30 | 4.98 | 318 |
| Effective number of years carrying the induced inflow (1/HHI) | 4.6 | 3.8 | 5.7 | 343 |
| *out of a window of* | 12 | 10 | 12 | |

*Notes:* Distribution across destinations of the three shock-structure measures defined in Section 4.2. All use predetermined weights, the origin shares of each destination's inflow in the three years preceding its recall window, census years −13 to −15. The pairwise correlation and the variance ratio are computed on origins with at least eight observed years, and the correlation additionally requires three such origins per destination, which is why it covers fewer. The effective number of years is $1/\mathrm{HHI}_d$, computed by census cycle. Table 20 interacts the first stage with these measures.

Table 20: The Spatio-Temporal Structure of Shocks Amplifies the First Stage

| | (1) Temporal concentration | (2) Variance ratio | (3) Pairwise correlation |
|---|---|---|---|
| Drought intensity | +0.00406*** | +0.00336*** | +0.00337*** |
| | (0.00070) | (0.00066) | (0.00067) |
| Drought × shock structure | +0.00216*** | +0.00077*** | +0.00033 |
| | (0.00070) | (0.00029) | (0.00053) |
| Amplification per SD, % | +53 | +23 | +10 |
| *Interaction coefficient under additional controls* | | | |
| + drought squared | +0.00213*** | +0.00078*** | +0.00035 |
| | (0.00070) | (0.00029) | (0.00054) |
| + drought × city-size class | +0.00209*** | +0.00089*** | +0.00033 |
| | (0.00069) | (0.00031) | (0.00054) |
| Observations | 235,922 | 216,990 | 216,990 |
| Destinations | 343 | 318 | 318 |
| OD pairs | 16,489 | 15,043 | 15,043 |

*Notes:* Outcome: the urban migration rate. Specification: the paper's main first stage interacted with the standardized shock-structure measure $C_d$ of Table 19, origin–destination pair and arrival-year fixed effects, distance-to-census control. The level of $C_d$ is absorbed by the origin–destination fixed effects, so the interaction reads as the change in the first-stage coefficient per standard deviation of $C_d$, and the amplification row expresses it as a percentage of the main effect. The last two rows repeat the interaction adding squared drought, and drought interacted with the destination's Africapolis size class. Column (3) covers fewer destinations because the pairwise correlation requires at least three origins with eight observed years. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

the two tests coincide.

The estimates move towards zero once the trends are included, with standard errors of the same order as in the baseline.

*Labor-Force Status -* Tables 22 and 23 report the reduced forms for the employment results, regressing the scale and propensity outcomes directly on origin drought for both instruments. On the scale margin, all six inflows rise significantly at the 1% level. On the propensity margin, drought shifts the composition of arrivals from inactivity toward employment under both instruments; since the reduced form does not rely on the first stage, the lagged columns provide valid inference where the lagged first stage of Table 33 is weak, and confirm the same reallocation.

Table 24 reports the three labour-force shares among adult arrivals under the same four specifications. In the baseline, contemporaneous drought raises the unemployed share and lowers the inactive share; both estimates move towards zero once destination-specific linear trends are added, with standard errors of the same order.

Appendix Figure 16 summarizes the reduced-form responses across housing and employ-

Table 21: Reduced Form: Drought Intensity and the Housing Composition of the Inflow

| | *Contemporaneous drought* | | | | *3-season mean (lagged)* | | | |
|---|---|---|---|---|---|---|---|---|
| | Share deprived | | Number of deprivations | | Share deprived | | Number of deprivations | |
| | Baseline | + Dest. trends | Baseline | + Dest. trends | Baseline | + Dest. trends | Baseline | + Dest. trends |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Drought intensity | 0.00731** | 0.00371 | 0.00571 | -0.00088 | 0.01995*** | 0.00773 | 0.03217** | 0.00480 |
| | (0.00325) | (0.00315) | (0.00637) | (0.00635) | (0.00687) | (0.00721) | (0.01360) | (0.01425) |
| Effect of 1-SD drought | +0.37 pp | +0.19 pp | +0.0029 | −0.0004 | +0.58 pp | +0.22 pp | +0.0093 | +0.0014 |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Destination linear trend | No | Yes | No | Yes | No | Yes | No | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.392 | 0.403 | 0.450 | 0.460 | 0.396 | 0.407 | 0.455 | 0.466 |
| Within $R^2$ | 0.007 | 0.002 | 0.002 | 0.001 | 0.007 | 0.002 | 0.002 | 0.001 |
| Mean outcome | 0.4064 | 0.4064 | 0.6620 | 0.6620 | 0.4042 | 0.4042 | 0.6564 | 0.6564 |
| Observations | 72,723 | 72,723 | 72,723 | 72,723 | 71,207 | 71,207 | 71,207 | 71,207 |
| OD pairs | 10,537 | 10,537 | 10,537 | 10,537 | 10,510 | 10,510 | 10,510 | 10,510 |

*Notes:* Outcomes: 'Share deprived' is the share of urban in-migrants living in low-quality housing under the paper's main definition, and 'Number of deprivations' the mean number of deprived housing dimensions among the same migrants, so a one-standard-deviation drought is reported in percentage points for the share and in dimensions for the count. Sample: both outcomes carry the migrant cohort in the denominator, so they are defined only on corridor-years receiving at least one urban in-migrant; that criterion responds to the instrument, so these estimates describe the housing composition of the cohorts corridors actually receive. 8 countries, 12-year window. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: the outcome is regressed directly on origin drought and the migration rate is not instrumented. Within each instrument and outcome the first column is the main specification and the second adds a linear time trend specific to each destination, as in column (2) of Table 1. Columns (1)–(4) use contemporaneous drought intensity, $\max(0, -\tilde{z})$, and columns (5)–(8) its three-season mean on the sample with complete lag coverage; the 1-SD effect uses each column's own sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

ment outcomes and compares the baseline estimates with specifications including destination-specific linear trends. The positive response of migrants who are simultaneously employed and living in low-quality housing remains statistically significant under these additional controls.

### 12.3.3 Identifying Variation

Figures 17 and 18 provide a graphical representation of the relationships underlying the identification strategy. Each panel plots mean outcomes within forty equal-sized bins of drought intensity at origin, after residualizing all variables on origin-destination pair and arrival-year fixed effects and the distance-to-census control; the fitted line is estimated on the underlying corridor-year data, so its slope equals the corresponding regression coefficient exactly, and the shaded area is the 95% confidence band based on standard errors clustered

Table 22: Reduced Form: Drought Intensity and the Labor-Force Status of the Inflow: Scale

| | *Contemporaneous drought* | | | *3-season mean (lagged)* | | |
|---|---|---|---|---|---|---|
| | Employed (1) | Unemployed (2) | Inactive (3) | Employed (4) | Unemployed (5) | Inactive (6) |
| Drought intensity | 1.547*** | 0.270*** | 1.284*** | 3.153*** | 0.894*** | 3.431*** |
| | (0.272) | (0.048) | (0.328) | (0.625) | (0.128) | (0.780) |
| Effect of 1-SD drought | +7.3% | +10.1% | +5.5% | +8.8% | +19.5% | +8.7% |
| $R^2$ | 0.474 | 0.298 | 0.377 | 0.486 | 0.303 | 0.387 |
| Within $R^2$ | 0.006 | 0.001 | 0.007 | 0.006 | 0.002 | 0.007 |
| Mean outcome | 10.6 | 1.34 | 11.8 | 10.6 | 1.36 | 11.8 |
| Observations | 241,540 | 241,540 | 241,540 | 235,920 | 235,920 | 235,920 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: employed, unemployed and inactive in-migrants per 100,000 urban residents, unwinsorized, with the 1-SD effect in percent of the mean outcome; undetermined labour-force status is excluded. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: each inflow is regressed directly on origin drought with no reliance on the first stage, so the $t$-test here is the Anderson–Rubin test of a zero effect in the 2SLS table. Columns (4)–(6) use the lagged instrument on the lag-restricted sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 23: Reduced Form: Drought Intensity and the Labor-Force Status of the Inflow: Propensity

| | *Contemporaneous drought* | | | *3-season mean (lagged)* | | |
|---|---|---|---|---|---|---|
| | Employed (1) | Unemployed (2) | Inactive (3) | Employed (4) | Unemployed (5) | Inactive (6) |
| Drought intensity | 0.00839*** | 0.00175 | -0.01014*** | 0.01403** | 0.00500* | -0.01904*** |
| | (0.00303) | (0.00137) | (0.00293) | (0.00599) | (0.00270) | (0.00576) |
| Effect of 1-SD drought | +0.42 pp | +0.09 pp | −0.51 pp | +0.41 pp | +0.14 pp | −0.55 pp |
| $R^2$ | 0.329 | 0.243 | 0.324 | 0.335 | 0.245 | 0.329 |
| Within $R^2$ | 0.003 | 0.000 | 0.005 | 0.003 | 0.000 | 0.005 |
| Mean outcome | 0.505 | 0.0578 | 0.437 | 0.507 | 0.0586 | 0.435 |
| Observations | 71,503 | 71,503 | 71,503 | 70,036 | 70,036 | 70,036 |
| OD pairs | 10,452 | 10,452 | 10,452 | 10,420 | 10,420 | 10,420 |

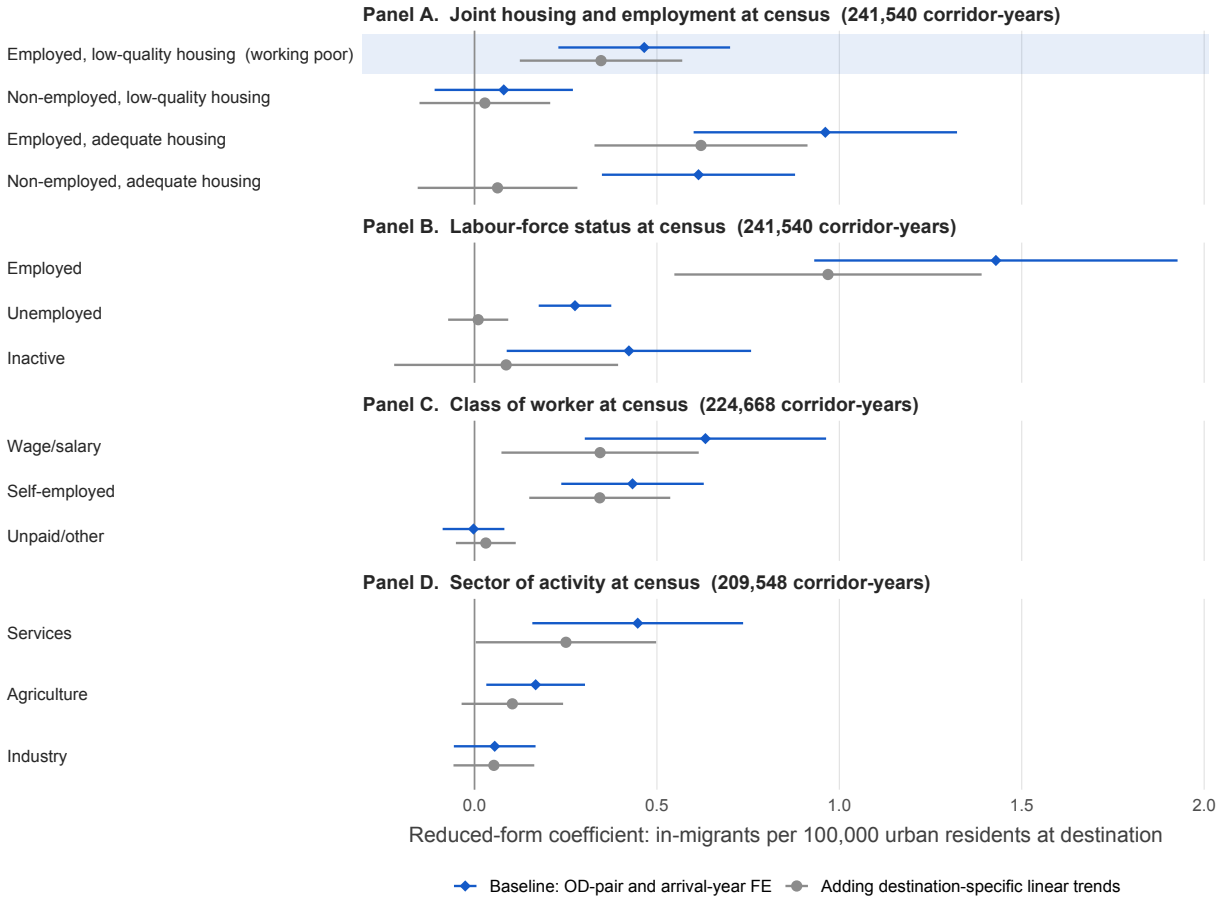
*Notes:* Outcomes: the employed, unemployed and inactive shares among known-status in-migrants; the three sum to one, so the coefficients sum to zero within each instrument, and the 1-SD effect is in percentage points. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: the shares are regressed directly on origin drought with no reliance on the first stage, so the $t$-test here is the Anderson–Rubin test of a zero effect in the 2SLS table, which is the honest inference given the weak lagged first stage there. Columns (4)–(6) use the lagged instrument on the lag-restricted sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 24: Reduced Form: Drought Intensity and the Labour-Force Composition of the Adult Inflow

| | *Contemporaneous drought* | | *3-season mean (lagged)* | |
|---|---|---|---|---|
| | Baseline | + Destination trends | Baseline | + Destination trends |
| | (1) | (2) | (3) | (4) |
| *Panel A. Share employed* | | | | |
| Drought intensity | 0.00296 | 0.00355 | -0.00192 | 0.00660 |
| | (0.00311) | (0.00317) | (0.00620) | (0.00649) |
| Effect of 1-SD drought | +0.15 pp | +0.18 pp | −0.06 pp | +0.19 pp |
| Mean outcome | 0.6381 | 0.6381 | 0.6392 | 0.6392 |
| Observations | 64,554 | 64,554 | 63,295 | 63,295 |
| *Panel B. Share unemployed* | | | | |
| Drought intensity | 0.00373** | 0.00054 | 0.01196*** | -0.00200 |
| | (0.00167) | (0.00170) | (0.00341) | (0.00362) |
| Effect of 1-SD drought | +0.19 pp | +0.03 pp | +0.34 pp | −0.06 pp |
| Mean outcome | 0.0680 | 0.0680 | 0.0689 | 0.0689 |
| Observations | 64,554 | 64,554 | 63,295 | 63,295 |
| *Panel C. Share inactive* | | | | |
| Drought intensity | -0.00669** | -0.00409 | -0.01004* | -0.00461 |
| | (0.00288) | (0.00293) | (0.00548) | (0.00547) |
| Effect of 1-SD drought | −0.34 pp | −0.21 pp | −0.29 pp | −0.13 pp |
| Mean outcome | 0.2939 | 0.2939 | 0.2918 | 0.2918 |
| Observations | 64,554 | 64,554 | 63,295 | 63,295 |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Destination linear trend | No | Yes | No | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |

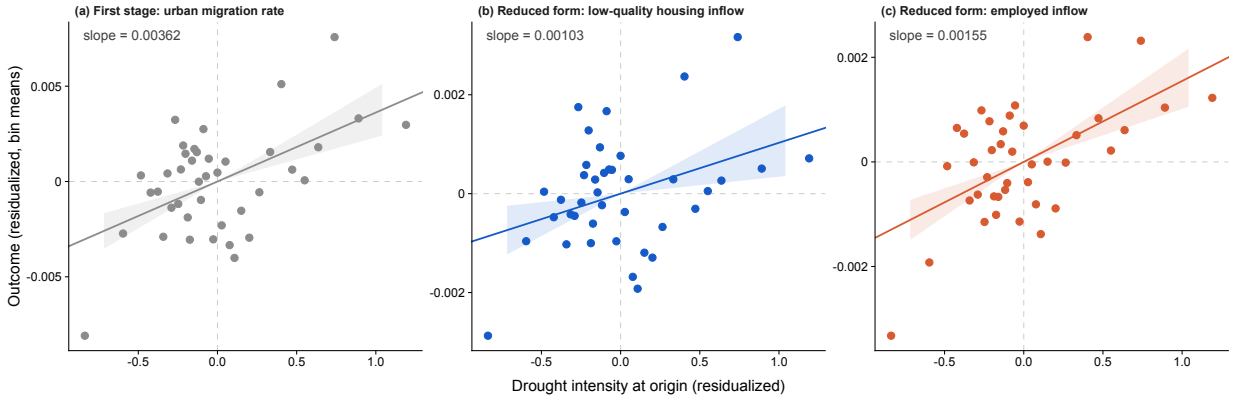
*Notes:* Outcomes: the employed, unemployed and inactive shares among adult in-migrants of known status; the three sum to one, so the coefficients sum to zero within each column. Sample: urban in-migrants aged 15 or over at arrival whose labour-force status is recorded at the census, as in Table 3. The ratio requires a positive denominator, so the sample is the corridor-years receiving at least one such migrant; that criterion responds to the instrument, so these estimates describe the composition of the cohorts corridors actually receive. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. This is the reduced form: the outcome is regressed directly on origin drought and the migration rate is not instrumented. Columns (1)–(2) use contemporaneous drought intensity, $\max(0, -\tilde{z})$, and columns (3)–(4) its three-season mean on the sample with complete lag coverage; columns (2) and (4) add a linear time trend specific to each destination. The 1-SD effect uses each column's own sample. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 16: Drought at Origin and the Working Poor: Reduced-Form Estimates




*Notes:* Each marker is a reduced-form coefficient on contemporaneous drought intensity at origin, with 95% confidence intervals; blue diamonds are the main specification, origin–destination-pair and arrival-year fixed effects with the distance-to-census control, and grey circles add a linear time trend specific to each destination. The outcome is the corresponding in-migrant inflow per 100,000 urban residents at destination, unwinsorized; the migration rate is not instrumented, so each coefficient is the response of that flow itself. The population is in-migrants aged 15 or over at arrival, whose employment and housing are recorded at the census, which may fall several years after the move. Panel A's four cells are mutually exclusive and exhaustive within the population whose employment and housing are both recorded and sum to its total inflow; the shaded row is the working poor, migrants who hold employment and live in low-quality housing. Panels B to D read the same population through one characteristic at a time: labour-force status on the full panel, then class of worker and sector on the census samples where IPUMS `CLASSWK` and `INDGEN` are populated for more than 20% of employed migrants, 10 and 9 of the 11 urban samples. Each panel heading reports its own sample size, so coefficients are comparable within a panel. Estimates and significance levels are in Tables 3, 4, 5 and 6. Standard errors clustered at the origin district. *Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Figure 17: First Stage and Reduced Form: Binned Scatterplots under the Main Specification




*Notes:* Each dot is the mean within one of forty equal-sized bins of residualized drought intensity at origin; the fitted line is estimated on the underlying corridor-year data, passes through the origin by construction and has a slope equal to the corresponding regression coefficient exactly. The shaded area is the 95% confidence band. All variables are residualized on origin–destination pair and arrival-year fixed effects and the distance-to-census control (Frisch–Waugh). Panel (a) plots the urban migration rate and panel (b) the low-quality-housing inflow rate, main definition, unwinsorized. Sample: urban migrants, all destinations, 8 countries, 241,540 corridor-years. Standard errors clustered at the origin district.
*Source:* Authors' computations from IPUMS census and CHIRPS climate data.

at the origin level. All relationships are positive, close to linear, and supported across the entire distribution of drought shocks rather than driven by a few extreme bins. Corridors experiencing unusually dry growing seasons send more migrants to urban areas, more migrants into low-quality housing, and more employed migrants (Figure 17), the visual counterparts of the first-stage and reduced-form estimates of Tables 1 and 2. The same holds for the propensity outcomes (Figure 18): among migrants, the shares in low-quality housing and in employment both rise with origin drought, the reduced-form counterpart of the sorting results.

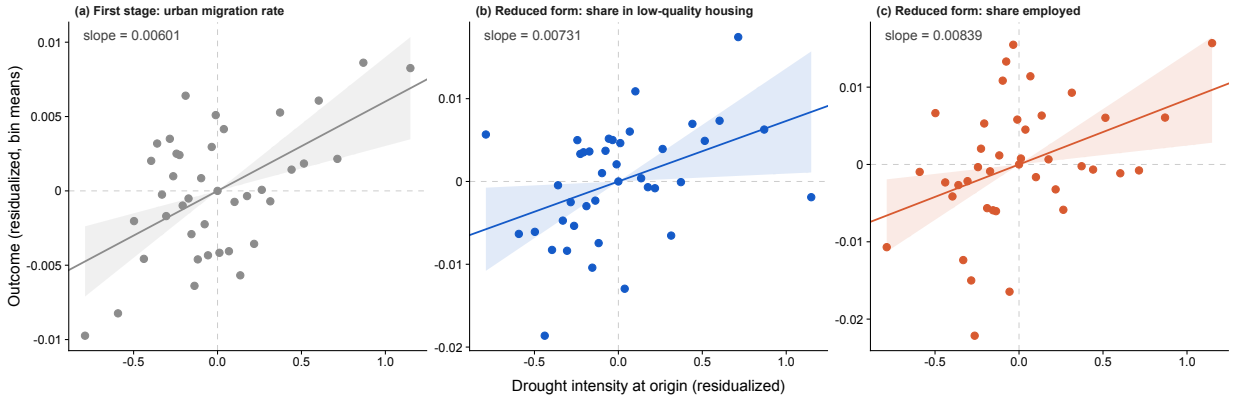
*Drought and the harvest at origin.* - Table 25 reports the reduced form of origin drought on the harvest itself, the first link of the agricultural-push channel discussed in Section 8.1.1. Crop statistics come from HarvestStat Africa v1.0 (Lee et al., 2025), matched to our origin districts through its areal crosswalk to IPUMS boundaries. Because production is the product of harvested area and yield, the three outcomes are an exact decomposition and their coefficients add up: the loss is entirely in yield per hectare, with harvested area unmoved. The sample is necessarily narrower than the estimation panel – 2,535 district-years, 214 districts and six countries – and the limits of the series are set out in the table notes.

## 12.4 Main Results

### 12.4.1 Residential Integration

*Scale*- Table 26 details the calculation behind the interpretation of the main estimate: for a representative corridor with 10,000 urban residents, it translates a one-standard-deviation drought at origin into additional migrants and additional residents in low-quality housing, whose ratio equals, by construction, the 2SLS coefficient of Table **??**. Table 27 replicates

Figure 18: First Stage and Reduced Form for the Propensity Outcomes: Binned Scatterplots




*Notes:* Each dot is the mean within one of forty equal-sized bins of residualized drought intensity at origin; the fitted line is estimated on the underlying corridor-year data, passes through the origin by construction and has a slope equal to the corresponding regression coefficient exactly. The shaded area is the 95% confidence band. All variables are residualized on origin–destination pair and arrival-year fixed effects and the distance-to-census control (Frisch–Waugh). Panel (a) plots the urban migration rate, panel (b) the share of migrants in low-quality housing under the main definition with migrants as denominator, and panel (c) the share employed among known-status migrants; shares are unwinsorized, as are all outcomes in the paper. Sample: urban migrants, all destinations, 8 countries; 72,821 corridor-years in panels (a)–(b), those with at least one migrant, and 71,587 in panel (c), those with at least one known-status migrant. Standard errors clustered at the origin district.

*Source:* Authors' computations from IPUMS census and CHIRPS climate data.

the main table under the two sample restrictions used in the paper - destinations limited to main cities, and rural origins moving to main cities - with the contemporaneous instrument, whose first stage remains strong in both samples.

To illustrate the magnitude of the 2SLS estimate, consider an origin–destination corridor with 10,000 urban residents at destination. At the baseline migration rate, this corridor receives approximately 2.68 migrants per year, of whom 1.20 are observed in low-quality housing. A one-standard-deviation increase in drought intensity at origin ($SD = 0.502$) raises the migration inflow by $3.618 \times 0.502 \times (10{,}000/100{,}000) = 0.18$ migrants, or 6.8% of its mean. The corresponding increase in low-quality housing inflows is $1.029 \times 0.502 \times (10{,}000/100{,}000) = 0.052$ migrants. The ratio of these two responses gives the IV estimate, $0.052/0.18 = 0.285$: for every 100 additional migrants associated with origin drought, approximately 28 are observed living in low-quality housing at the census.

### 12.4.2 Propensity

*Propensity* - Table 29 re-estimates the propensity outcomes - the share of in-migrants in low-quality housing under the main definition and the mean number of deprived dimensions - in the two restricted samples, with the contemporaneous instrument. The propensity subsamples are considerably smaller than for the scale outcome, since they require corridors with observed housing characteristics, and both first stages are moderate ($F = 6.1$ in the main-cities sample, $F = 10.4$ in the rural-to-cities sample), so the Anderson-Rubin intervals carry the inference. The 2SLS point estimates on the extensive margin (2.22 and 2.41) are larger than the full-sample estimate of 1.15, but both Anderson-Rubin intervals are very wide and include zero. These subsamples therefore neither confirm nor contradict the main

Table 25: Drought and the Harvest at Origin: The First Link of the Agricultural Channel

| | log Production | | log Harvested area | | log Yield | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Drought intensity at origin | −0.064* | −0.059 | 0.002 | −0.016 | −0.065*** | −0.044** |
| | (0.036) | (0.042) | (0.030) | (0.036) | (0.018) | (0.022) |
| Effect of 1-SD drought (%) | −3.03 | −2.81 | +0.09 | −0.74 | −3.11 | −2.07 |
| District FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | — | Yes | — | Yes | — |
| Country × year FE | — | Yes | — | Yes | — | Yes |
| Mean of dependent variable (level) | 30,916 t | | 20,244 ha | | 1.43 t/ha | |
| Districts | | | 214 | | | |
| Countries | | | 6 | | | |
| District-years | | | 2,535 | | | |

*Notes:* Outcomes: log maize yield, log harvested area and log production at origin. Sample: 2,535 district-years, 214 districts and 6 countries, 1981–2018, from HarvestStat Africa v1.0 (Lee et al., 2025), mapped onto the paper's origin districts by areal crosswalk; yield is recomputed as production over harvested area after aggregation, and the top and bottom yield percentiles within each country are trimmed. Mali enters HarvestStat at the first administrative level and the panel at the second, so its drought intensity is averaged over the districts sharing one crop observation. Specification: drought intensity is the instrument of the main specification, $\max(0, -z)$ standardized within district; columns (2), (4) and (6) compare districts of the same country in the same year. Since log Production = log Harvested area + log Yield, the three outcomes are an exact decomposition and the coefficients add up within each fixed-effect structure; 'Effect of 1-SD drought' is $b \times SD(drought_int_z) = 0.476$ on this sample, in percent of the dependent variable. HarvestStat records area harvested and not area planted, so abandoning the worst plots would raise measured yield: it does not. This table is the first link of the agricultural-push channel, with Figure 5 the second. A year-by-year mediation of migration through yields is not identified here: two independent yield indices for the same district-year correlate at 0.30 net of fixed effects, the series covers 69% of the estimation panel, and the pooled estimate rests largely on Kenya. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

propensity result; they are too imprecise to be informative on their own.

### 12.4.3 Labor Market Integration

Table 33 re-estimates the employment results of Tables 30 and 32 with the 3-season cumulative drought instrument. On the scale margin the first stage is strong ($F = 27.4$) and the employed, unemployed, and inactive inflows all rise significantly. On the propensity margin the first stage is weak ($F = 7.1$), so inference relies on the Anderson-Rubin intervals, which confirm the contemporaneous pattern: the employed share of arrivals rises, the inactive share falls, and the unemployed share is unchanged.

Table 26: Interpreting the Main Estimate: A Representative Corridor

*Panel A. Step-by-step calculation*

| | Formula | Value |
|---|---|---|
| *Baseline (no drought)* | | |
| Migrants | $\overline{\text{MigRate}}/100 \times 10{,}000$ | $26.82003 \times 0.1 = 2.68$ |
| Migrants in low-quality housing | $\overline{S}/100 \times 10{,}000$ | $11.96444 \times 0.1 = 1.20$ |
| *Effect of a 1-SD drought shock* | | |
| $\Delta$ Migration rate | $\beta_{FS} \times SD(\text{drought})$ | $3.61787 \times 0.502 = 1.81565$ |
| $\Delta$ Migrants | $\Delta\text{MigRate}/100 \times 10{,}000$ | $+0.18$ |
| $\Delta$ Low-quality housing inflow ($S$) | $\beta_{2SLS} \times \beta_{FS} \times SD(\text{drought})$ | $0.285 \times 3.61787 \times 0.502 = 0.51664$ |
| $\Delta$ Migrants in low-quality housing | $\Delta S/100 \times 10{,}000$ | $+0.052$ |
| *Share of new migrants in low-quality housing* | | |
| $\Delta$ Low-quality / $\Delta$ Migrants | $\beta_{2SLS}$ | $0.052/0.18 = 28\%$ |

*Panel B. Summary*

| | Without drought | With 1-SD drought | Change | Share of new migrants in low-quality housing |
|---|---|---|---|---|
| Number of migrants | 2.68 | 2.86 | +0.18 (+6.8%) | – |
| Migrants in low-quality housing | 1.20 | 1.25 | +0.052 (+4.3%) | 28% |

*Notes:* Illustration for a single origin–destination corridor with 10,000 urban residents at destination, all-destinations sample, main definition of low-quality housing. Notation as in the text: MigRate is the urban migration rate and $S$ the low-quality housing inflow rate, both per 100 urban residents, and bars denote sample means. Because $S$ and MigRate share the destination urban population as denominator, $\beta_{2SLS}$ reads directly as the share of marginal, drought-induced migrants who settle in low-quality housing. Inputs: $\beta_{FS} = 3.61787$, $\beta_{2SLS} = 0.28455$ and $SD(\text{drought}) = 0.502$, all from Table **??**.

### 12.4.4 Interpreting the effect

### 12.4.5 Decomposing the Effect: Scale versus Propensity

Drought fills low-quality housing through two channels that can operate at once: it sends more migrants (the scale channel), and it may change how those migrants are housed relative to the ones a corridor sends in ordinary years (the propensity channel). On the sample where both margins are defined, a one-standard-deviation drought raises the number of migrants by 3.4% while raising their propensity for low-quality housing by 0.37 percentage points, from 40.6 to 41.0 percent of arrivals.

*Volume accounts for* 80% *of the response, composition for* 20%. - Because the scale outcome and the migration rate share the same urban-population denominator, the scale outcome is exactly the product of the two margins: $S_{odt} = P_{odt}M_{odt}$. Expanding this product around corridor-specific means yields

$$S_{odt} = \overline{P}_{od}M_{odt} + \overline{M}_{od}P_{odt} + \left(P_{odt} - \overline{P}_{od}\right)\left(M_{odt} - \overline{M}_{od}\right) + \text{const}_{od}, \tag{6}$$

Table 27: Drought-Induced Migration and Low-Quality Housing Inflows: Main Cities and Rural-Origin Samples

| | *Main cities* | | | *Rural origins → main cities* | | |
|---|---|---|---|---|---|---|
| | OLS | FS | 2SLS | OLS | FS | 2SLS |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Urban migration rate | 0.342*** | | 0.131* | 0.327*** | | 0.139* |
| | (0.025) | | (0.078) | (0.028) | | (0.074) |
| Drought intensity | | 3.122*** | | | 2.541*** | |
| | | (0.916) | | | (0.653) | |
| First-stage $F$ | | | 11.61 | | | 15.16 |
| AR 95% CI | | | [-0.123, 0.267] | | | [-0.066, 0.273] |
| Effect of 1-SD drought (% mean) | +4.6% | +4.8% | +1.8% | +4.7% | +5.2% | +2.0% |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes | Yes | Yes |
| $R^2$ | 0.879 | 0.770 | 0.814 | 0.856 | 0.757 | 0.804 |
| Within $R^2$ | 0.589 | 0.025 | 0.365 | 0.528 | 0.042 | 0.355 |
| Mean outcome | | 11.634 | | | 8.890 | |
| Mean migration rate | | 33.169 | | | 24.660 | |
| SD(drought intensity) | | 0.507 | | | 0.505 | |
| Observations | | 27,700 | | | 24,280 | |
| OD pairs | | 1,948 | | | 1,704 | |

*Notes:* Outcome: the low-quality-housing inflow rate; OLS and 2SLS columns report the effect of the urban migration rate on it, FS columns the sample-specific first stage. Sample: columns (1)–(3) restrict destinations to main cities, a census urban share of at least 0.55, and (4)–(6) additionally to rural origins. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. The cumulative instrument is omitted, its first stage collapsing here. First-stage $F$ is the cluster-robust Wald statistic and AR 95% CI the Anderson–Rubin interval. Effect of 1-SD drought: coefficient × SD(instrument) ÷ mean outcome × 100, multiplied by the first-stage coefficient in the OLS and 2SLS columns. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

where the corridor-specific constant is absorbed by the corridor fixed effect. Estimating Equation (4) with each of the three remaining terms as the outcome therefore provides an exact decomposition of the scale coefficient.

On the sample for which $P_{odt}$ is defined, the scale coefficient is 0.320, compared with 0.285 in the full panel. Its three components are 0.299, 0.063, and −0.042, respectively. The first captures the additional low-quality-housing inflow mechanically generated when drought increases migration while holding the corridor's prevailing housing composition fixed. The second captures the contribution of changes in migrant composition. The third is the within-corridor covariance between the two deviations, and it belongs to the volume channel rather than to composition: grouped with the first term it comes to 0.257, which is 80% of the

Table 28: Propensity for Low-Quality Housing: Main Results

| | *Contemporaneous drought* | | | | *3-season mean drought (lagged)* | | | |
|---|---|---|---|---|---|---|---|---|
| | Deprived Dummy - Main def. | | Number of deprivations Continuous | | Deprived Dummy - Main def. | | Number of deprivations Continuous | |
| | OLS | 2SLS | OLS | 2SLS | OLS | 2SLS | OLS | 2SLS |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Urban migration rate | 0.00003** | 0.00122** | 0.00002 | 0.00095 | 0.00003** | 0.00198** | 0.00003 | 0.00319* |
| | (0.00002) | (0.00060) | (0.00004) | (0.00107) | (0.00002) | (0.00093) | (0.00004) | (0.00164) |
| First-stage $F$ | | 15.50 | | 15.50 | | 8.24 | | 8.24 |
| AR 95% CI | | [0.00, 0.00] | | [-0.00, 0.00] | | [0.00, 0.01] | | [0.00, 0.01] |
| Effect of 1-SD drought | +0.01 pp | +0.37 pp | +0.0001 | +0.0029 | +0.01 pp | +0.58 pp | +0.0001 | +0.0093 |
| $R^2$ | 0.392 | 0.295 | 0.450 | 0.436 | 0.396 | 0.141 | 0.455 | 0.303 |
| Within $R^2$ | 0.007 | -0.151 | 0.002 | -0.023 | 0.007 | -0.411 | 0.002 | -0.278 |
| Mean outcome (average migrant) | 0.406 | 0.406 | 0.662 | 0.662 | 0.404 | 0.404 | 0.656 | 0.656 |
| Mean migration rate | 89.079 | 89.079 | 89.079 | 89.079 | 88.806 | 88.806 | 88.806 | 88.806 |
| Implied marginal migrant | | 0.515 | | 0.747 | | 0.580 | | 0.940 |
| Observations | 72,723 | 72,723 | 72,723 | 72,723 | 71,207 | 71,207 | 71,207 | 71,207 |
| OD pairs | 10,537 | 10,537 | 10,537 | 10,537 | 10,510 | 10,510 | 10,510 | 10,510 |

*Notes:* Outcomes: the share of urban in-migrants in low-quality housing under the main UN-Habitat definition, at least one deprived dimension among floor, wall, roof, sanitation and crowding with metal roofs durable, and the mean number of deprived dimensions. Water is excluded: the harmonized IPUMS water-supply codes cannot reliably separate deficient from adequate access. Sample: each column's own estimation sample, columns (5)–(8) lag-restricted. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought in columns (1)–(4) and by its three-season mean in columns (5)–(8), whose first stage is weak ($F \approx 7$), so the lagged point magnitudes are not interpretable and the Anderson–Rubin intervals carry the inference. The 1-SD effect is in percentage points of the share and in dimension units for the count. 'Implied marginal migrant' is $\overline{P} + \beta\overline{M}$: since $S = P \times M$, the 2SLS coefficient equals $(p_{\text{marg}} - \overline{P})/\overline{M}$, and both means come from the column's own sample, the full-panel mean migration rate being three times smaller and shrinking the implied gap by the same factor. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

coefficient, against 0.063, or 20%, for composition; the two shares sum to the whole. Drought therefore increases the inflow into low-quality housing primarily by sending more migrants, and only secondarily by changing their housing composition.

*Decomposing the Effect for housing* - Table 36 details the decomposition of the additional low-quality housing inflow from a one-standard-deviation drought into its scale and propensity components, reporting the formula, inputs, and implied contribution of each channel - roughly 90% through the number of migrants and 10% through their propensity to live in low-quality housing.

## 12.5 Individual Heterogeneity: Migrant Profiles

This section asks two distinct questions about the migrants behind the main result: who the drought-induced migrants are (Table 37), and whether the effect varies with their profile - for low-quality housing (Figure 19) and for employment (Figure 20).

Table 29: Propensity for Low-Quality Housing: Main Cities and Rural-Origin Samples

| | *Main cities* | | | | *Rural origins → main cities* | | | |
|---|---|---|---|---|---|---|---|---|
| | Deprived Dummy - Main def. | | Number of deprivations Continuous | | Deprived Dummy - Main def. | | Number of deprivations Continuous | |
| | OLS | 2SLS | OLS | 2SLS | OLS | 2SLS | OLS | 2SLS |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Urban migration rate | 0.07472*** | 2.21915 | 0.10170** | 0.66641 | 0.14962** | 2.40963 | 0.22845** | -0.75675 |
| | (0.02551) | (1.80309) | (0.04359) | (2.61755) | (0.05970) | (2.23782) | (0.10708) | (3.49205) |
| First-stage $F$ | | 6.07 | | 6.07 | | 10.44 | | 10.44 |
| AR 95% CI | | [-0.85, 12.82] | | [-6.93, 9.90] | | [-1.83, 9.44] | | [-9.88, 7.20] |
| Effect of 1-SD drought | +0.01 pp | +0.36 pp | +0.0002 | +0.0011 | +0.02 pp | +0.33 pp | +0.0003 | −0.0010 |
| $R^2$ | 0.361 | 0.229 | 0.384 | 0.381 | 0.364 | 0.305 | 0.383 | 0.379 |
| Within $R^2$ | 0.011 | -0.193 | 0.004 | -0.001 | 0.011 | -0.080 | 0.004 | -0.002 |
| Mean outcome | 0.345 | 0.345 | 0.472 | 0.472 | 0.346 | 0.346 | 0.470 | 0.470 |
| Observations | 17,273 | 17,273 | 17,273 | 17,273 | 14,720 | 14,720 | 14,720 | 14,720 |
| OD pairs | 1,633 | 1,633 | 1,633 | 1,633 | 1,414 | 1,414 | 1,414 | 1,414 |

*Notes:* Outcomes: within each sample, the first pair of columns is the share of in-migrants in low-quality housing under the main definition and the second pair the mean number of deprived dimensions; the denominator is migrants, unweighted and unwinsorized. Sample: columns (1)–(4) restrict destinations to main cities, a census urban share of at least 0.55, and columns (5)–(8) additionally to rural origins. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. The cumulative instrument has no first stage in these samples and is omitted. First-stage $F$ is the cluster-robust Wald statistic on each column's own sample and AR 95% CI the Anderson–Rubin interval. Effect of 1-SD drought: coefficient × first-stage coefficient × SD(instrument). Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

*Who Are the Drought-Induced Migrants?* - Table 37 profiles the drought-induced migrants on gender, age at arrival, and education, comparing the naive read - migrants arriving in high- versus no-drought years - with the complier characterization of Abadie (2003), which describes the marginal migrants identified by the instrument, the same population as the main result. The two readings agree: drought-induced migrants are observationally similar to other migrants - the same gender balance and comparable or slightly higher education - but younger, with an over-representation of children under fifteen, consistent with whole-household displacement rather than the selective migration of young workers. Their higher propensity for low-quality housing is therefore not accounted for by observably lower human capital.

*Does the Effect Vary by Profile?* - Figure 19 estimates the effect separately by subgroup, on both margins, with subgroup-specific inflows and migration rates under the main specification. Effects and propensity shifts are broadly uniform across gender and age - every identified estimate sits close to the aggregate anchor, and no within-dimension contrast is significant. The only gradient is educational: on the scale margin, the effect is twice as strong for primary-educated adults as for those with secondary education or above ($p = 0.036$), suggestive since education is measured at census. Full estimates and contrasts are in Sup-

Table 30: Drought-Induced Migration and the Labor-Force Status of the Inflow: Scale

| | *Employed* | | *Unemployed* | | *Inactive* | |
|---|---|---|---|---|---|---|
| | OLS | 2SLS | OLS | 2SLS | OLS | 2SLS |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Urban migration rate | 0.530*** | 0.676*** | 0.066*** | 0.130*** | 0.401*** | 0.200*** |
| | (0.016) | (0.056) | (0.007) | (0.022) | (0.015) | (0.057) |
| First-stage $F$ | | 28.84 | | 28.84 | | 28.84 |
| AR 95% CI | | [0.572, 0.808] | | [0.092, 0.186] | | [0.058, 0.299] |
| Effect of 1-SD drought (% mean) | +5.8% | +7.4% | +6.1% | +12.1% | +7.9% | +4.0% |
| $R^2$ | 0.828 | 0.801 | 0.374 | 0.277 | 0.687 | 0.593 |
| Within $R^2$ | 0.672 | 0.621 | 0.142 | 0.009 | 0.550 | 0.414 |
| Mean outcome | 9.721 | 9.721 | 1.139 | 1.139 | 5.371 | 5.371 |
| Observations | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 |

*Notes:* Outcomes: employed, unemployed and inactive in-migrants per 100 urban residents at destination, unwinsorized. Sample: in-migrants aged 15 or over *at arrival*, age minus years since migration, all destinations, 8 countries, 12-year window, undetermined labour-force status excluded from the known-status denominator. EMPSTAT is not asked below age 5 and records 5–14 year-olds as inactive in over 80% of cases, so the restriction is what makes the inactive category meaningful: status is known for 99.8% of this population against 94.6% overall. The housing tables carry no such restriction, housing being an attribute of the dwelling. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. The endogenous regressor is the adult migration rate, so each coefficient reads per adult migrant and the first stage differs from that of the housing tables, its value for this table being reported above. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

plementary Appendix Tables S7 and S8.

Figure 20 repeats the exercise for the employment margins. The mirror image completes the picture: employment absorption is uniform where housing is not - there is no educational gradient in the employed inflow ($p = 0.787$), while the housing effect is twice as strong for the primary-educated. Full estimates are in Supplementary Appendix Tables S9 and S10.

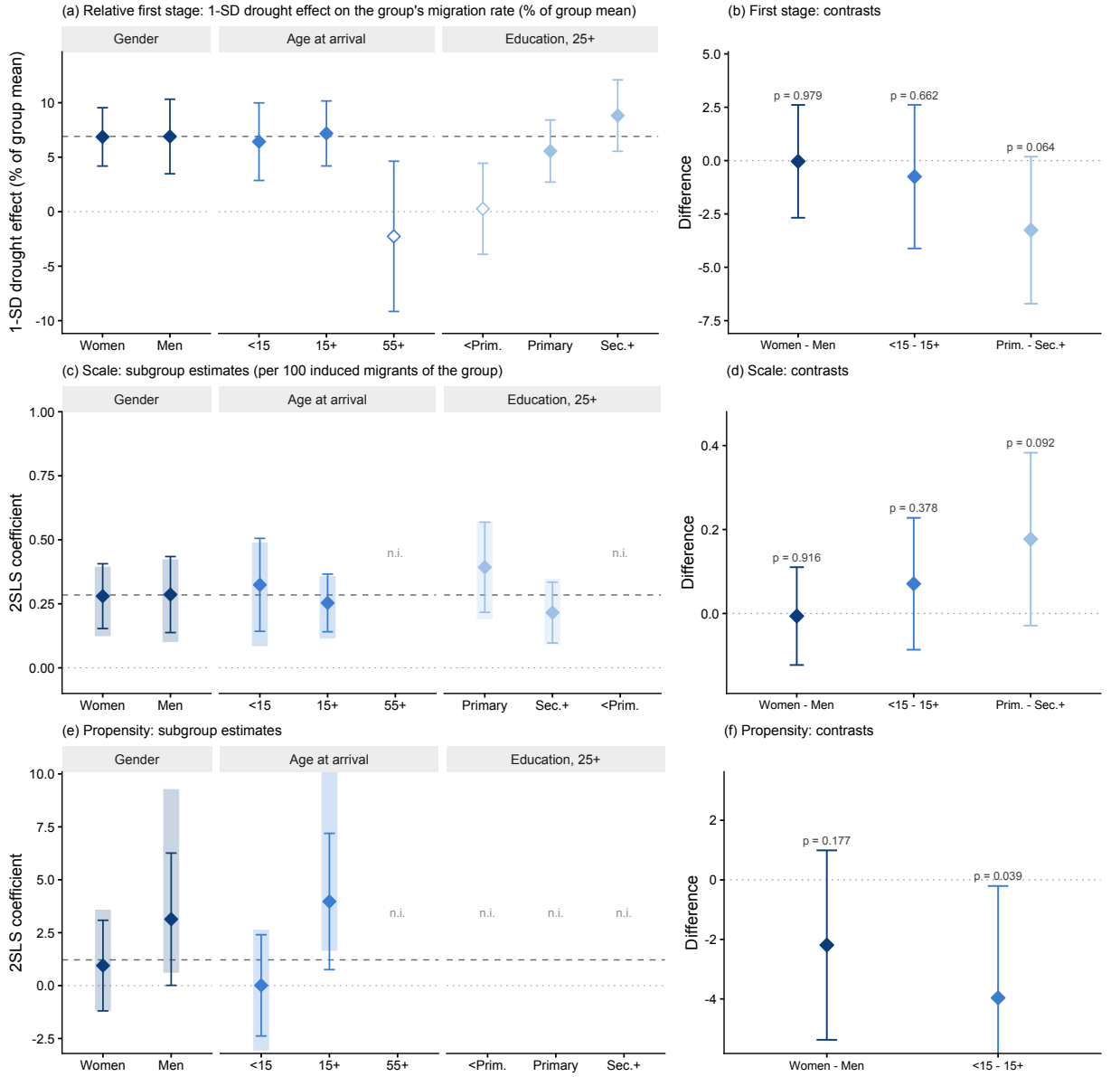
## 12.6 Geography of the Drought-Induced Inflows

Figure 21 maps the destination side of the first stage: for each destination, the share of its observed urban in-migration that the first stage attributes to drought. It illustrates the destination heterogeneity of Section 8.1.1 geographically – the share is highest in intermediate destinations and lowest in the largest metropolises – and is an illustration of where the estimated inflows land, not an identified result.

Table 31: Employment Outcomes per 100 Additional Adult Arrivals: Estimates and Weak-Instrument-Robust Intervals

| | Baseline | + Destination trends |
|---|---|---|
| | (1) | (2) |
| *Panel A. Labor-Force Status* | | |
| Employed | 67.63*** | 91.24*** |
| | (5.62) | (14.45) |
| 95% CI | [56.62, 78.64] | [62.91, 119.57] |
| AR 95% CI | [57.22, 81.01] | [70.41, 151.77] |
| Unemployed | 13.03*** | 0.95 |
| | (2.21) | (3.81) |
| 95% CI | [8.69, 17.37] | [-6.53, 8.42] |
| AR 95% CI | [9.15, 18.62] | [-12.31, 7.56] |
| Inactive | 20.03*** | 8.16 |
| | (5.65) | (12.94) |
| 95% CI | [8.95, 31.10] | [-17.20, 33.53] |
| AR 95% CI | [5.80, 29.94] | [-43.59, 27.69] |
| First stage | 2.113 (0.394) | 1.062 (0.329) |
| First-stage $F$ | 28.84 | 10.40 |
| Observations | 241,540 | |
| *Panel B. Class of Worker* | | |
| Wage/salaried | 44.14*** | 46.78*** |
| | (8.23) | (16.16) |
| 95% CI | [28.00, 60.28] | [15.11, 78.45] |
| AR 95% CI | [28.91, 67.66] | [18.79, 223.63] |
| Self-employed | 30.20*** | 46.65*** |
| | (5.90) | (16.56) |
| 95% CI | [18.64, 41.76] | [14.20, 79.11] |
| AR 95% CI | [20.45, 49.05] | [25.41, 291.39] |
| Unpaid/other | -0.19 | 4.21 |
| | (3.03) | (5.45) |
| 95% CI | [-6.13, 5.75] | [-6.47, 14.89] |
| AR 95% CI | [-9.12, 5.25] | [-17.24, 30.46] |
| First stage | 1.434 (0.384) | 0.736 (0.337) |
| First-stage $F$ | 13.92 | 4.78 |
| Observations | 224,668 | |
| *Panel C. Sector of Employment* | | |
| Services | 39.80*** | 56.20 |
| | (10.90) | (36.96) |
| 95% CI | [18.44, 61.17] | [-16.23, 128.64] |
| AR 95% CI | [20.46, 82.89] | Unbounded |
| Agriculture | 14.91*** | 23.26 |
| | (5.19) | (15.39) |
| 95% CI | [4.73, 25.09] | [-6.90, 53.41] |
| AR 95% CI | [4.35, 32.83] | Unbounded |
| Industry | 4.92 | 11.90 |
| | (4.41) | (11.05) |

Figure 19: Heterogeneous Effects by Migrant Profile




*Notes:* Whiskers are 95% Wald intervals and shaded bars Anderson–Rubin 95% intervals; dashed lines mark the aggregate anchors and shades of blue index the dimension. Panels (a)–(b) report the relative first stage, the effect of a 1-SD drought on the group's migration rate as a percentage of the group's own mean, by OLS and so valid regardless of instrument strength, the dashed line being the aggregate +6.9%. Panels (c)–(f) are subgroup-specific 2SLS under the main specification: the subgroup's low-quality-housing inflow per 100 urban residents in panel (c), unwinsorized and reading per 100 induced migrants of the group, and the share of its migrants in low-quality housing in panel (e), on positive-flow corridor-years, with the subgroup's migration rate instrumented by drought at origin. Subgroups with a conditional first-stage $F$ below 8 appear as empty positions and are reported AR-only in Supplementary Appendix Tables S7–S8; hollow diamonds mark $F < 10$, and propensity first stages are weak throughout, so inference rests on the AR intervals. Panels (b), (d) and (f) are within-dimension contrasts from the saturated stacked interaction. Education is measured at census and may post-date arrival. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Table 32: Drought-Induced Migration and the Labor-Force Status of the Inflow: Propensity

| | *Employed* | | *Unemployed* | | *Inactive* | |
|---|---|---|---|---|---|---|
| | OLS | 2SLS | OLS | 2SLS | OLS | 2SLS |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Urban migration rate | -0.00008*** | 0.00094 | 0.00002** | 0.00119* | 0.00006*** | -0.00214* |
| | (0.00002) | (0.00105) | (0.00001) | (0.00064) | (0.00002) | (0.00116) |
| First-stage $F$ | | 11.27 | | 11.27 | | 11.27 |
| AR 95% CI | | [-0.001, 0.004] | | [0.000, 0.003] | | [-0.006, -0.000] |
| Effect of 1-SD drought | −0.01 pp | +0.15 pp | +0.00 pp | +0.19 pp | +0.01 pp | −0.34 pp |
| $R^2$ | 0.348 | 0.322 | 0.265 | 0.122 | 0.349 | 0.218 |
| Within $R^2$ | 0.009 | -0.030 | 0.000 | -0.196 | 0.011 | -0.189 |
| Mean outcome (average migrant) | 0.638 | 0.638 | 0.068 | 0.068 | 0.294 | 0.294 |
| Mean migration rate | 60.842 | 60.842 | 60.842 | 60.842 | 60.842 | 60.842 |
| Implied marginal migrant | | 0.695 | | 0.140 | | 0.164 |
| Observations | 64,554 | 64,554 | 64,554 | 64,554 | 64,554 | 64,554 |

*Notes:* Outcomes: the employed, unemployed and inactive shares among known-status in-migrants; the three sum to one, so the 2SLS coefficients sum to zero by construction. Sample: in-migrants aged 15 or over *at arrival*, age minus years since migration, all destinations, 8 countries, 12-year window, undetermined status excluded from the denominator. EMPSTAT is not asked below age 5 and records 5–14 year-olds as inactive in over 80% of cases, so the restriction is what makes the inactive category meaningful: status is known for 99.8% of this population against 94.6% overall. The housing tables carry no such restriction, housing being an attribute of the dwelling. Specification: origin–destination pair and arrival-year fixed effects, distance-to-census control, migration instrumented by contemporaneous drought intensity at origin; the endogenous regressor is the adult migration rate, so each coefficient reads per adult migrant and the first stage differs from that of the housing tables, its value being reported above. Means are those of this table's own estimation sample. 'Implied marginal migrant' is $\overline{P} + \beta\overline{M}$, a truncation of the identity $\beta_1 = \overline{P} + \overline{M}\beta_P +$ cross whose third term exceeds the second in absolute value, reported for comparability only. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## 12.7 Robustness

### 12.7.1 Leave-One-Country-Out

### 12.7.2 Housing Quality Gradient

## 12.8 Model

Figure 23 illustrates the two-state housing integration framework of Section 9.

Table 33: Drought-Induced Migration and the Labor-Force Status of the Inflow: Lagged Instrument

| | *Scale* | | | *Propensity* | | |
|---|---|---|---|---|---|---|
| | Inflows per 100 urban residents | | | Shares of known-status in-migrants | | |
| | Employed 2SLS | Unemployed 2SLS | Inactive 2SLS | Employed 2SLS | Unemployed 2SLS | Inactive 2SLS |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Urban migration rate | 0.52274*** | 0.16607*** | 0.31710*** | -0.30566 | 1.90688** | -1.60122 |
| | (0.04253) | (0.02191) | (0.03569) | (0.98895) | (0.84324) | (1.05507) |
| First-stage $F$ | 32.93 | 32.93 | 32.93 | 7.46 | 7.46 | 7.46 |
| AR 95% CI | [0.433, 0.609] | [0.130, 0.222] | [0.237, 0.384] | [-3.223, 2.282] | [0.769, 6.755] | [-6.683, 0.104] |
| Effect of 1-SD drought | +9.2% | +24.6% | +10.2% | −0.06 pp | +0.34 pp | −0.29 pp |
| $R^2$ | 0.831 | 0.161 | 0.676 | 0.353 | -0.098 | 0.280 |
| Within $R^2$ | 0.672 | -0.157 | 0.525 | 0.008 | -0.498 | -0.105 |
| Mean outcome | 0.00977 | 0.00116 | 0.00536 | 0.639 | 0.0689 | 0.292 |
| Observations | 235,920 | 235,920 | 235,920 | 63,295 | 63,295 | 63,295 |
| OD pairs | 16,846 | 16,846 | 16,846 | 9,945 | 9,945 | 9,945 |

*Notes:* Outcomes: columns (1)–(3) are employed, unemployed and inactive in-migrants per 100 urban residents, unwinsorized, with the 1-SD effect in percent of the mean; columns (4)–(6) are their shares among known-status in-migrants, which sum to one so that the coefficients sum to zero by construction, with the 1-SD effect in percentage points. Sample: urban migrants, all destinations, 8 countries, 12-year window, lag-restricted; undetermined labour-force status excluded. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by the three-season cumulative drought instrument. The propensity-side first stage is weak ($F \approx 7$), so the point magnitudes of columns (4)–(6) are not interpretable and inference rests on the Anderson–Rubin intervals. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 34: What Kind of Job Do Drought-Induced Migrants Find? Class of Worker

| | *All* | Wage/salary | Self-employed | Unpaid/other |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Urban migration rate | 0.74143*** | 0.44137*** | 0.30199*** | -0.00194 |
| | (0.09949) | (0.08234) | (0.05898) | (0.03031) |
| First-stage $F$ | 13.92 | 13.92 | 13.92 | 13.92 |
| AR 95% CI | [0.574, 1.048] | [0.292, 0.676] | [0.205, 0.490] | [-0.090, 0.052] |
| Of 100 marginal migrants | 74.1 | 44.1 | 30.2 | -0.2 |
| Share, marginal migrant (%) | 100.0 | 59.5 | 40.7 | -0.3 |
| Share, average arrival (%) | 100.0 | 59.3 | 33.4 | 7.4 |
| Gap (pp) | – | +0.3 | +7.4 | -7.6* |
| AR 95% CI (gap) | – | [-15.3, +15.8] | [-4.7, +25.5] | [-20.7, -0.6] |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |
| Mean outcome | 0.008772 | 0.004531 | 0.003440 | 0.000802 |
| Observations | 224,668 | 224,668 | 224,668 | 224,668 |

*Notes:* Outcome: in-migrants of each class of worker per 100 urban residents at destination, unwinsorized. Sample: urban in-migrants in employment aged 15 or over *at arrival*, as in Tables 30 and 32; IPUMS `CLASSWK` is carried by 10 of the 11 urban census samples. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. The endogenous regressor is the migration rate of the same adult population, so each coefficient reads per adult migrant and the first stage differs from that of the housing tables. Outcome and migration rate share a denominator, so each coefficient is the number out of 100 marginal migrants in that category and column (1) is the sum of (2)–(4). 'Share, marginal migrant' divides by that total, 'Share, average arrival' is the migrant-weighted share among all urban in-migrants of the sample, and 'Gap' is their difference, estimated on the excess outcome and rescaled by the total. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 35: What Kind of Job Do Drought-Induced Migrants Find? Sector

| | *All* | Agriculture | Industry | Services |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Urban migration rate | 0.59636*** | 0.14910*** | 0.04924 | 0.39803*** |
| | (0.11682) | (0.05195) | (0.04411) | (0.10900) |
| First-stage $F$ | 8.43 | 8.43 | 8.43 | 8.43 |
| AR 95% CI | [0.392, 1.062] | [0.044, 0.328] | [-0.094, 0.143] | [0.207, 0.829] |
| Of 100 marginal migrants | 59.6 | 14.9 | 4.9 | 39.8 |
| Share, marginal migrant (%) | 100.0 | 25.0 | 8.3 | 66.7 |
| Share, average arrival (%) | 100.0 | 7.8 | 15.9 | 76.3 |
| Gap (pp) | – | +17.2** | -7.6 | -9.6 |
| AR 95% CI (gap) | – | [-0.5, +43.7] | [-36.9, +6.3] | [-32.7, +20.1] |
| Origin-Destination FE | Yes | Yes | Yes | Yes |
| Arrival-year FE | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes |
| Mean outcome | 0.008146 | 0.001313 | 0.001293 | 0.005541 |
| Observations | 209,548 | 209,548 | 209,548 | 209,548 |

*Notes:* Outcome: in-migrants of each sector per 100 urban residents at destination, unwinsorized. Sample: urban in-migrants in employment aged 15 or over *at arrival*, as in Tables 30 and 32; IPUMS `INDGEN` is carried by 9 of the 11 urban census samples, and the split is three-way because the services subdivision is not comparable across censuses. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin, the endogenous regressor being the migration rate of the same adult population, so each coefficient reads per adult migrant and the first stage differs from that of the housing tables. Outcome and migration rate share a denominator, so each coefficient is the number out of 100 marginal migrants in that category and column (1) is the sum of (2)–(4). 'Share, marginal migrant' divides by that total, 'Share, average arrival' is the migrant-weighted share among all urban in-migrants of the sample, and 'Gap' is their difference, estimated on the excess outcome and rescaled by the total. The first stage is weak here, so inference rests on the Anderson–Rubin intervals and the point magnitudes are indicative. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 36: Decomposition of the Drought Effect on Low-Quality Housing Inflows: Scale versus Propensity

| | Low-quality housing inflow rate (scale) (1) | Share of migrants in low-quality housing (propensity) (2) |
|---|---|---|
| *Panel A: Both channels on the common (positive-flow) sample* | | |
| Urban migration rate | 0.32022*** | 1.21591** |
| | (0.10891) | (0.59790) |
| First-stage $F$ | 15.50 | 15.50 |
| AR 95% CI | [0.032, 0.525] | [0.170, 2.940] |
| Mean outcome | 0.03974 | 0.4064 |
| Observations | 72,723 | 72,723 |
| *Panel B: Decomposition of a one-standard-deviation drought* | | |
| | Calculation | Share |
| Scale: $\Delta\text{MigRate} \times \overline{P}$ *(more migrants, same propensity)* | $0.00302 \times 0.4064 = 0.00123$ | 79% |
| Propensity: $\overline{\text{MigRate}} \times \Delta P$ *(same migrants, higher propensity)* | $0.08908 \times 0.00368 = 0.00033$ | 21% |
| Total | 0.00156 | 100% |

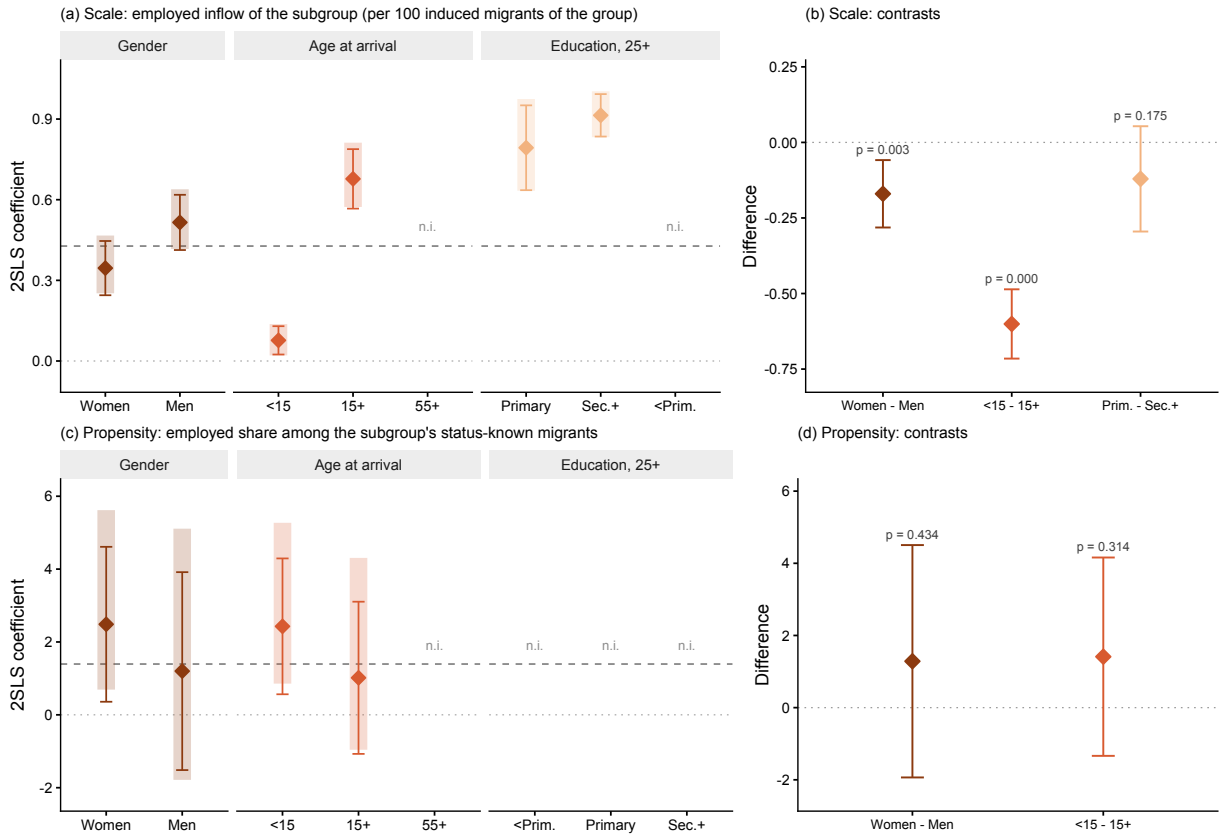
*Notes:* Outcome: the additional low-quality housing inflow caused by a one-standard-deviation drought, per 100 urban residents at destination, split as $\Delta S \approx \Delta\text{MigRate} \times \overline{P} + \overline{\text{MigRate}} \times \Delta P$. Sample: the identity is an accounting split only if all four inputs come from one sample, so both channels are estimated on the corridor-years with a positive urban migrant flow ($N$ = 72,723), where the propensity is defined. Specification: origin–destination pair and arrival-year fixed effects, distance-to-census control, migration instrumented by contemporaneous drought intensity at origin, with cluster-robust Anderson–Rubin intervals. Column (1) of Panel A is therefore not the paper's headline scale estimate of 0.28455 on the full panel (Table **??**): restricting to positive-flow corridors conditions on the endogenous outcome and serves here only to give the two channels one denominator. Panel B reports the four inputs, all from that subsample. The implied total of 0.00156 exceeds the direct estimate of 0.00097 on the same sample because the identity evaluates the scale channel at the average migrant's propensity, 40.6%, and not at the marginal migrant's, which is the scale coefficient itself, 32.0%: the decomposition measures the relative weight of the two channels and not the level of the effect. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 37: Who Are the Drought-Induced Migrants? Selection on Observables

| | *Arrival-year drought (descriptive)* | | | *Complier characterization (2SLS)* | | |
|---|---|---|---|---|---|---|
| | No drought | High drought | Diff. | Drought-induced | All migrants | Diff. |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Share female (%) | 50.5 | 50.9 | +0.4 | 51.8 | 50.4 | +1.4 |
| Mean age at arrival (years) | 21.0 | 20.3 | -0.7*** | 16.4 | 21.1 | -4.8*** |
| Share below 15 at arrival (%) | 29.1 | 30.3 | +1.2 | 41.6 | 28.5 | +13.1*** |
| Share 15–29 at arrival (%) | 52.5 | 52.2 | -0.3 | 41.9 | 53.1 | -11.2** |
| Share below primary, 25+ (%) | 15.7 | 14.7 | -1.0 | 11.2 | 15.2 | -4.0 |
| Share secondary or above, 25+ (%) | 43.9 | 45.1 | +1.2 | 47.0 | 44.7 | +2.3 |
| Migrants represented (millions) | 5.72 | 1.29 | | 11.60 | | |

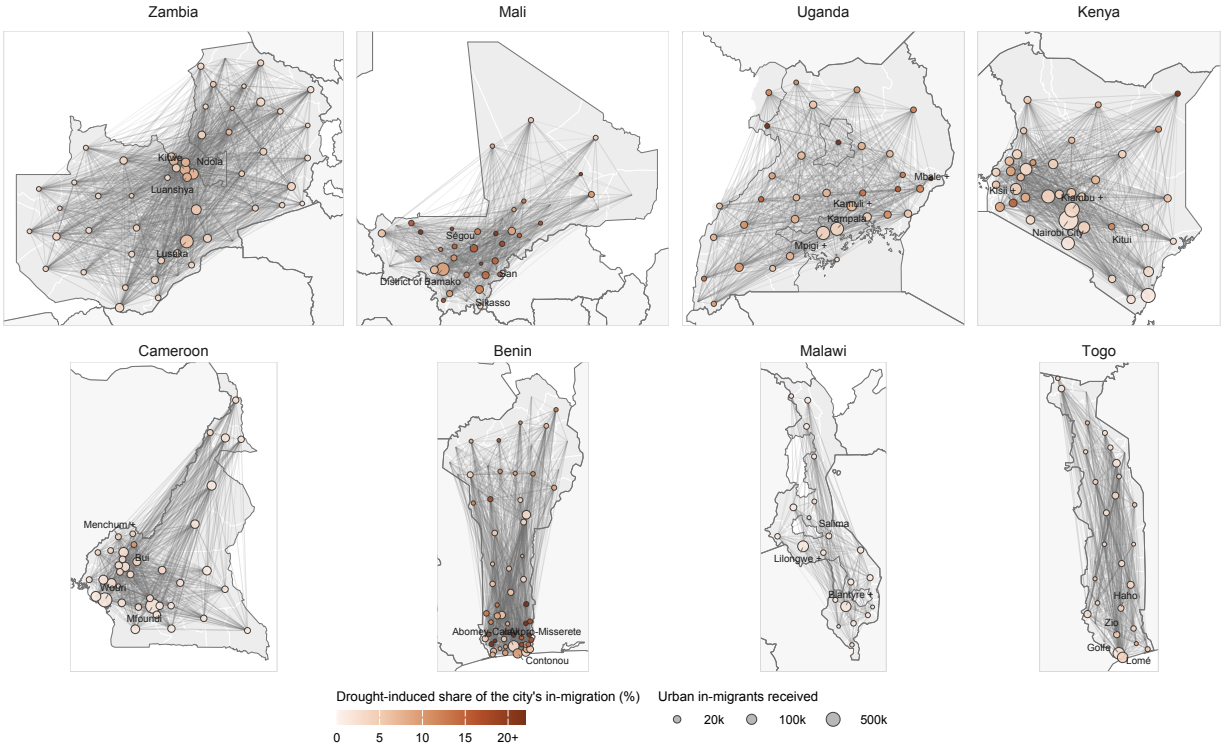
*Notes:* Outcome: characteristics of urban in-migrants, canonical panel, $N = 241{,}540$ corridor-years. Sample: age at arrival is census age minus years since migration; education is measured at census, may post-date arrival, and covers adults aged 25 and over with known attainment, 40 to 55% of migrants. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Columns (1)–(3) are descriptive: migrant-weighted means for arrivals in no-drought ($z = 0$) against high-drought corridor-years, the top quartile of positive $z$, differenced by a migrant-weighted WLS contrast. Columns (4)–(6) characterise the compliers (Abadie, 2003): each characteristic-specific inflow rate is regressed by 2SLS on the migration rate instrumented by drought, column (4) being the ratio of coefficients, which describes the marginal drought-induced migrants, and column (6) the difference, tested on the excess outcome. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 20: Heterogeneous Effects by Migrant Profile: Employment




*Notes:* As Figure 19, for the employment margins, the first stage being common to both figures and shown in Figure 19(a). Panel (a) is the subgroup's employed inflow per 100 urban residents and panel (c) the employed share among its status-known migrants; dashed lines mark the aggregate anchors and shades of orange index the dimension. Display rule as in Figure 19: empty positions where the conditional first-stage $F$ falls below 8, hollow markers between 8 and 10, and Anderson–Rubin inference for the propensity margins, whose first stages are weak throughout. The age contrast in panel (b) is mechanical, children being rarely employed, and the under-15 column of panel (c) is child employment, conceptually thin. Full estimates in Supplementary Appendix Tables S9–S10.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

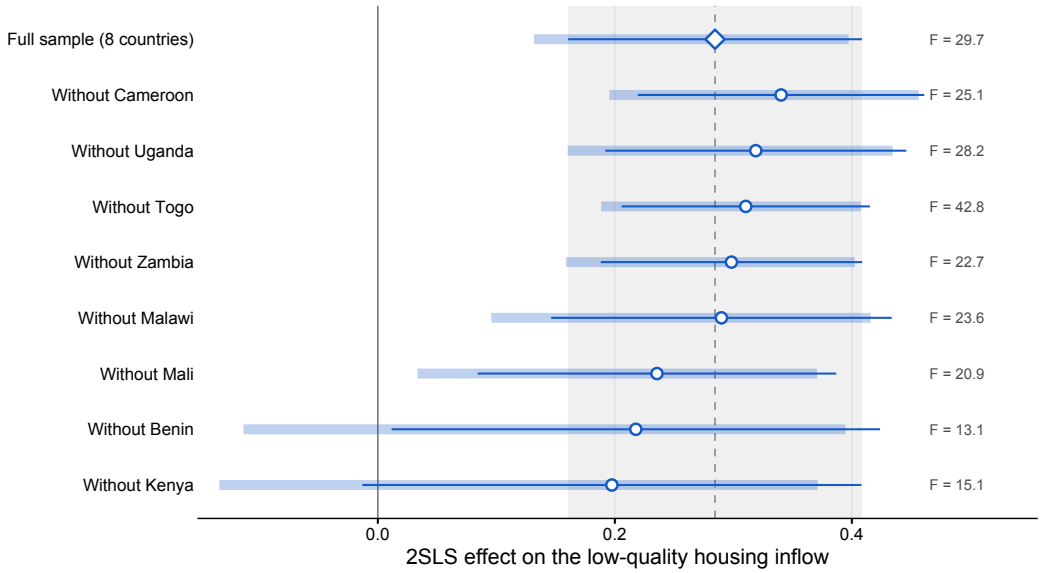
Figure 21: Drought-Induced Share of Urban In-Migration by Destination, by Country




*Notes:* Circles mark destinations receiving at least 1,000 urban in-migrants, 299 of 352; area is proportional to total in-migration and shade to the drought-induced share of it, the first-stage-implied inflow $\sum_{o,t} \hat{\beta}_{k(o,d)}\, \text{drought}_{o,t}\, \text{pop}_d^{urb}/100$ divided by the observed one. That share is bounded at destination level, with a median of 4.5%, and capped at 22% on the scale; the corridor-level analogue is not usable. Grey lines show all 10,399 corridors with a positive urban flow and the darker arcs the 189 largest, the top 25 per country. The first stage is estimated by within-country distance tercile rather than pooled, since one coefficient would give short and long corridors the same per-capita response; the mapped pattern is robust to this, with a rank correlation of 0.935 against the pooled version, and the corrected allocation totals 430,494 drought-induced migrants against 400,139 pooled. The figure illustrates the geography of estimated inflows: the identified result is the bin-level 2SLS analysis of Figure S4. Centroids are interior points of administrative polygons and not city locations.

*Source:* Authors' computations from IPUMS census microdata, CHIRPS data and IPUMS harmonized shapefiles.

Figure 22: Leave-One-Country-Out Robustness of the Main Effect




*Notes:* The diamond is the full eight-country estimate and the shaded band its 95% confidence interval; each row re-estimates on the sample with one country removed, thin whiskers being cluster-robust Wald intervals, thick bars the weak-instrument-robust Anderson–Rubin 95% intervals, and the annotation the first-stage $F$ of that subsample. The outcome is the low-quality-housing inflow per 100 urban residents, main UN-Habitat definition, unwinsorized; migration is instrumented by drought intensity at origin under the main specification, origin–destination-pair and arrival-year fixed effects with the distance-to-census control. The estimate stays positive in every subsample and all eight fall inside the full-sample confidence interval, so no single country moves the magnitude. Precision depends on the countries that carry the first stage: dropping Benin or Kenya lowers the first-stage $F$ to about 11 and 12, and in Benin, Kenya and Mali the Anderson–Rubin interval then includes zero, which reflects lost power and not a change in the estimate, those point estimates sitting inside the full-sample interval. Standard errors clustered at the origin district.

*Source:* Authors' computations from IPUMS census microdata and CHIRPS climate data.

Table 38: Robustness: Scale and Propensity across Definitions of Low-Quality Housing

| | *Metal roofs deprived* | | *Metal roofs durable* | | |
|---|---|---|---|---|---|
| | ≥1 dim. | ≥2 dim. (conservative) | ≥1 dim. (main) | ≥2 dim. | Nb. of deprived dimensions |
| | (1) | (2) | (3) | (4) | (5) |
| *Panel A. Scale: definition-specific inflow per 100,000 urban residents* | | | | | |
| Urban migration rate | 0.941*** | 0.301*** | 0.285*** | 0.132*** | -0.004 |
| | (0.042) | (0.052) | (0.063) | (0.051) | (0.005) |
| First-stage $F$ | | | 29.65 | | |
| AR 95% CI | [0.86, 1.04] | [0.18, 0.40] | [0.13, 0.40] | [0.01, 0.23] | [-0.02, 0.00] |
| Effect of 1-SD drought (% mean) | +7.5% | +5.3% | +4.3% | +4.2% | −1.1% |
| Mean outcome | 22.75470 | 10.33987 | 11.96444 | 5.65762 | 0.64710 |
| Observations | | | 241,540 | | |
| *Panel B. Propensity: outcome measured among in-migrants* | | | | | |
| Urban migration rate | 0.000 | 0.001* | 0.001** | 0.000 | 0.001 |
| | (0.000) | (0.001) | (0.001) | (0.000) | (0.001) |
| First-stage $F$ | | | 15.50 | | |
| AR 95% CI | [-0.00, 0.00] | [0.00, 0.00] | [0.00, 0.00] | [-0.00, 0.00] | [-0.00, 0.00] |
| Effect of 1-SD drought | +0.12 pp | +0.32 pp | +0.37 pp | +0.01 pp | +0.0029 |
| Mean outcome | 0.8508 | 0.3475 | 0.4064 | 0.1637 | 0.6620 |
| Observations | | | 72,723 | | |
| Origin-Destination FE | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes |
| Dist. to census control | Yes | Yes | Yes | Yes | Yes |

*Notes:* Outcomes: Panel A is the inflow per 100,000 urban residents, unwinsorized, Panel B the share among in-migrants, unweighted; column (5) uses the mean number of deprived dimensions, so its coefficients and 1-SD effects are in dimension units. Each column re-estimates both channels under one definition of low-quality housing, column (3) being the paper's main one, at least one deprivation with metal roofing durable. Water is excluded from every definition: the harmonized IPUMS water-supply codes cannot reliably separate deficient from adequate access. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. The first stage is common to all columns within a panel. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 39: Crowding-Out: Does Drought-Induced Migration Degrade the Housing of Earlier Cohorts?

| | Concurrent cohort | | Earlier cohorts | |
|---|---|---|---|---|
| | Full sample | Restricted | Low-quality housing | Total inflow |
| | (1) | (2) | (3) | (4) |
| Urban migration rate | 0.28455*** | 0.19956 | -1.01066 | 0.24737 |
| | (0.06323) | (0.12759) | (0.82838) | (1.38003) |
| First-stage $F$ | 29.65 | 10.80 | 10.80 | 10.80 |
| AR 95% CI | [0.130, 0.398] | [-0.287, 0.396] | [-4.499, 0.138] | [-4.944, 2.403] |
| Mean outcome | 0.01196 | 0.01252 | 0.04644 | 0.10137 |
| Observations | 241,540 | 219,589 | 219,589 | 219,589 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 |

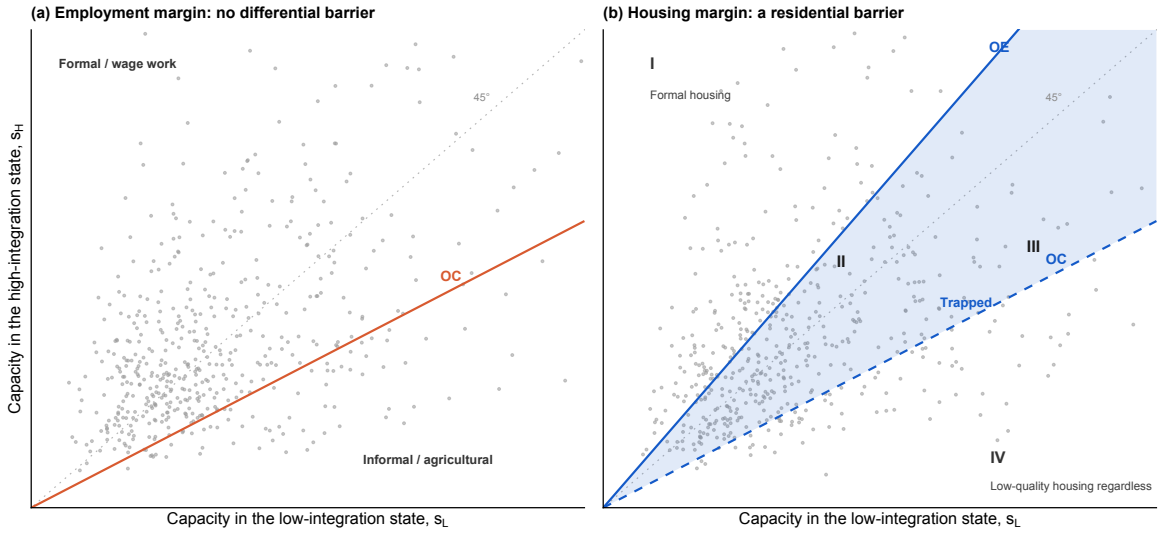
*Notes:* Outcome: whether drought-induced migration crowds previously settled migrants out of decent housing. Column (1) is the headline scale estimate (Table **??**); column (2) repeats it on the corridors with at least one earlier arrival cohort, the sample of columns (3)–(4); column (3) replaces the outcome with the low-quality-housing count of migrants who arrived on the same corridor in earlier years, per 100 urban residents, those migrants having arrived before the shock and so unable to be its compliers, which makes a positive coefficient crowding-out; column (4) is a placebo on their total number, which no mechanism should move. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. This channel does not enter the headline outcome, which counts only the concurrent cohort: a null here shows that crowding-out does not spill onto settled migrants and not that it is absent within the arrival cohort (Table 40). Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 40: Crowding-Out within the Arrival Cohort: Evidence from Migrants Whose Inflow Does Not Respond to Drought

| | Less than primary (1) | Primary (2) | Secondary or more (3) | All migrants (4) |
|---|---|---|---|---|
| *Panel A: Does drought change how many of them arrive?* | | | | |
| Drought intensity | 0.000142 | 0.000532*** | 0.000597*** | 0.003618*** |
| | (0.000102) | (0.000107) | (0.000119) | (0.000664) |
| $p$-value | 0.163 | 0.000 | 0.000 | 0.000 |
| *Panel B: Does their low-quality-housing propensity rise with drought-induced migration?* | | | | |
| Urban migration rate | 1.25066 | 1.52444* | 1.72953* | 1.21591** |
| | (0.79559) | (0.82700) | (1.01358) | (0.59790) |
| First-stage $F$ | 13.78 | 11.57 | 6.15 | 15.50 |
| AR 95% CI | [-0.210, 3.526] | [0.192, 4.448] | [0.324, 9.009] | [0.178, 2.942] |
| Baseline share | 0.531 | 0.395 | 0.247 | 0.406 |
| Observations | 28,734 | 37,688 | 34,071 | 72,723 |

*Notes:* Outcome: crowding-out within the arrival cohort, the channel that would bias the headline estimate. Sample: education is measured at census among adults aged 25 and over. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Panel A regresses each group's arrival rate per 100 urban residents on drought intensity at origin: the inflow of the least-educated group does not respond, so its arrivals in a drought year are almost entirely always-takers. Panel B regresses each group's share in low-quality housing on the total urban migration rate, instrumented by drought; crowding-out would make column (1) positive. It is positive and of the order of the aggregate propensity effect, and its Anderson–Rubin interval contains both zero and substantial crowding-out, so the test is inconclusive and we do not read it as a null. Column (4) reproduces the aggregate estimate of Table 28. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 23: The Two-Margin Integration Model: Employment versus Housing




*Notes:* The figure is an illustration and not an estimate: the scattered points are simulated draws, the slopes are chosen for legibility, no parameter is taken from the data, and the width of the shaded region carries no quantitative content. Each panel is a two-state Roy diagram for one integration margin, in the space of a migrant's capacities $(s_L, s_H)$ in the low- and high-integration states, the panels showing distinct margin-specific draws that are positively correlated across margins. Absent barriers a migrant chooses the high state whenever $w_H s_H > w_L s_L$, above the ray $OC$ of slope $w_L/w_H$; a proportional barrier $\tau$ rotates the cutoff to the steeper ray $OE$ of slope $w_L/[(1-\tau)w_H]$. In Panel (a) the estimates find no differential employment barrier across destinations, so the cutoff does not rotate; in Panel (b) a residential barrier opens the shaded fan between $OC$ and $OE$, containing the migrants it misallocates, who clear the employment margin and remain in low-quality housing, the working poor of Table 8. The 45° line splits that fan into Region II, whose comparative advantage lies in the high state, and Region III, whose does not. Two features are conventions and not findings: placing Region II above the 45° line presumes $(1-\tau)w_H < w_L < w_H$, and reading the fan as the working poor presumes capacities are positively correlated across margins, which the cross-tabulation supports and the figure does not impose. In the paper's estimates the barrier is indexed by the destination city and by nothing else.

*Source:* Authors' illustration.

# S1 Supplementary Appendix

## S1.1 Exclusion Restriction: Further Tables

Table S1: Exclusion Restriction: Destination Shocks and the Propensity Margin

| | Destination drought | | Destination conflict | | Destination crop prices | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | No control | With control | No control | With control | No control | With control |
| *Share of urban in-migrants in low-quality housing* | | | | | | |
| Urban migration rate | 1.21591** | 1.00212 | 1.83083** | 1.82955** | 1.13951* | 1.16502* |
| | (0.59790) | (0.72609) | (0.90150) | (0.90228) | (0.59312) | (0.60277) |
| First-stage $F$ | 15.50 | 13.47 | 9.47 | 9.46 | 15.70 | 15.64 |
| AR 95% CI | [0.17, 2.94] | [-0.44, 2.93] | [0.49, 5.69] | [0.49, 5.69] | [0.08, 2.81] | [0.11, 2.86] |
| Observations | 72,723 | 72,723 | 53,168 | 53,168 | 72,109 | 72,109 |

*Notes:* Outcome: the propensity margin, the share of in-migrants in low-quality housing, bounded in $[0, 1]$, so no ratio to the outcome mean is reported. Sample and controls as in Table 10: one subsample per control group, the odd column of each pair reporting that subsample without the control. Specification: 2SLS of the urban migration rate instrumented by contemporaneous drought intensity at origin, origin–destination-pair and arrival-year fixed effects, distance-to-census control, with cluster-robust Anderson–Rubin 95% intervals; the instrument is origin drought in every column and every control is measured at the destination. This is the paper's more fragile margin and the paired columns show why: within each group the control changes little, 1.216 to 1.002 under destination drought, 1.831 to 1.830 under conflict and 1.140 to 1.165 under crop prices, while the no-control columns differ sharply across groups, from 1.216 on the full panel to 1.831 on the ACLED subsample, so most of the apparent movement is coverage. The first-stage $F$ lies between 9 and 16 in every column, below the scale sample throughout. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## S1.2 Multiple-Hypothesis Correction

## S1.3 Robustness: Level of Clustering

## S1.4 The Structure of Drought Shocks

Figure S1 documents the shock process behind Section 4.2 country by country. Panel (a) shows how many origin districts a drought reaches in a given year and how hard; panels (b) and (c) give the full distribution across destinations of the two synchrony measures summarised by their medians in Table 19.

Table S2: Exclusion Restriction: All Three Destination Controls Entered Jointly

| | All three controls jointly | |
|---|---|---|
| | (1) | (2) |
| | No control | With controls |
| *Panel A. Low-quality housing inflow* | | |
| Urban migration rate | 0.24498*** | 0.17473** |
| | (0.05553) | (0.08422) |
| First-stage $F$ | 18.78 | 11.38 |
| AR 95% CI | [0.11, 0.35] | [-0.11, 0.32] |
| Observations | 168,820 | 168,820 |
| *Panel B. Employed inflow* | | |
| Urban migration rate | 0.54363*** | 0.58392*** |
| | (0.05424) | (0.07979) |
| First-stage $F$ | 18.78 | 11.38 |
| AR 95% CI | [0.45, 0.69] | [0.46, 0.87] |
| Observations | 168,820 | 168,820 |
| *Panel C. Share in low-quality housing* | | |
| Urban migration rate | 1.75015** | 3.05267* |
| | (0.88962) | (1.77002) |
| First-stage $F$ | 9.58 | 5.25 |
| AR 95% CI | [0.40, 5.47] | [0.66, 20.72] |
| Observations | 52,593 | 52,593 |

*Notes:* As Table 10, with destination drought, destination conflict and destination crop prices entered together on the subsample where all three are observed, $N = 168{,}820$, the binding constraint being ACLED's 1997 start. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Migration is instrumented by contemporaneous drought intensity at origin. This is a stress test and not an informative control. The three destination shocks are correlated with origin drought, rainfall being spatially correlated and cropping shares not independent of climate, so entered together they absorb a sizeable share of the identifying variation and trade significance among themselves: the first-stage $F$ drops, and in the reduced form of Supplementary Appendix Table 11 the instrument loses significance on this specification while destination drought gains it, having been indistinguishable from zero on its own. The employment result is unaffected. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S3: Multiple-Hypothesis Correction on the Secondary Family of Destination Contrasts

| | Contrast | Raw $p$ | Romano–Wolf $p$ | Bonferroni $p$ |
|---|---|---|---|---|
| *Panel A. First stage: volume of drought-induced migration ($\times 10^{-3}$)* | | | | |
| Past built-up expansion, 1975–1990 | +3.05 | 0.0002 | 0.0050*** | 0.0038 |
| Corridor distance | -4.83 | 0.0003 | 0.0075*** | 0.0054 |
| Housing saturation at destination | -2.48 | 0.0083 | 0.0920* | 0.1485 |
| Baseline migration share at destination | +1.93 | 0.0155 | 0.1484 | 0.2784 |
| Pre-period corridor network | +2.67 | 0.0383 | 0.2809 | 0.6896 |
| Industrial employment share at destination | +1.89 | 0.0486 | 0.3098 | 0.8739 |
| *Panel B. 2SLS conversion into low-quality housing (per 100 marginal migrants)* | | | | |
| Industrial employment share at destination | +0.189 | 0.2350 | 0.8406 | 1.0000 |
| Baseline migration share at destination | -0.052 | 0.5932 | 0.9890 | 1.0000 |
| Corridor distance | +0.105 | 0.7255 | 0.9900 | 1.0000 |
| Pre-period corridor network | -0.041 | 0.7658 | 0.9900 | 1.0000 |
| Housing saturation at destination | +0.028 | 0.8527 | 0.9900 | 1.0000 |
| Past built-up expansion, 1975–1990 | -0.005 | 0.9615 | 0.9900 | 1.0000 |
| *Panel C. 2SLS conversion into employment (per 100 marginal migrants)* | | | | |
| Pre-period corridor network | +0.234 | 0.0256 | 0.1934 | 0.4600 |
| Past built-up expansion, 1975–1990 | -0.140 | 0.1232 | 0.5977 | 1.0000 |
| Corridor distance | -0.194 | 0.2320 | 0.8406 | 1.0000 |
| Baseline migration share at destination | +0.060 | 0.4231 | 0.9715 | 1.0000 |
| Housing saturation at destination | -0.057 | 0.5483 | 0.9890 | 1.0000 |
| Industrial employment share at destination | +0.054 | 0.5793 | 0.9890 | 1.0000 |

*Notes:* Outcomes: 18 tests, six destination characteristics × three targets, the first stage and the two 2SLS conversions. Each contrast is the difference between the two bins of the saturated interacted IV of Section 8.1.3, split at the within-country median. The city-size contrast is excluded by design: it is the paper's pre-specified primary hypothesis (Section 8.1.1), tested on one dimension chosen ex ante, and it does not belong to the exploratory family a stepdown correction addresses. Specification: Romano–Wolf $p$-values from a stepdown procedure on 2,000 pairs-bootstrap draws resampling the 354 origin-district clusters, the same cluster draw applied to every test within a replication so that the correlation across tests is preserved, which is what makes the correction less punitive than the Bonferroni values reported alongside. None of the twelve conversion contrasts survives, which reinforces the paper's reading: the conversion of arrivals into low-quality housing and into employment does not vary with any destination characteristic in this family, and correcting for multiplicity can only strengthen a null. Stars refer to the corrected $p$-value: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S4: Robustness of the Main Results to the Level of Clustering

| | (1) admin_origin (main) | (2) Origin × year | (3) Two-way (origin + year) | (4) od_pair |
|---|---|---|---|---|
| *Panel A. Scale: inflow per 100 urban residents at destination* | | | | |
| Low-quality housing inflow | 0.28455*** | 0.28455*** | 0.28455*** | 0.28455*** |
| | (0.06323) | (0.05909) | (0.08450) | (0.04932) |
| | [< 0.0001] | [< 0.0001] | [0.0018] | [< 0.0001] |
| Employed inflow | 0.42760*** | 0.42760*** | 0.42760*** | 0.42760*** |
| | (0.04330) | (0.04976) | (0.09043) | (0.03607) |
| | [< 0.0001] | [< 0.0001] | [< 0.0001] | [< 0.0001] |
| First-stage $F$ | 29.65 | 24.66 | 8.08 | 90.48 |
| Observations | 241,540 | | | |
| *Panel B. Propensity: share among urban in-migrants* | | | | |
| Share in low-quality housing | 1.21591** | 1.21591* | 1.21591* | 1.21591** |
| | (0.59790) | (0.64169) | (0.70369) | (0.55078) |
| | [0.0427] | [0.0582] | [0.0923] | [0.0273] |
| Share employed | 1.39452** | 1.39452** | 1.39452* | 1.39452** |
| | (0.61688) | (0.63927) | (0.71685) | (0.58343) |
| | [0.0244] | [0.0292] | [0.0594] | [0.0169] |
| First-stage $F$ | 15.50 | 12.44 | 7.53 | 32.49 |
| Observations | 72,723 / 71,503 | | | |
| Number of clusters | 354 | 5,410 | 354 / 38 | 16,846 |

*Notes:* Outcomes: the paper's four main results, two scale margins and two propensity margins, each on its own table's sample. Specification: 2SLS of the urban migration rate instrumented by contemporaneous drought intensity at origin, origin–destination-pair and arrival-year fixed effects, distance-to-census control. Only the level of clustering varies across columns, so the coefficients are invariant by construction; column (1) is the level used throughout the paper. Column (2) clusters at the origin-district × year cell, finer than column (1): it permits arbitrary correlation within an origin-year while imposing independence across years within an origin, which rainfall autocorrelation makes difficult to defend. Column (3) is the demanding check, allowing arbitrary correlation across all origins within a year, the spatial synchrony documented in Section 4.2; its first-stage $F$ of 8.08 reflects the 38 clusters the year dimension supplies and not a weaker instrument. Column (4) clusters at the corridor and is a lower bound on the standard errors. The scale results hold at $p < 0.0001$ under every level; the housing propensity is the most fragile margin, moving from $p = 0.043$ to $p = 0.092$ under two-way clustering, and the employment propensity holds at the 10% level throughout. Standard errors in parentheses, $p$-values in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

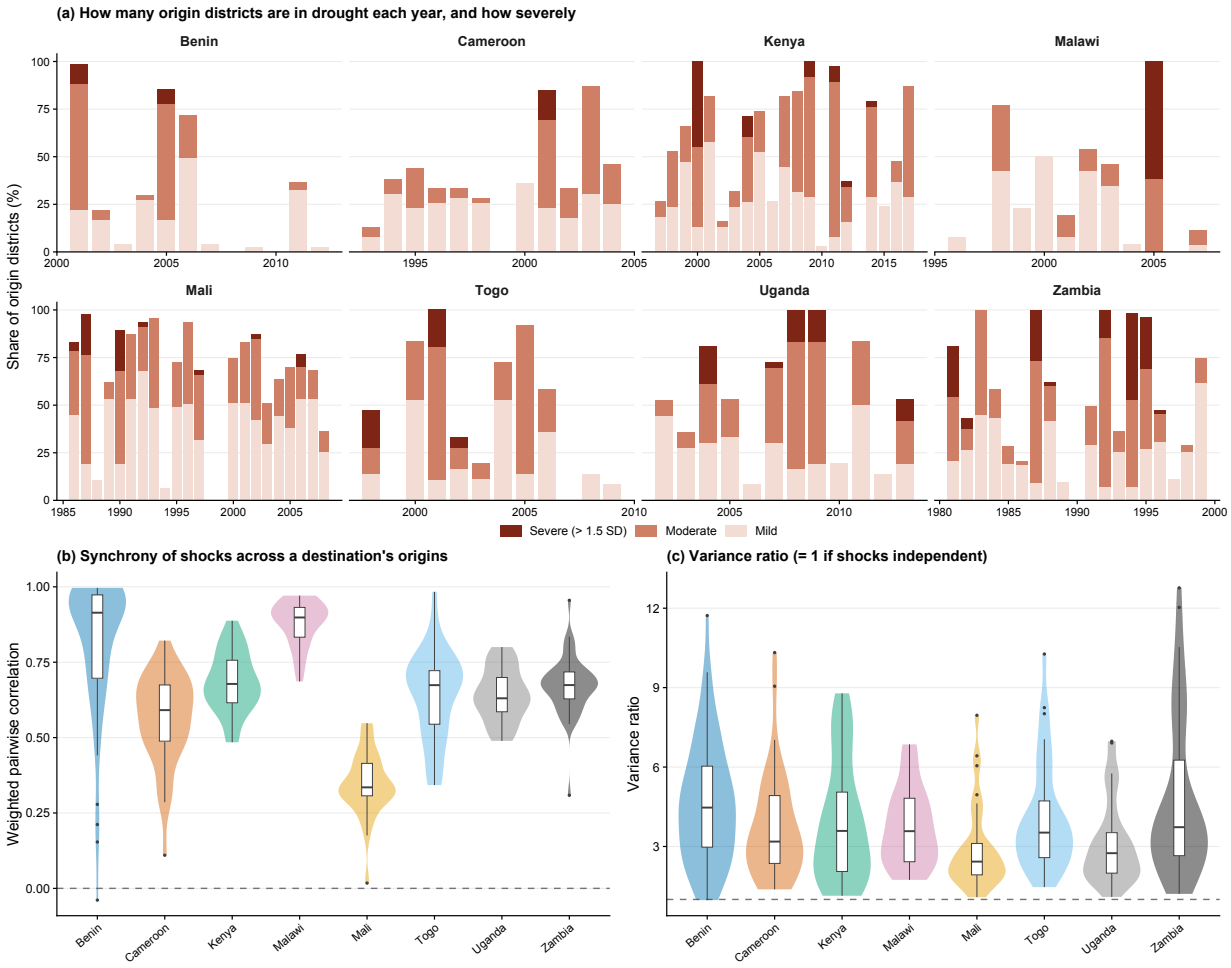



Figure S1: The Extent, Severity and Synchrony of Drought Shocks Across a Country's Origin Districts

*Notes:* Panel (a) reports, for each country and each year of its retrospective window, the share of origin districts recording a rainfall deficit, split by severity: *mild* is $0 < \text{drought}_{ot} \leq 0.643$, the median of the strictly positive values of the instrument; *moderate* is $0.643 < \text{drought}_{ot} \leq 1.5$; *severe* is $\text{drought}_{ot} > 1.5$, the threshold already used to define severe drought in the construction of the climate table. Bar height is therefore the extent of the shock and the stacking its intensity. The complement of each bar is the share of origins at or above normal rainfall, which is 50.1% of origin-years across the panel because the instrument is censored at zero; mild, moderate and severe account for 25%, 21% and 4%. Nine country-years reach 100% of origins in drought. Panels (b) and (c) report, by country, the distribution across destinations of the two synchrony measures of Table 19: the $w_{od}w_{o'd}$-weighted average pairwise correlation of drought across a destination's origins, and the variance ratio $\text{Var}_t(\sum_o w_{od}\,\text{drought}_{ot}) / \sum_o w_{od}^2\,\text{Var}_t(\text{drought}_{ot})$, which equals one when origin shocks are independent; boxes are interquartile ranges with medians, violins are kernel densities of equal width, and the dashed lines mark the independence benchmarks of 0 and 1. Every country lies well above both benchmarks. Mali is the least synchronous on both measures, which is why its drought-attributable migration is the least temporally concentrated in Figure 1.

## S1.5 Descriptive Statistics

### S1.5.1 Migration Data: Fifteen-Country Sample

Figure S2 reports average in-migration rates over time for the full set of fifteen IPUMS countries, using five-year migration brackets for cross-country comparability; the yearly rates for the eight-country estimation sample are in Appendix Figure 7.

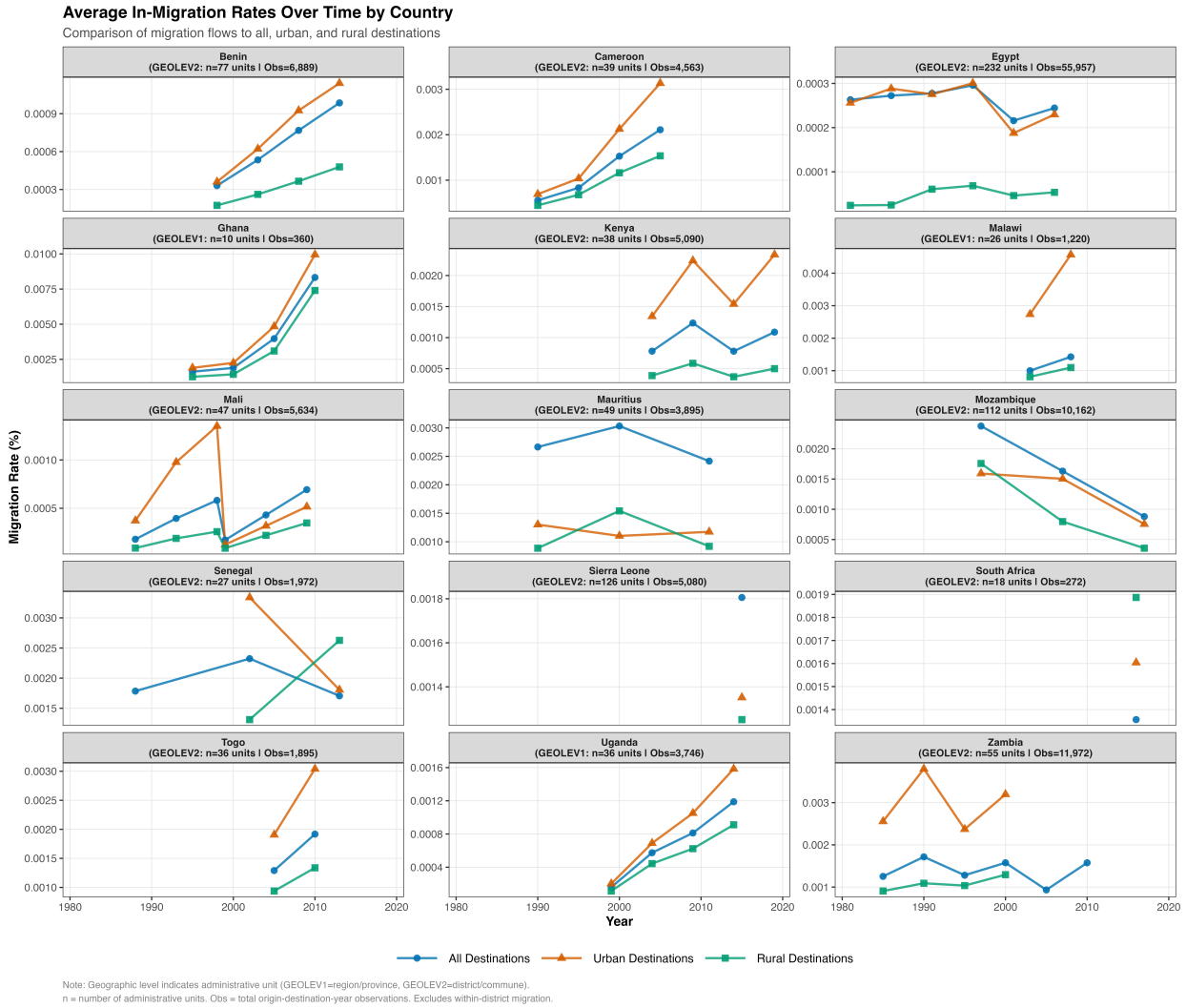


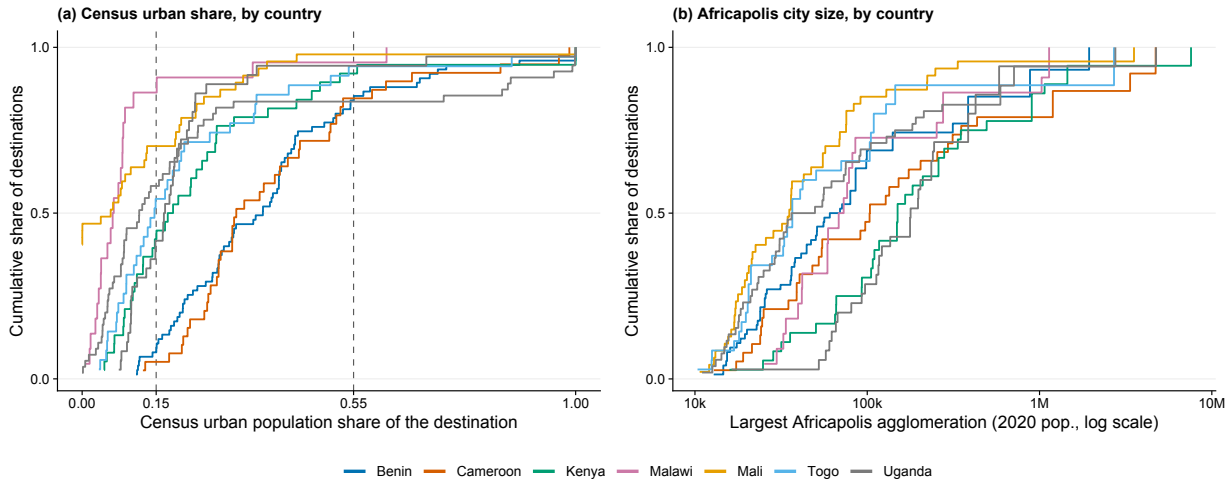

Figure S2: Average In-Migration Rates Over Time by Country.

*Notes:* Mean in-migration rates across administrative units for 15 African countries, by destination type: in-migrants weighted by `perwt` divided by the destination population, aggregated into 5-year brackets, from IPUMS census microdata. Panel titles give the geographic level, GEOLEV1 for region or province and GEOLEV2 for district or commune, the number of administrative units and the total observations. Y-axis scales vary across countries, to allow within-country comparison, and countries with a single census year, Sierra Leone and South Africa, provide limited temporal variation. *Source:* Authors' elaboration based on IPUMS census data.

### S1.5.2 Urbanization Data

Figure S3 reports the cumulative distributions of the census urban share and of Africapolis city size at destination, separately by country. The height of a country's curve at the 0.15 and 0.55 thresholds gives the share of its destinations in each urbanization bin; the curves confirm that the lower bins are populated by all countries rather than driven by any single one, with Malawi and Mali most concentrated at low urban shares.

Figure S3: Distributions of Census Urban Share and Africapolis City Size, by Country




*Notes:* Empirical cumulative distribution functions across destinations, separately by country. Panel (a) is the destination's urban population share in the most recent census: at each dashed threshold, 0.15 and 0.55, the height of a country's curve gives the proportion of its destinations below it. Panel (b) is the 2020 population of the largest agglomeration in the Africapolis 2024 database located within, or within 10 km of, each destination unit, on a logarithmic scale. Destinations without a matched Africapolis agglomeration of at least 10,000 inhabitants are excluded, and four Africapolis records reporting conurbation-cluster totals are corrected as described in the notes to Table 16.

*Source:* Authors' computations using IPUMS census data and the Africapolis 2024 database.

## S1.6 Heterogeneity by the Census Urban Share

This section repeats the destination heterogeneity of Section 8.1.1 on the census urban share instead of Africapolis agglomeration size, as a complementary check. Destinations are split into three bins by their urban population share at the most recent census: rural-like ($< 0.15$), urbanizing cities ($[0.15, 0.55)$) and main cities ($\geq 0.55$). The specification is identical to the one in the body.

Figure S4 and Table S5 report the estimates. The direction matches the size-based result: the coefficient declines as destinations urbanize, from 0.324 in rural-like destinations to 0.258 in urbanizing cities and 0.131 in main cities, the last significant only at the 10% level. The magnitudes are also consistent with the body, since the rural-like and urbanizing bins together cover much of what Africapolis classifies as small towns and secondary cities. What this classification does not deliver is inference: no pairwise contrast along the gradient is significant – urbanizing versus rural-like $p = 0.515$, urbanizing versus main cities $p = 0.128$, rural-like versus main cities $p = 0.119$ – and the urbanizing bin does not differ from the other

Table S5: Heterogeneity by Destination Urbanization: 3-Bin Interacted IV Estimates

| | *Low-quality housing inflow* | | | *Employed inflow* | | |
|---|---|---|---|---|---|---|
| | Rural-like† | Urbanizing cities | Main cities | Rural-like† | Urbanizing cities | Main cities |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Urban migration rate | 0.3239*** | 0.2579*** | 0.1305* | 0.4732*** | 0.3651*** | 0.4947*** |
| | (0.0994) | (0.0629) | (0.0779) | (0.0714) | (0.0497) | (0.0602) |
| SW conditional $F$ | 14.01 | 24.17 | 11.62 | 14.01 | 24.17 | 11.62 |
| AR 95% CI | [0.067, 0.524] | [0.102, 0.372] | [-0.123, 0.267] | [0.358, 0.703] | [0.263, 0.474] | [0.381, 0.674] |
| Mean outcome | 0.0153 | 0.0096 | 0.0116 | 0.0122 | 0.0086 | 0.0139 |
| Observations | 91,019 | 122,821 | 27,700 | 91,019 | 122,821 | 27,700 |
| Urbanizing − rural-like | | $p = 0.515$ | | | $p = 0.152$ | |
| Urbanizing − main cities | | $p = 0.128$ | | | $p = 0.043$ | |
| Urbanizing − rest | | $p = 0.596$ | | | $p = 0.092$ | |

*Notes:* Outcomes: per 100 urban residents at destination, unwinsorized. Sample: corridors with non-missing destination urbanization, $N$ = 193,290; zones are the census share urban, rural-like below 0.15, urbanizing cities in $[0.15, 0.55)$, main cities above. Specification: three-bin interacted IV, the migration rate interacted with each bin and instrumented by drought interacted with the same indicators, origin–destination and arrival-year × bin fixed effects, bin-specific `dist_census` control. SW $F$ is the Sanderson–Windmeijer conditional statistic and † marks a weak first stage, where the Anderson–Rubin interval carries the inference. The first stage is common to both outcomes, the sample being identical, with per-bin coefficients 4.23 / 2.75 / 3.12 ($\times 10^{-3}$); contrasts come from the same system's clustered covariance and 'rest' pools the rural-like and main-cities bins. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

two pooled ($p = 0.596$). For employment the ordering is different, with absorption lowest in urbanizing cities (0.365 against 0.473 and 0.495), and the urbanizing–main-cities contrast is the one significant difference here ($p = 0.043$).

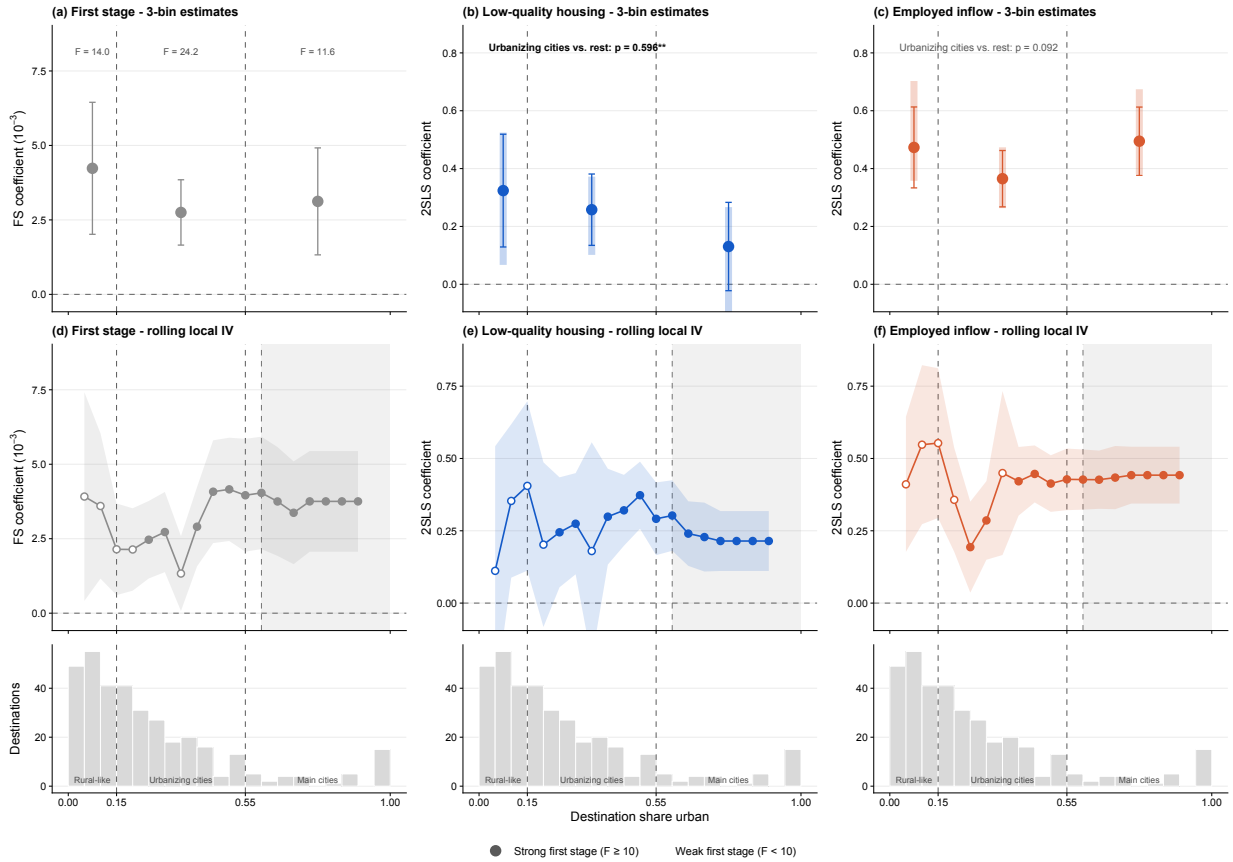
The reason we do not use this classification in the body is that its inference depends on where the rural/urban cutoff is placed, and the urban-share distribution offers no break to place it at. Across 351 destinations the share rises smoothly to a mode near 0.09–0.10 and declines slowly thereafter; the largest gap between consecutive destination shares below 0.35 is 0.014, against a median gap of 0.0008, so there is no separation anywhere in the range. Moving the lower cutoff therefore moves the contrasts without any data-driven reason to prefer one position:

| Lower cutoff | 0.05 | 0.10 | 0.15 | 0.20 | 0.25 |
|---|---|---|---|---|---|
| Secondary bin ($N$) | 186,843 | 151,097 | 122,821 | 93,884 | 71,652 |
| Housing: secondary − main cities | +0.167 | +0.183 | +0.127 | +0.146 | +0.117 |
| $p$-value | 0.065 | 0.040 | 0.128 | 0.099 | 0.194 |
| Employment: secondary − main cities | −0.068 | −0.077 | −0.130 | −0.170 | −0.139 |
| $p$-value | 0.325 | 0.220 | 0.043 | 0.006 | 0.026 |

The housing contrast crosses the 5% level at one cutoff only, and the employment contrast does so at a different and disjoint set of cutoffs. We therefore report the census-urban-share bins at the conventional 0.15 threshold, treat the declining gradient as directional

evidence consistent with the body, and rest the inference on the size-based classification, which requires no such choice. The rural-like bin is also weakly identified here (Sanderson–Windmeijer conditional $F = 14.0$ against 24.2 in the urbanizing bin), so its Anderson–Rubin interval carries the inference.

Figure S4: Heterogeneity by Destination Urbanization: Low-Quality Housing and Employment

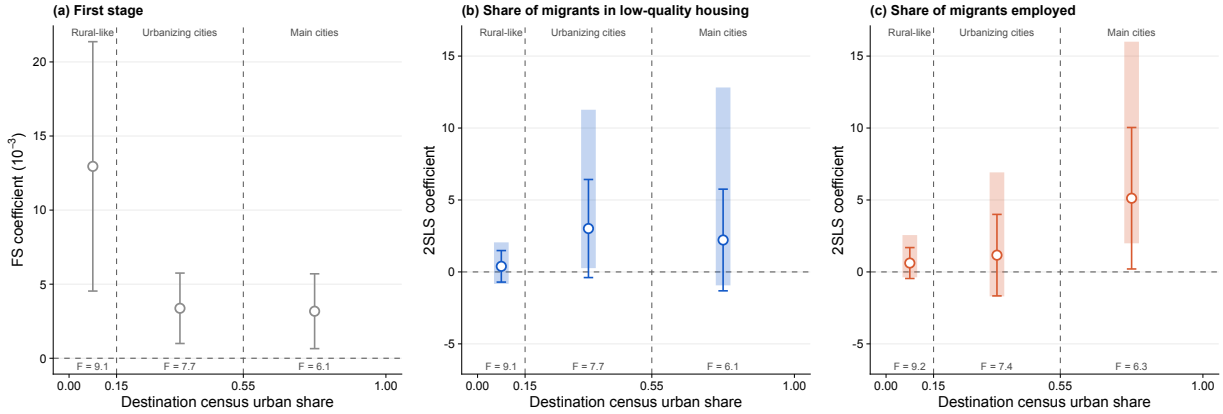



*Notes:* Markers are plotted at bin midpoints, whiskers are 95% Wald intervals and shaded bars Anderson–Rubin 95% intervals; hollow points flag a weak first stage, the Sanderson–Windmeijer conditional $F$ taking bin values 14.0 / 24.2 / 11.6. Outcomes in panels (b)–(c) are the low-quality-housing and employed inflows per 100 urban residents. Destinations are split by urban population share at the most recent census: rural-like below 0.15, urbanizing cities in $[0.15, 0.55)$, main cities above. Panels (a)–(c) come from one specification saturated by bin, the migration rate interacted with each bin indicator and instrumented by drought interacted with the same indicators, with origin–destination and arrival-year × bin fixed effects and bin-specific distance-to-census controls. For housing no pairwise contrast is significant and the urbanizing bin does not differ from the other two pooled; for employment the urbanizing bin lies significantly below main cities. Panels (d)–(f) are a rolling-window local IV, the main specification re-estimated on corridors into the 60 destinations nearest each grid point, with pointwise 95% Wald bands, indicative only since windows overlap; the shaded area marks where the window no longer slides, too few destinations lying above 0.55, and the binned estimates carry the inference there. $N = 241{,}540$ corridor-years, 91,019 / 122,821 / 27,700 across bins; the bottom row is the distribution of destination urban shares. Standard errors clustered at the origin district.

Figure S5 reports the urbanization heterogeneity for the propensity (sorting) margin.

*Job Type and Sector of Drought-Induced Migrants* - Figure S6 plots the composition gaps of Tables 34 and 35: for each class of worker and each sector, the difference in percentage points between the job-type composition of the marginal drought-induced migrants and that

Figure S5: Heterogeneity by Destination Urbanization: Migrant Propensity




*Notes:* Whiskers are conventional 95% Wald intervals and shaded bars weak-instrument-robust 95% Anderson–Rubin intervals; intervals extending beyond the displayed axis are truncated, and hollow markers flag a Sanderson–Windmeijer conditional $F$ below 10, which is the case in every bin, so panels (b) and (c) are to be read with caution. Outcomes are the propensity margins, defined only on corridor-years with positive migrant flows: $N = 55{,}698$ for housing and $N = 55{,}323$ for employment. Specification: the three-bin fully interacted IV of Figure S4, with panel (a) reporting the first stage. The Anderson–Rubin interval excludes zero only for the share of migrants in low-quality housing in urbanizing cities, which is consistent with the scale heterogeneity of Figure S4, although no pairwise contrast is significant. Standard errors clustered at the origin district.

*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

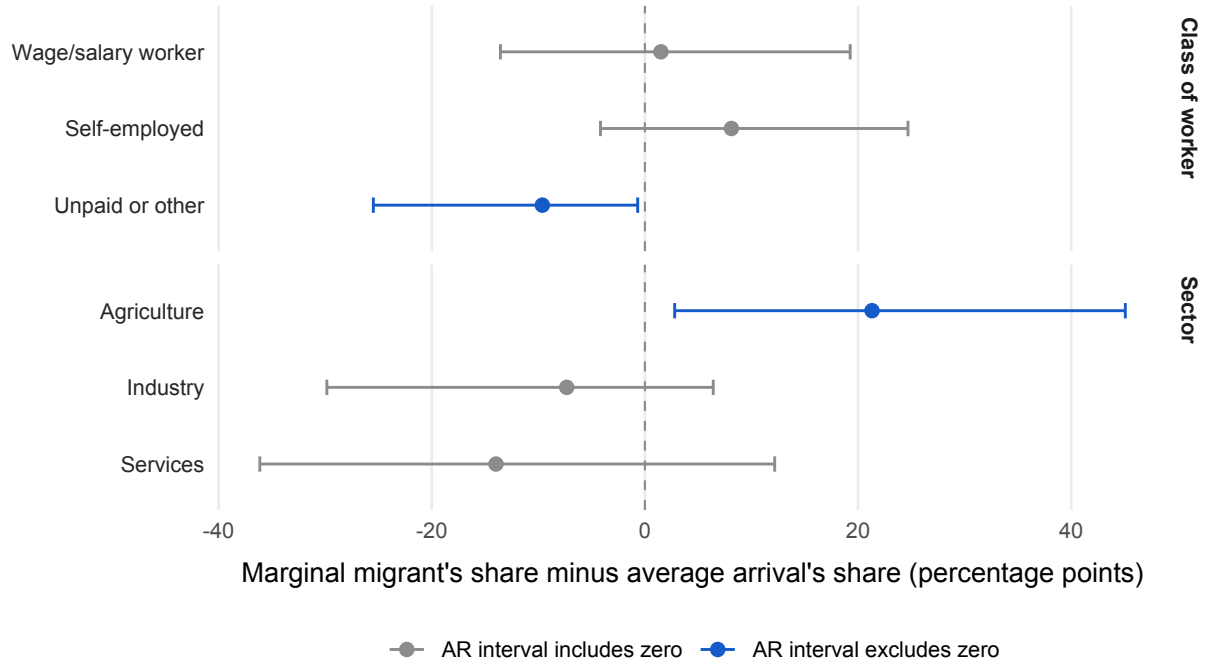
of the average urban in-migrant of the same sample. Two gaps are distinguishable from zero – less unpaid family work and more agriculture – and the wage/self-employed split is not.

*Heterogeneity by Destination Characteristic: Full Estimates* - Table S6 reports, for each of the six destination characteristics summarised in Figure 6, the first-stage and 2SLS coefficients in each bin and the contrast between them. Figures 3–S15 then give the full estimates characteristic by characteristic, each with its first stage, its two reduced forms, a rolling local-window counterpart and the distribution of destinations.

*Conversion by Destination Characteristic* - Figures S11–S13 report the 2SLS counterparts of the reduced forms of Section 8.1.3: for each of the three destination characteristics, the number out of 100 marginal drought-induced migrants who enter low-quality housing and who are in employment, by half of the within-country distribution. No contrast is significant on either outcome. The industrial split is the informative null, both halves having a first stage above ten; the saturated half of the housing split and the long-distance half of the distance split are weakly identified and their Anderson–Rubin intervals are correspondingly wide, so those two are an absence of evidence rather than evidence of absence. Specification, rolling window and conventions are those of Figures S7–S9; the vertical range of each panel is set from the Wald intervals and from the rolling windows whose first stage exceeds ten, so bands from weakly identified windows extend beyond the plotting range and are truncated.

*Heterogeneous Effects by Migrant Profile* - Tables S7–S10 report the full estimates behind Appendix Figures 19 and 20: the main effect separately by migrant subgroup, on the scale

Figure S6: Do Drought-Induced Migrants Hold a Different Kind of Job?



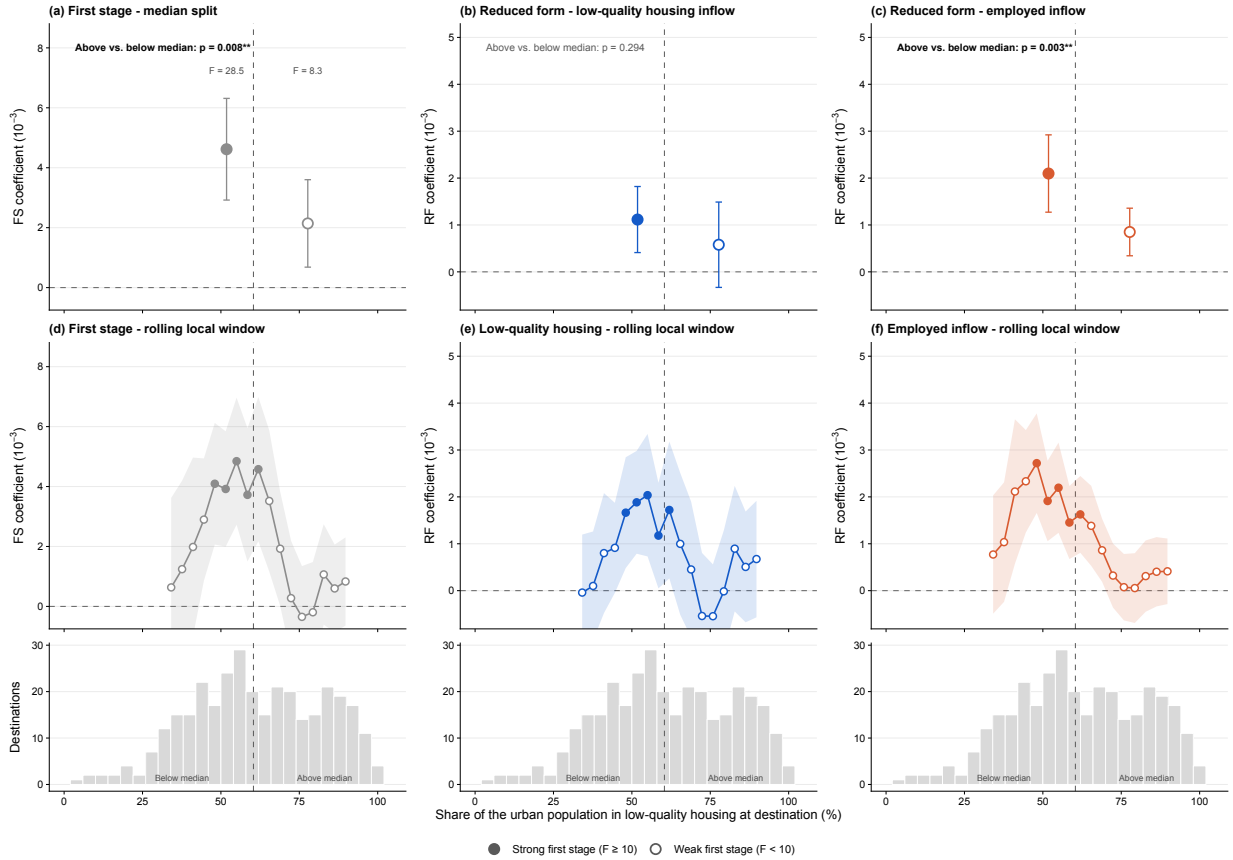

*Notes:* Difference, in percentage points, between the job-type composition of the marginal drought-induced migrants and that of the average urban in-migrant. The marginal share is the ratio of the category's 2SLS coefficient to the coefficient on the category-known total, both from the regressions of Tables 34 and 35, and the average share is the migrant-weighted share among all urban in-migrants of the same sample. Migration is instrumented by contemporaneous drought intensity at origin, with origin–destination-pair and arrival-year fixed effects and the `dist_census` control. Class of worker rests on 10 census samples, $N = 224{,}668$, and sector on 8, $N = 209{,}548$. The sector panel is suggestive: dropping Zambia or Mali leaves the implied composition unchanged and reduces the first stage to $F = 2.66$ and 3.71. Standard errors clustered at `admin_origin`. *Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Table S6: Destination Characteristics: Volume Moves, Conversion Does Not

| | First stage ($\times 10^{-3}$) | | | | 2SLS, low-quality housing (per 100) | | |
|---|---|---|---|---|---|---|---|
| Destination characteristic | Low | High | Contrast | $p$ | Low / High | Contrast | $p$ |
| City size: metropolises vs. rest | 3.35 | 5.21 | +1.86* | (0.095) | 30.6 / 10.0 | -20.6** | (0.032) |
| Distance: far vs. near | 5.79 | 0.95 | -4.83*** | (0.000) | 18.6 / 29.1 | +10.5 | (0.726) |
| Housing saturation: high vs. low | 4.62 | 2.14 | -2.48*** | (0.008) | 24.2 / 27.0 | +2.8 | (0.853) |
| Industrial base: high vs. low | 2.64 | 4.53 | +1.89** | (0.049) | 16.9 / 35.8 | +18.9 | (0.236) |
| Past built-up growth: fast vs. slow | 2.11 | 5.16 | +3.05*** | (0.000) | 28.8 / 28.3 | -0.5 | (0.962) |
| Baseline migration share: high vs. low | 2.67 | 4.59 | +1.93** | (0.016) | 31.7 / 26.5 | -5.2 | (0.594) |

*Notes:* Each row is a two-bin saturated interacted IV on one destination characteristic: instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control. Splits are at the within-country median, except city size, where the pre-specified contrast is metropolises of at least 1M in Africapolis against the other two zones pooled. Columns 2–5 report the first-stage coefficient in each bin and their difference, how much drought-induced migration a destination of that type receives; columns 6–8 report the 2SLS coefficient in each bin and their difference, how many of 100 marginal migrants enter low-quality housing there. The employment conversion, not shown, is insignificant in every row. Samples, in the order of the table: $N = 240{,}340$; 157,556; 230,676; 215,556; 241,540; 241,540. Standard errors clustered at the origin district, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

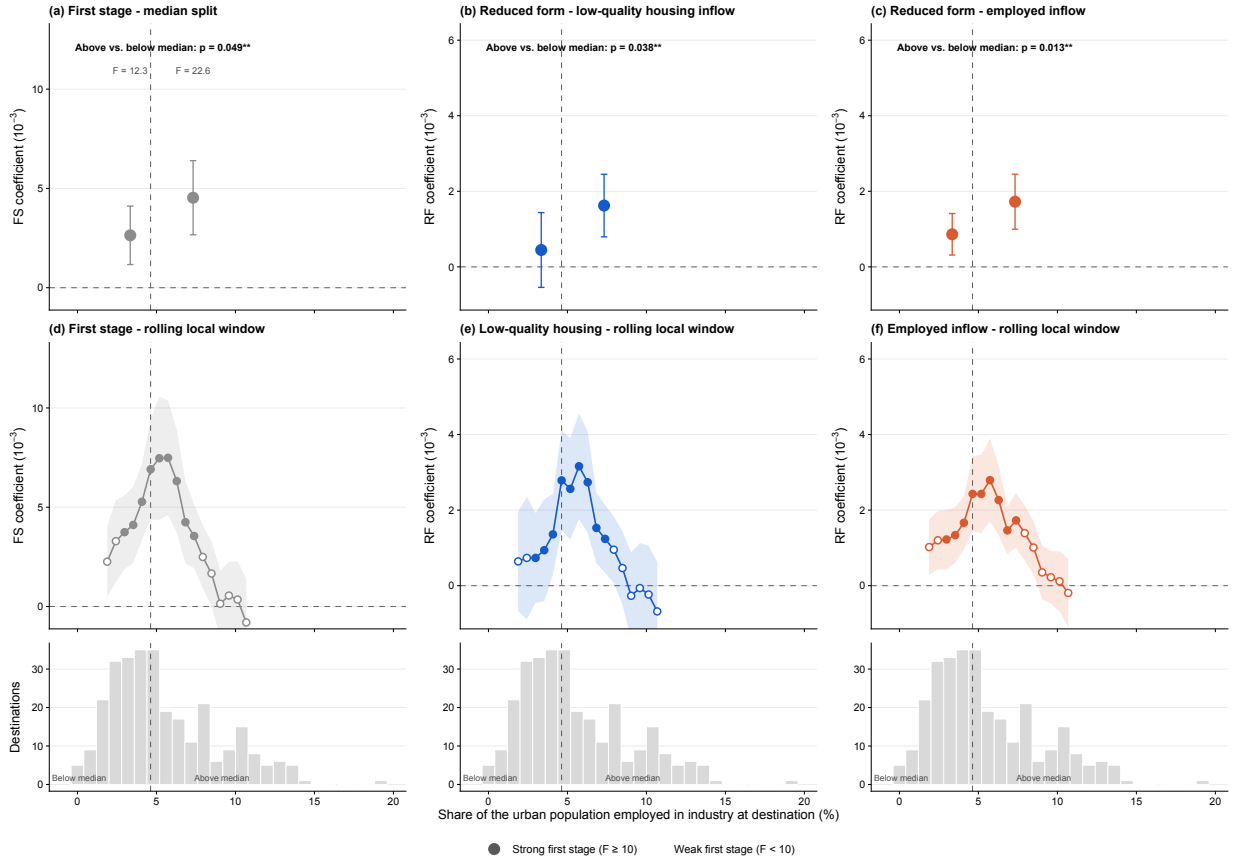
Figure S7: Where Drought Sends Migrants: Destinations by Housing Saturation




*Notes:* Markers are coefficients $\times 10^{-3}$ with 95% Wald whiskers; hollow points flag a weak first stage ($F < 10$), reported once per half in panel (a), the Sanderson–Windmeijer conditional $F$ coinciding with it under disjoint bin indicators. Panel (a) is the first stage, the effect of drought intensity at origin on the urban migration rate; panels (b) and (c) are reduced forms on the inflow of urban in-migrants in low-quality housing and on the inflow of employed urban in-migrants, both per 100 urban residents at destination. Destinations are split at the within-country median of the share of their urban population in low-quality housing, measured on all urban residents at the last usable census of each country, Zambia 2000 rather than 2010 whose urban variable is missing. All three come from a two-bin saturated interacted model: instrument × bin, origin–destination and arrival-year × bin fixed effects, bin-specific distance-to-census control. Panels (d)–(f) re-estimate the same quantities on a rolling local window, the 60 destinations whose saturation rate is closest to each grid point, with pointwise 95% Wald bands; these are indicative only, the inference being carried by the binned contrasts. The dashed vertical line marks the pooled median and is indicative, the split being within country. $N = 230{,}676$ corridor-years, 331 destinations, 8 countries; the bottom row is the distribution of the classifier. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

margin (the subgroup's low-quality-housing inflow) and the propensity margin (the share of the subgroup's migrants in low-quality housing), each with subgroup-specific migration rates instrumented by drought under the main specification; the reference column reproduces the aggregate anchors. Both margins are broadly uniform across gender and age - no within-dimension contrast is significant. The only gradient is educational: on the scale margin, the effect is twice as strong for primary-educated adults as for those with secondary education or above ($p = 0.036$), suggestive given that attainment is measured at census. The oldest and least-educated adults are AR-only on both margins - drought barely moves their migration, itself informative - and the propensity first stages are weak throughout ($F$ between 8 and 13), so propensity inference rests on the Anderson–Rubin intervals: the sorting shift is detected

Figure S8: Where Drought Sends Migrants: Destinations by Industrial Employment Share



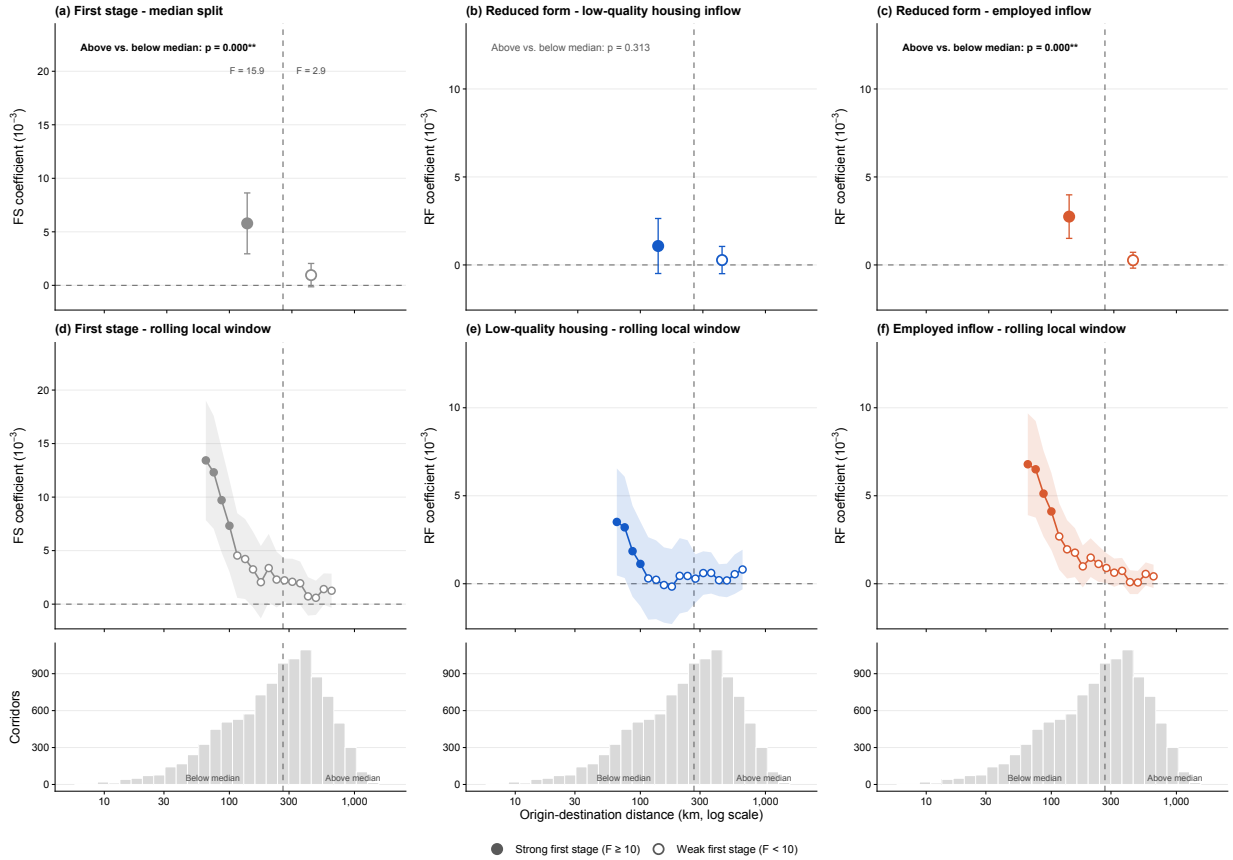

*Notes:* Panels, specification, rolling window and conventions as in Figure S7; neither half is weakly identified. Destinations are split at the within-country median of the share of their urban population employed in industry, IPUMS `INDGEN` codes 20/30/40/50 among those with known employment status, at the last usable census of each country. Uganda is absent, its 2014 census carrying a class-of-worker variable but no industry code, and Zambia 2000 has industry coverage for 81% of employed urban residents against 97 to 101% elsewhere. $N = 215{,}556$ corridor-years, 295 destinations, 7 countries. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

among men and adults, and imprecise for women and children.

*Saturated First Stage by Profile* - Table S11 reports the fully saturated first stage across the twelve gender × age-at-arrival × education cells - the additivity check behind Section 8.2: selection into migration runs along the educational gradient alone, with all less-than-primary cells near zero across gender and age.

*Does Drought Select on Destination Type?* - Table S12 reports the linear projection of a causal forest fitted on the first stage, a data-driven check on whether drought directs migrants toward a particular kind of destination. Because the treatment is rainfall at origin, unconfoundedness holds and the forest recovers how the migration response varies with corridor and destination characteristics; the same exercise would be invalid on the second stage, where the migration rate is endogenous. The response does vary across corridors – the calibration test rejects homogeneity – but only corridor distance and origin urban

Figure S9: Where Drought Sends Migrants: Corridors by Origin–Destination Distance

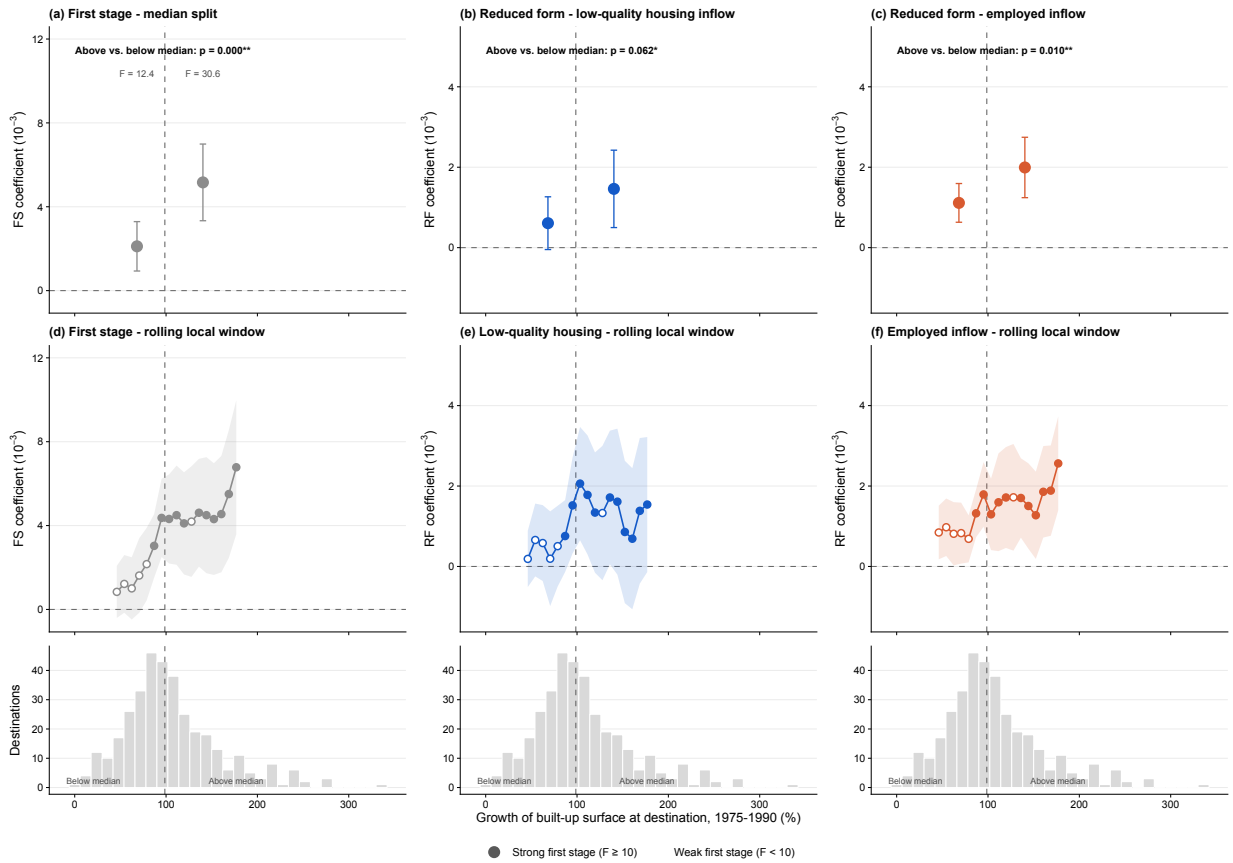



*Notes:* Corridors are split at the within-country median of the great-circle distance between the centroids of the origin and destination units, so this classifier is at the corridor rather than the destination level and the bottom row counts corridors; the horizontal axis is on a logarithmic scale. Distance is available for 10,375 of 16,846 corridors, $N = 157{,}556$ of 241,540 corridor-years; on this subsample the pooled low-quality-housing reduced form is already insignificant while the employment margin is unchanged, so panel (b) carries little information. The rolling window retains, at each grid point, the 2,075 corridors nearest to it, the same 20% of units as the 60-destination window of Figure S7. Other panels, specification and conventions as in Figure S7. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

share have projection coefficients distinguishable from zero. No destination characteristic does, including the urban share that defines the bins of Figure S4. The forest also recovers, without being told, the distance decay imposed by hand elsewhere in the paper: the predicted response falls from 0.0061 in the nearest within-country distance tercile to 0.0032 and 0.0019.

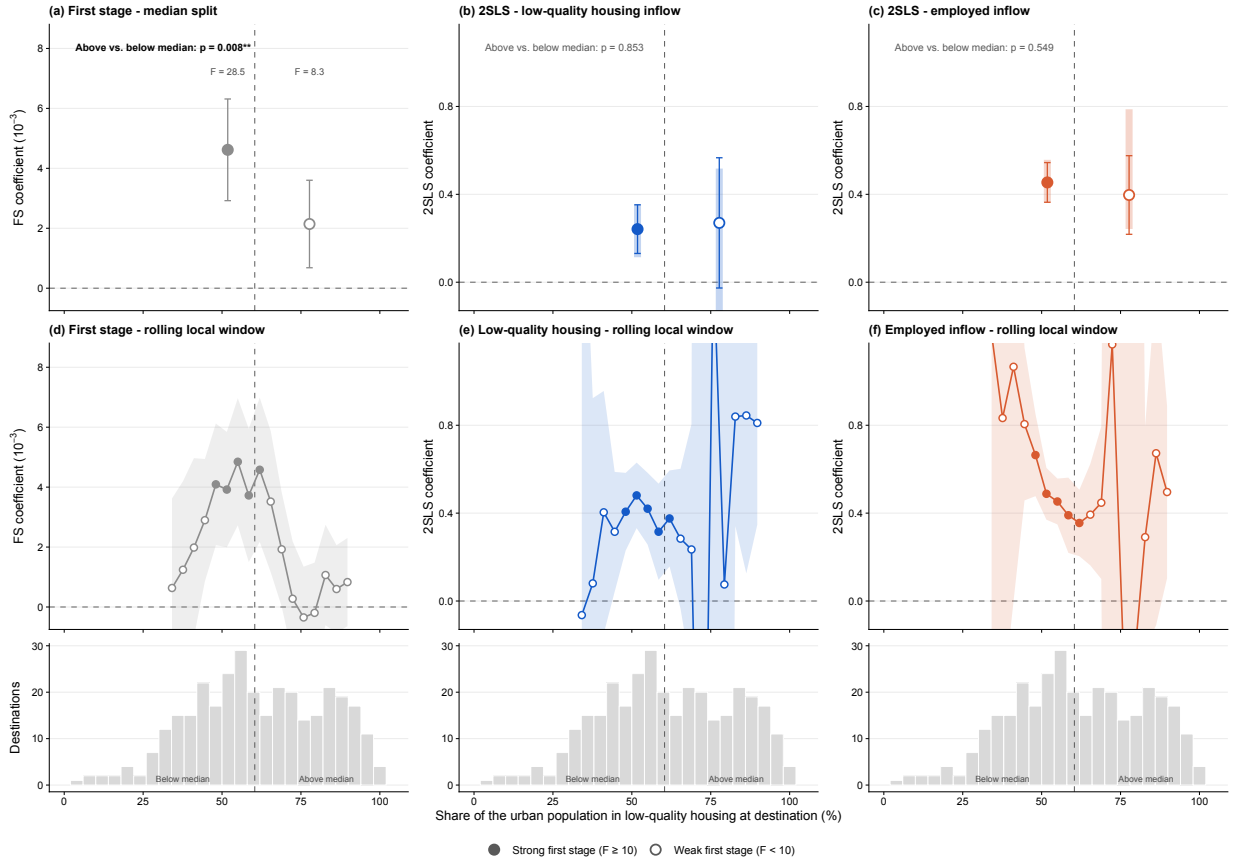
*Composition of the Housing Index* - Figures S17 to S20 open up the index dimension by dimension. Figure S17 pools the eight countries and shows which dimensions actually drive the measure; Figures S18 and S19 report, country by country, which material and facility categories fall on the deprived side of each rule; Figure S20 shows the distribution of persons per habitable room against the three-person threshold.

Figure S10: Where Drought Sends Migrants: Destinations by Past Built-Up Expansion



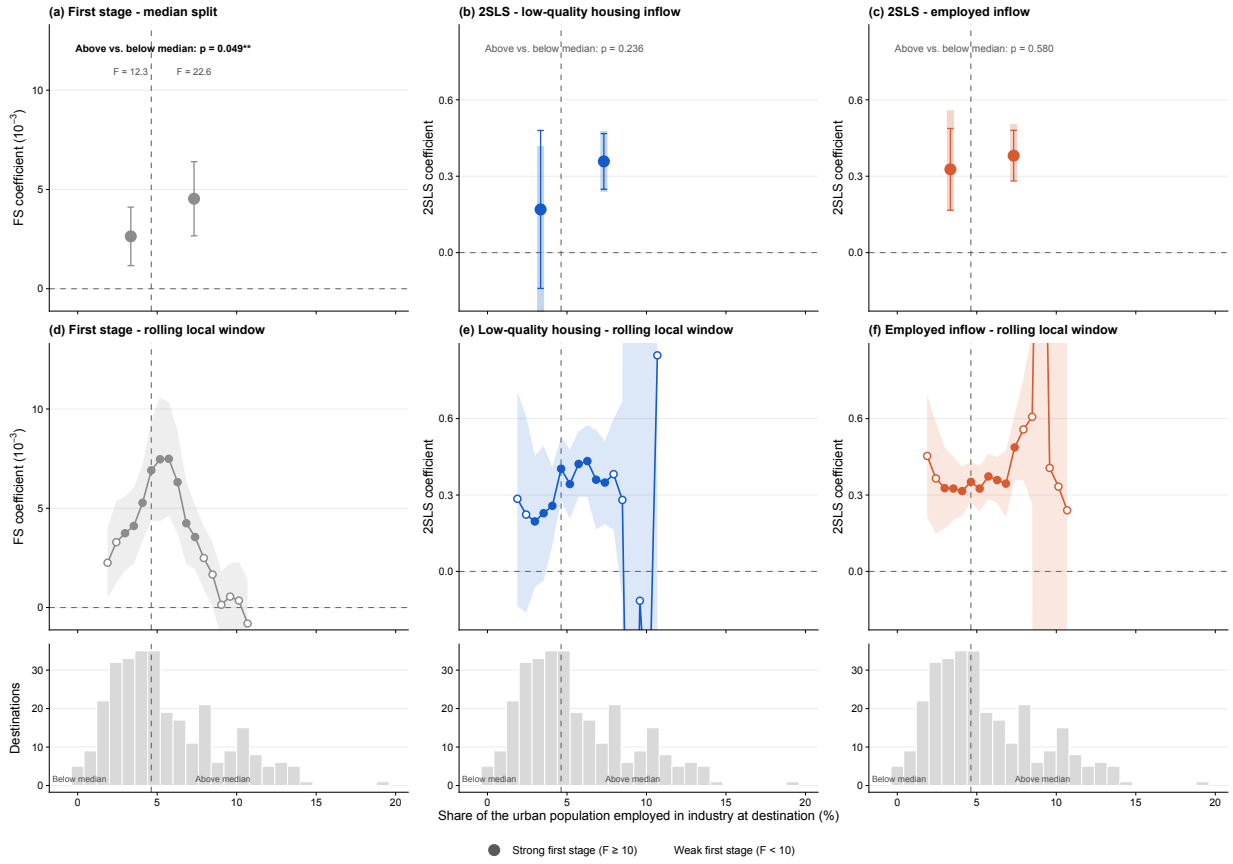

*Notes:* Panels, specification, rolling window and conventions as in Figure S7; neither half is weakly identified. Destinations are split at the within-country median of the growth of their built-up surface between 1975 and 1990, from the Global Human Settlement Layer (GHS-BUILT-S, release R2023A; Pesaresi and Politis, 2023), supplied already aggregated to IPUMS administrative units; for Malawi and Uganda, which enter the panel at the first administrative level, the nested second-level units are summed, an aggregation that reproduces the published first-level totals to within 0.001%. All 351 destinations and all $N = 241{,}540$ corridor-years are matched, so this is the only one of the four splits estimated on the full canonical panel. The variable is a proxy for the destination's past capacity to build, measured before the migration flows and before the drought shocks that instrument them; it is not a supply elasticity in the sense of Saiz (2010), and it mixes construction capacity with the demand that was already being accommodated. The effect runs opposite to a supply-constraint prediction: both the volume and the housing inflow are larger where past expansion was faster. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census, CHIRPS climate and GHSL built-up data.

Figure S11: Conversion into Low-Quality Housing and Employment, by Destination Housing Saturation



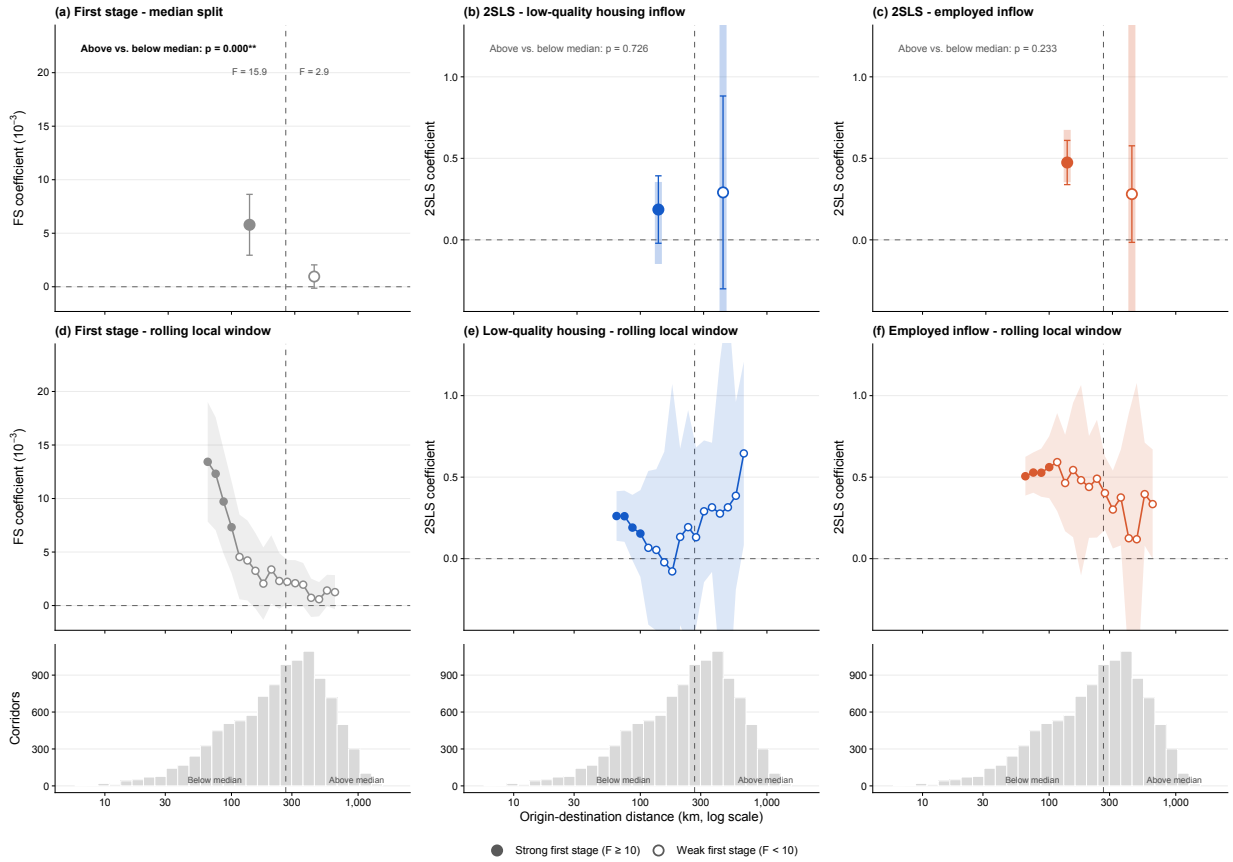

*Notes:* Whiskers are 95% Wald intervals and shaded bars cluster-robust Anderson–Rubin 95% intervals computed on the just-identified subsample; hollow points mark a weak first stage, reported once per half in panel (a). Panel (a) reproduces the first stage of Figure S7, and panels (b) and (c) report 2SLS coefficients from the two-bin saturated interacted model, reading as the number out of 100 marginal drought-induced migrants in low-quality housing and in employment. Panels (d)–(f) are the rolling local-window counterparts. Neither contrast is significant, and the above-median half is weakly identified, its housing Anderson–Rubin interval containing both zero and twice the below-median estimate. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Figure S12: Conversion into Low-Quality Housing and Employment, by Destination Industrial Employment Share



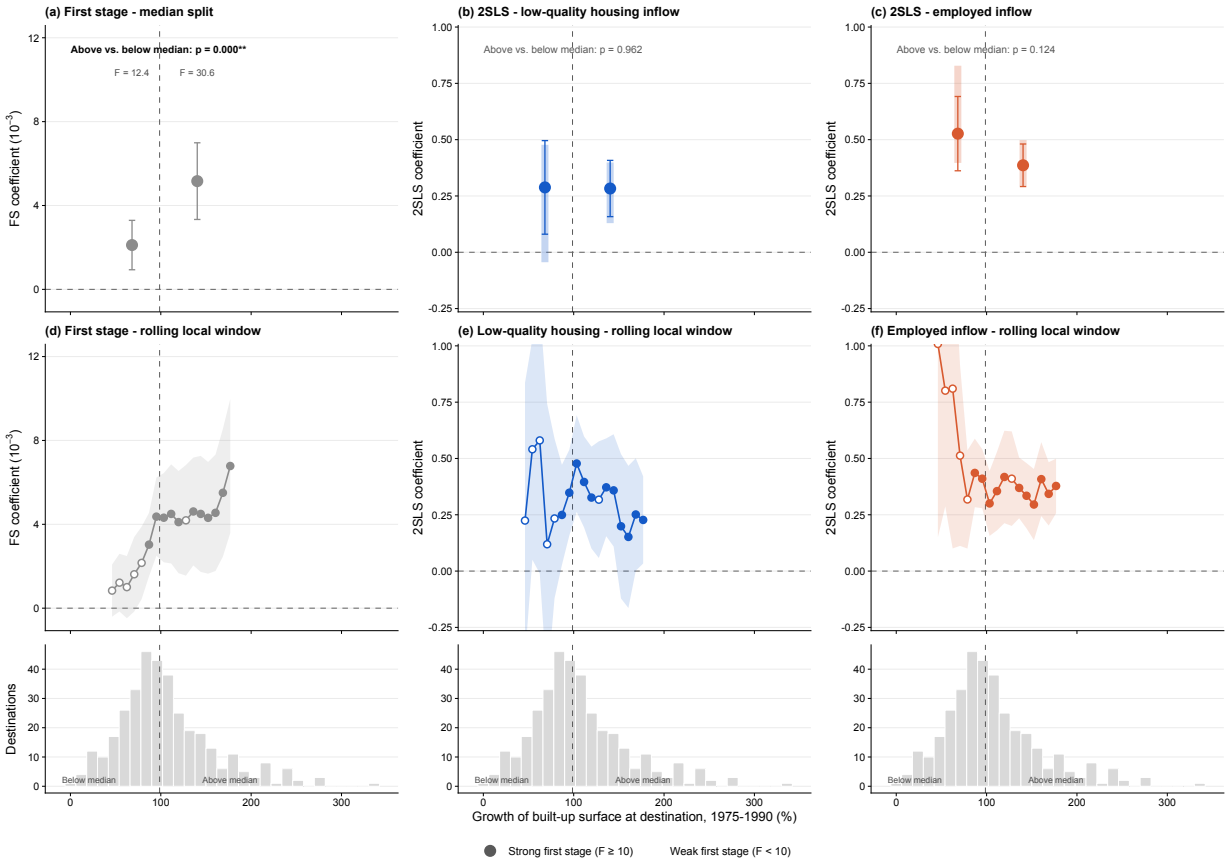

*Notes:* Panels, specification and conventions as in Figure S11; the classifier is that of Figure S8. Both halves are strongly identified and neither contrast is significant, which makes this the most informative of the three conversion nulls. The below-median Anderson–Rubin interval on housing, $[-0.40, +0.42]$, nonetheless remains compatible with a difference of the plotted size. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Figure S13: Conversion into Low-Quality Housing and Employment, by Origin–Destination Distance



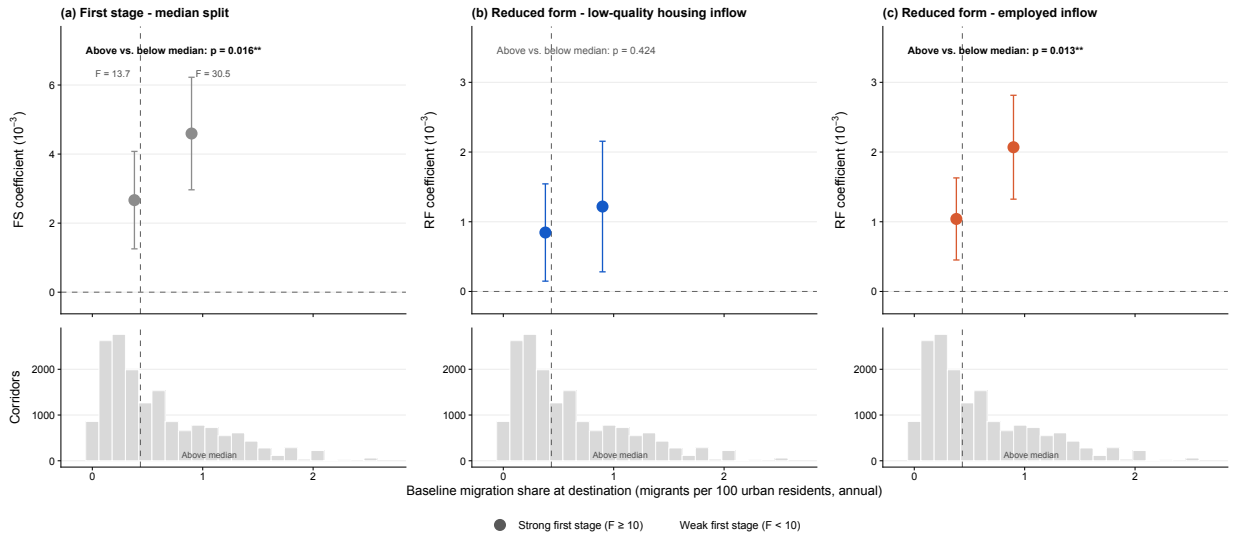

*Notes:* Panels, specification and conventions as in Figure S11; the classifier is that of Figure S9. The above-median half has a first stage of 2.9 and Anderson–Rubin intervals of $[-3.32, +3.91]$ on housing and $[-1.53, +2.09]$ on employment, so its contrasts are mechanically non-rejectable and carry no information. The below-median half is well identified, and its housing Anderson–Rubin interval includes zero, consistent with the pooled housing result already being unidentified on the 65% of corridor-years for which distance is available. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Figure S14: Conversion into Low-Quality Housing and Employment, by Past Built-Up Expansion



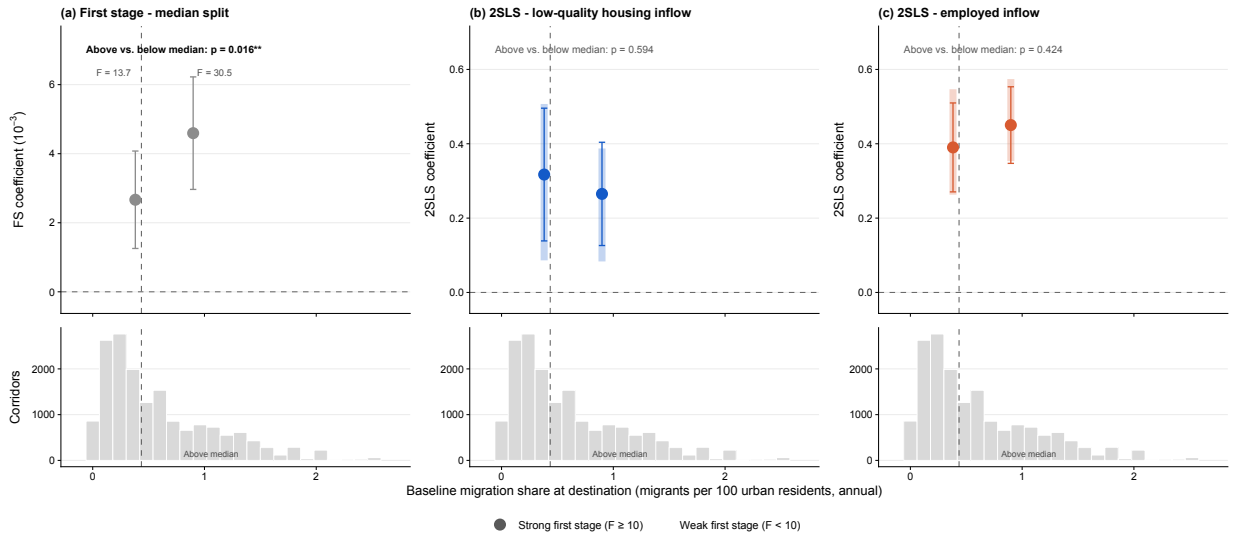

*Notes:* Panels, specification and conventions as in Figure S11; the classifier is that of Figure S10. Both halves are strongly identified and the housing conversion is all but identical across them, with Anderson–Rubin intervals of $[-0.04, +0.48]$ and $[+0.13, +0.40]$. The employment conversion is lower where past expansion was faster, and that contrast is not significant. Together with the first stage of Figure S10, two and a half times larger in the fast-expanding half, this is the clearest case of the pattern described in Section 8.1.3: the destination moves the number of arrivals and not what becomes of them. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census, CHIRPS climate and GHSL built-up data.

Figure S15: Where Drought Sends Migrants: Corridors by the Destination's Baseline Migration Share




*Notes:* Corridors are split at the within-country median of their destination's baseline migration share, the number of urban in-migrants the destination received from other origins over the three earliest years of its recall window, per year and per 100 of its urban residents. The leave-own-origin-out construction means a corridor never enters its own classifier, so those three years remain in the estimation sample, and a single value per corridor from its earliest census cycle is applied throughout, which keeps the origin–destination fixed effects nested within bins. This is a descriptive and not a predetermined characteristic: a destination's baseline migration share is itself the product of everything that makes it attractive. $N = 241{,}540$ corridor-years, 16,846 corridors. Panels and specification as in Figure S7, without the rolling local-window row; both halves are strongly identified. Drought sends 1.7 times more migrants to destinations that were already migrant-intensive, and the employed-inflow reduced form follows, while the low-quality-housing reduced form does not separate. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Figure S16: Conversion into Low-Quality Housing and Employment, by the Destination's Baseline Migration Share




*Notes:* Corridors are split at the within-country median of their destination's baseline migration share, the number of urban in-migrants the destination received from other origins over the three earliest years of its recall window, per year and per 100 of its urban residents. The leave-own-origin-out construction means a corridor never enters its own classifier, so those three years remain in the estimation sample, and a single value per corridor from its earliest census cycle is applied throughout, which keeps the origin–destination fixed effects nested within bins. This is a descriptive and not a predetermined characteristic: a destination's baseline migration share is itself the product of everything that makes it attractive. $N = 241{,}540$ corridor-years, 16,846 corridors. Panels, specification and conventions as in Figure S11. Both halves are strongly identified, so these are informative nulls: neither the housing conversion nor the employment conversion separates across the two bins, and both Anderson–Rubin intervals exclude zero. Standard errors clustered at the origin district.
*Source:* Authors' computations using IPUMS census and CHIRPS climate data.

Table S7: Heterogeneous Effects by Migrant Profile: Low-Quality Housing Inflows (Scale)

| | | *Gender* | | *Age at arrival* | | | *Education, 25+* | | |
|---|---|---|---|---|---|---|---|---|---|
| | All | Women | Men | <15 | 15+ | 55+ | <Prim. | Primary | Sec.+ |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Subgroup urban migration rate | 0.3106*** | 0.2799*** | 0.2862*** | 0.3242*** | 0.2535*** | AR-only | AR-only | 0.3928*** | 0.2158*** |
| | (0.0858) | (0.0645) | (0.0757) | (0.0926) | (0.0574) | | | (0.0898) | (0.0606) |
| First-stage $F$ (SW cond.) | 23.86 | 29.70 | 25.88 | 19.96 | 28.66 | 0.10 | 1.95 | 24.71 | 25.00 |
| AR 95% CI | [0.08, 0.46] | [0.12, 0.40] | [0.10, 0.42] | [0.08, 0.49] | [0.11, 0.36] | [-16.9, 17.6] | [-1.80, 3.37] | [0.19, 0.57] | [0.09, 0.35] |
| Contrast $p$ (within dimension) | | 0.916 | | | 0.378 | | | 0.092 | |
| Share of migrants (%) | 100.0 | 50.4 | 49.6 | 28.5 | 71.5 | 1.4 | 7.8 | 20.5 | 22.8 |
| Dist. control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| OD-pair FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean outcome | 0.0123 | 0.0061 | 0.0059 | 0.0050 | 0.0070 | 0.0003 | 0.0019 | 0.0021 | 0.0010 |
| Observations | 245,292 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 |
| OD pairs | 16,888 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: the subgroup's low-quality-housing inflow per 100 urban residents, main definition, unwinsorized. Sample: by migrant profile; age at arrival is census age minus years since migration, and education is measured at census and may post-date arrival. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Each column is a separate 2SLS in which the subgroup's migration rate is instrumented by drought at origin, so coefficients read as the number per 100 additional drought-induced migrants of the subgroup settling in low-quality housing; column (1) is the aggregate reference and reproduces the published anchor of 0.228. Contrast $p$-values come from the fully saturated stacked interaction. The only gradient is educational, which stays suggestive given how education is measured. Columns whose conditional first-stage $F$ falls below 8, the 55+ and less-than-primary cells, are AR-only: drought barely moves their migration, which is itself informative. The propensity-margin counterpart is Table S8. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S8: Heterogeneous Effects by Migrant Profile: Share in Low-Quality Housing (Propensity)

| | | *Gender* | | *Age at arrival* | | | *Education, 25+* | | |
|---|---|---|---|---|---|---|---|---|---|
| | All | Women | Men | <15 | 15+ | 55+ | <Prim. | Primary | Sec.+ |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Subgroup urban migration rate | 1.2159** | 0.9462 | 3.1375** | 0.0117 | 3.9751** | AR-only | AR-only | AR-only | AR-only |
| | (0.5979) | (1.0923) | (1.5943) | (1.2202) | (1.6423) | | | | |
| First-stage $F$ (SW cond.) | 15.50 | 19.48 | 10.11 | 12.53 | 11.27 | 2.33 | 0.70 | 5.16 | 7.27 |
| AR 95% CI | [0.17, 2.94] | [-1.19, 3.59] | [0.61, 9.29] | [-3.06, 2.64] | [1.6, 10.6] | [-450.4, 568.2] | [-367.9, 430.1] | [1.6, 91.4] | [1.7, 38.9] |
| Contrast $p$ (within dimension) | | 0.177 | | | 0.039 | | | – | |
| Share of migrants (%) | 100.0 | 50.4 | 49.6 | 28.5 | 71.5 | 1.4 | 7.8 | 20.5 | 22.8 |
| Dist. control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| OD-pair FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean outcome | 0.406 | 0.413 | 0.394 | 0.424 | 0.395 | 0.438 | 0.531 | 0.395 | 0.247 |
| Observations | 72,723 | 57,284 | 58,921 | 46,618 | 64,621 | 8,305 | 28,734 | 37,688 | 34,071 |
| OD pairs | 10,537 | 9,337 | 9,573 | 8,478 | 9,981 | 2,948 | 6,657 | 7,272 | 6,712 |

*Notes:* Outcome: the share of a subgroup's migrants in low-quality housing, unweighted and unwinsorized, on corridor-years with a positive subgroup flow, so $N$ varies by column. Sample: by migrant profile; education is measured at the census and may post-date arrival. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Each column is a separate 2SLS in which the subgroup's migration rate is instrumented by drought at origin. Column (1) is the aggregate reference and reproduces the published anchor of 1.081; contrast $p$-values come from the fully saturated stacked interaction, and the education contrast is not estimable, both columns falling below the $F$ gate. First stages are weak throughout, so inference rests on the cluster-robust Anderson–Rubin intervals, and columns with a conditional $F$ below 8 are AR-only: 55+ and less than primary are not identified on this margin, drought barely moving their migration, which is itself informative. The scale-margin counterpart is Table S7. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S9: Heterogeneous Effects by Migrant Profile: Employed Inflows (Scale)

| | | *Gender* | | *Age at arrival* | | | *Education, 25+* | | |
|---|---|---|---|---|---|---|---|---|---|
| | All | Women | Men | <15 | 15+ | 55+ | <Prim. | Primary | Sec.+ |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Subgroup urban migration rate | 0.4276*** | 0.3455*** | 0.5155*** | 0.0768*** | 0.6775*** | AR-only | AR-only | 0.7932*** | 0.9139*** |
| | (0.0433) | (0.0515) | (0.0524) | (0.0269) | (0.0564) | | | (0.0805) | (0.0403) |
| First-stage $F$ (SW cond.) | 29.65 | 29.70 | 25.88 | 19.96 | 28.66 | 0.10 | 1.95 | 24.71 | 25.00 |
| AR 95% CI | [0.35, 0.53] | [0.25, 0.47] | [0.42, 0.64] | [0.02, 0.14] | [0.57, 0.81] | [-54.6, 52.1] | [-1.84, 3.31] | [0.63, 0.97] | [0.83, 1.00] |
| Contrast $p$ (within dimension) | | 0.003 | | | 0.000 | | | 0.175 | |
| Share of migrants (%) | 100.0 | 50.4 | 49.6 | 28.5 | 71.5 | 1.4 | 7.8 | 20.5 | 22.8 |
| Dist. control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| OD-pair FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean outcome | 0.0106 | 0.0042 | 0.0064 | 0.0009 | 0.0097 | 0.0002 | 0.0017 | 0.0032 | 0.0031 |
| Observations | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 | 241,540 |
| OD pairs | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 | 16,846 |

*Notes:* Outcome: the subgroup's employed inflow per 100 urban residents, unwinsorized, $N = 245{,}292$ per row. Sample: by migrant profile; education is measured at census and may post-date arrival. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Each column is a separate 2SLS in which the subgroup's migration rate is instrumented by drought at origin; column (1) reproduces the published anchor of 0.339, and contrast $p$-values come from the fully saturated stacked interaction. The age contrast is mechanical, children being rarely employed so that their employed inflow is near zero by construction, and should not be read as economic heterogeneity. Employment absorption is otherwise uniform, with no significant gender contrast and no educational gradient, in contrast with the housing margin (Table S7). Columns whose conditional first-stage $F$ falls below 8, the 55+ and less-than-primary cells, are AR-only: drought barely moves their migration, which is itself informative. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S10: Heterogeneous Effects by Migrant Profile: Employed Share (Propensity)

| | | *Gender* | | *Age at arrival* | | | *Education, 25+* | | |
|---|---|---|---|---|---|---|---|---|---|
| | All | Women | Men | <15 | 15+ | 55+ | <Prim. | Primary | Sec.+ |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Subgroup urban migration rate | 1.3945** | 2.4854** | 1.2001 | 2.4288** | 1.0167 | AR-only | AR-only | AR-only | AR-only |
| | (0.6169) | (1.0848) | (1.3858) | (0.9511) | (1.0656) | | | | |
| First-stage $F$ (SW cond.) | 15.51 | 19.29 | 10.17 | 16.33 | 11.17 | 2.29 | 0.67 | 4.98 | 7.26 |
| AR 95% CI | [0.40, 3.32] | [0.69, 5.62] | [-1.78, 5.11] | [0.86, 5.27] | [-0.96, 4.31] | [-665.5, 526.8] | [-284.0, 249.7] | [-3.0, 66.6] | [-4.4, 17.4] |
| Contrast $p$ (within dimension) | | 0.434 | | | 0.314 | | | – | |
| Share of migrants (%) | 100.0 | 50.4 | 49.6 | 28.5 | 71.5 | 1.4 | 7.8 | 20.5 | 22.8 |
| Dist. control | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| OD-pair FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean outcome | 0.505 | 0.385 | 0.614 | 0.143 | 0.638 | 0.461 | 0.592 | 0.702 | 0.796 |
| Observations | 71,503 | 56,113 | 57,537 | 42,415 | 64,593 | 8,300 | 28,717 | 37,671 | 34,057 |
| OD pairs | 10,452 | 9,248 | 9,459 | 8,047 | 9,979 | 2,945 | 6,652 | 7,271 | 6,710 |

*Notes:* Outcome: the employed share among the subgroup's migrants of known labour-force status, unweighted and unwinsorized, on corridor-years with a positive status-known subgroup flow, so $N$ varies by column. Sample: by migrant profile; education is measured at census and may post-date arrival, and the under-15 column is child employment, conceptually thin. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Each column is a separate 2SLS in which the subgroup's migration rate is instrumented by drought at origin; column (1) reproduces the published anchor of 1.319, and contrast $p$-values come from the fully saturated stacked interaction, the education contrast being not estimable with both columns below the $F$ gate. First stages are weak throughout, so inference rests on the cluster-robust Anderson–Rubin intervals, and columns with a conditional $F$ below 8 are AR-only. The scale counterpart is Table S9. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S11: Saturated First Stage by Migrant Profile: Who Does Drought Displace?

| | Share of migrants (%) | Relative first stage (1-SD drought, % of cell mean) | (SE) |
|---|---|---|---|
| *Panel A: Secondary education or above* | | | |
| Women, arrived <30 | 13.7 | 8.38*** | (1.99) |
| Women, arrived 30+ | 4.6 | 13.56*** | (3.59) |
| Men, arrived <30 | 16.9 | 5.92*** | (2.01) |
| Men, arrived 30+ | 9.4 | 8.16*** | (2.25) |
| *Panel B: Primary completed* | | | |
| Women, arrived <30 | 13.7 | 5.62*** | (1.71) |
| Women, arrived 30+ | 6.0 | 7.30*** | (2.69) |
| Men, arrived <30 | 12.5 | 5.06*** | (1.70) |
| Men, arrived 30+ | 8.0 | 5.14** | (2.03) |
| *Panel C: Less than primary (the trapped margin)* | | | |
| Women, arrived <30 | 4.6 | 2.77 | (2.03) |
| Women, arrived 30+ | 4.2 | 3.82 | (2.68) |
| Men, arrived <30 | 3.2 | 1.14 | (2.93) |
| Men, arrived 30+ | 3.2 | 0.86 | (2.66) |
| Equality of the 12 responses (Wald) | | $p = 0.289$ | |
| Interactions beyond additive margins (Wald, 7 terms) | | $p = 0.955$ | |

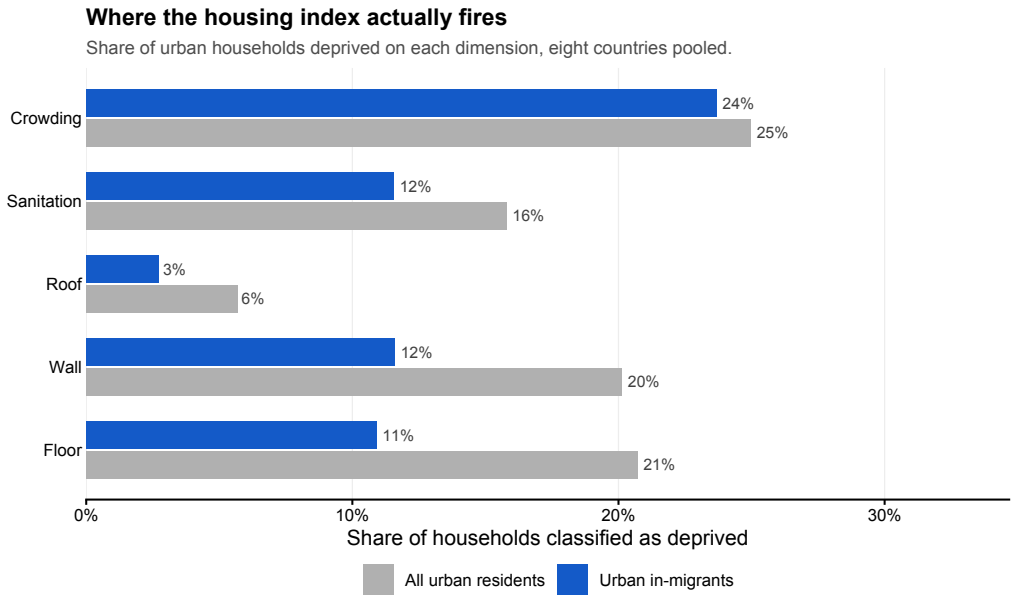
*Notes:* Outcome: the corridor-year migration rate of each of the twelve gender × age-at-arrival × education cells, regressed on drought intensity at origin. Sample: urban migrants aged 25 or over at census with known attainment, $N$ = 245,292 corridor-years per cell; education is measured at census and may post-date arrival, and the top cell, women arrived at 30 or over with secondary or more, is 4.6% of migrants. Main specification: origin–destination pair and arrival-year fixed effects, distance-to-census control. Because rainfall is exogenous, each cell's OLS first stage is valid regardless of instrument strength. The reported effect is the relative first stage, a one-standard-deviation drought as a percentage of the cell's own mean migration rate, which is the scale comparable across cells, raw coefficients mostly reflecting cell size; the aggregate value is +6.9%. Joint tests come from the stacked model on the standardized outcome with shared clusters. The four less-than-primary cells are the trapped-population result, robust across gender and age; its formal test is the pooled contrast reported in the main text. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table S12: Does Drought Select on Destination Type? Linear Projection of the First-Stage Causal Forest

| | Projection coefficient (per 1 SD, $\times 10^{-3}$) | (SE) | $p$-value |
|---|---|---|---|
| *Corridor* | | | |
| Distance, within-country percentile | -2.424** | (1.119) | 0.030 |
| Distance (km) | 0.017 | (0.900) | 0.985 |
| *Origin* | | | |
| Urban share | -1.691** | (0.738) | 0.022 |
| Growing-season rainfall (mm) | -1.268 | (0.968) | 0.190 |
| *Destination* | | | |
| Urban share | -0.847 | (0.549) | 0.123 |
| Baseline low-quality housing share | -1.081* | (0.561) | 0.054 |
| Baseline employed share | 0.186 | (0.582) | 0.749 |
| Growing-season rainfall (mm) | -0.061 | (0.805) | 0.940 |
| Population density (per km$^2$) | 0.055 | (0.247) | 0.823 |
| Mean $\hat{\tau}(x)$ | 0.00369 (pooled first stage 0.00362) | | |
| Calibration test, differential forest prediction | 1.105 ($p = 1.4 \times 10^{-22}$) | | |
| Corridor-years / corridors / origin clusters | 241,540 / 16,846 / 354 | | |
| Trees | 1000 | | |

*Notes:* Outcome: the conditional average partial effect $\hat{\tau}(x)$ of drought intensity at origin on the corridor's urban migration rate, from a causal forest (`grf`, 1000 trees), projected linearly on the covariates. Sample: the estimation panel; the two destination composition covariates are the low-quality-housing and employed shares of in-migrants in the destination's earliest census window, leave-one-origin-out with a 30-observation floor, the panel carrying no resident population. Specification: the forest has no fixed effects, so the migration rate and drought intensity are residualized on origin–destination pair and arrival-year fixed effects with the *dist_census* control and the forest is fitted on the residuals, keeping the identifying variation within corridor while the time-invariant covariates carry the heterogeneity across corridors; the mean of $\hat{\tau}(x)$ reproduces the pooled first stage, 0.00369 against 0.00362. Coefficients are per one standard deviation of the covariate and in units of $10^{-3}$, a monotone rescaling that leaves $p$-values unchanged. The projection, and not the forest's variable-importance ranking, is the object to interpret: importance splits credit arbitrarily between correlated covariates. The calibration test rejects the absence of heterogeneity ($p = 1.4 \times 10^{-22}$). This is a first-stage validation exercise, drought being exogenous, and says which destinations drought directs migrants toward. Standard errors clustered at `admin_origin`, in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

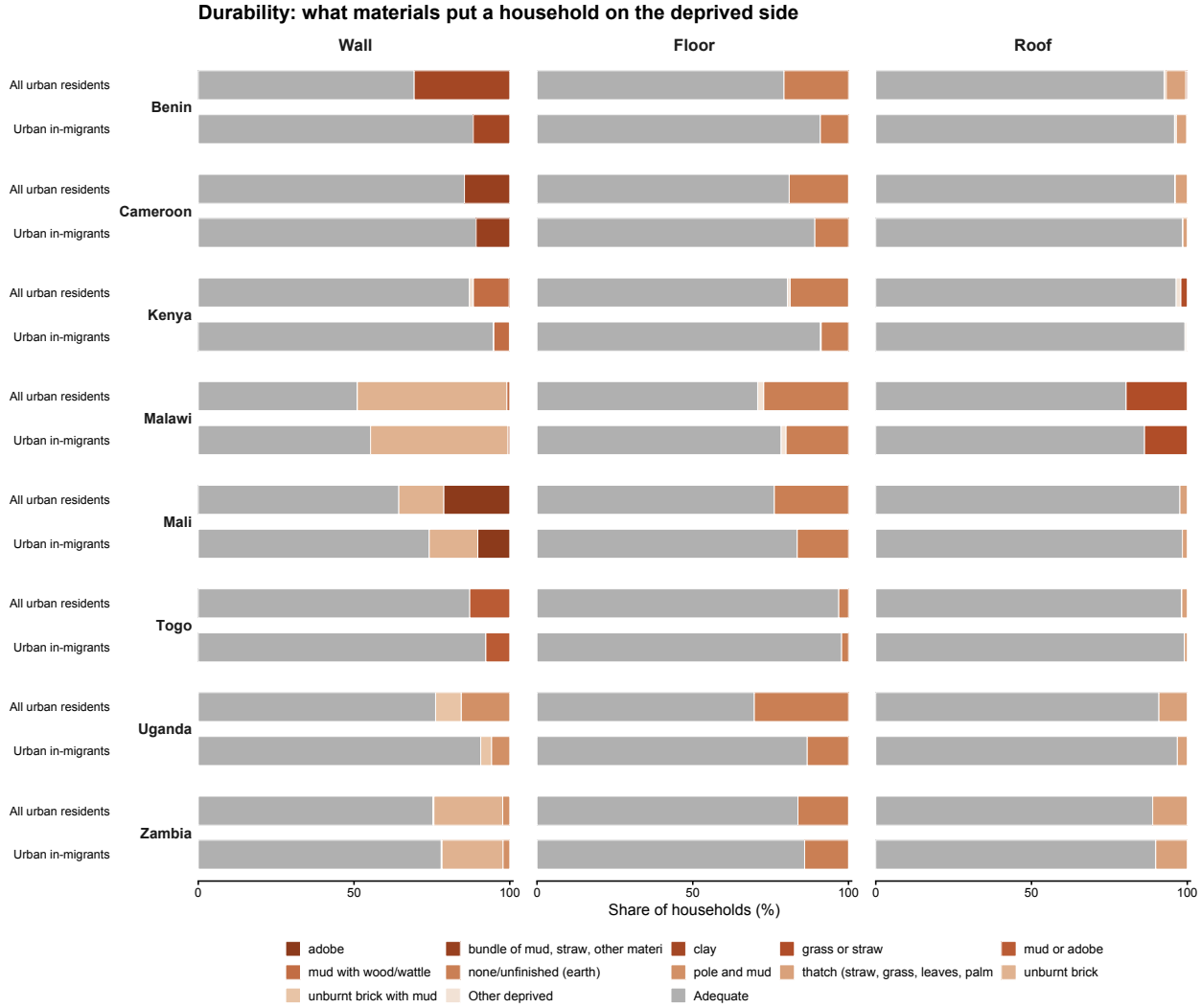
Figure S17: Which Dimensions Drive the Housing Index




*Notes:* Share of urban households deprived on each dimension of the housing index, eight countries pooled. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window; urban households, person-weighted. Water is omitted throughout, deprivation being nil in every sample: these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. Crowding is the only dimension on which in-migrants and residents coincide; on every material dimension in-migrants are about half as often deprived.

*Source:* Authors' computations from IPUMS census microdata.

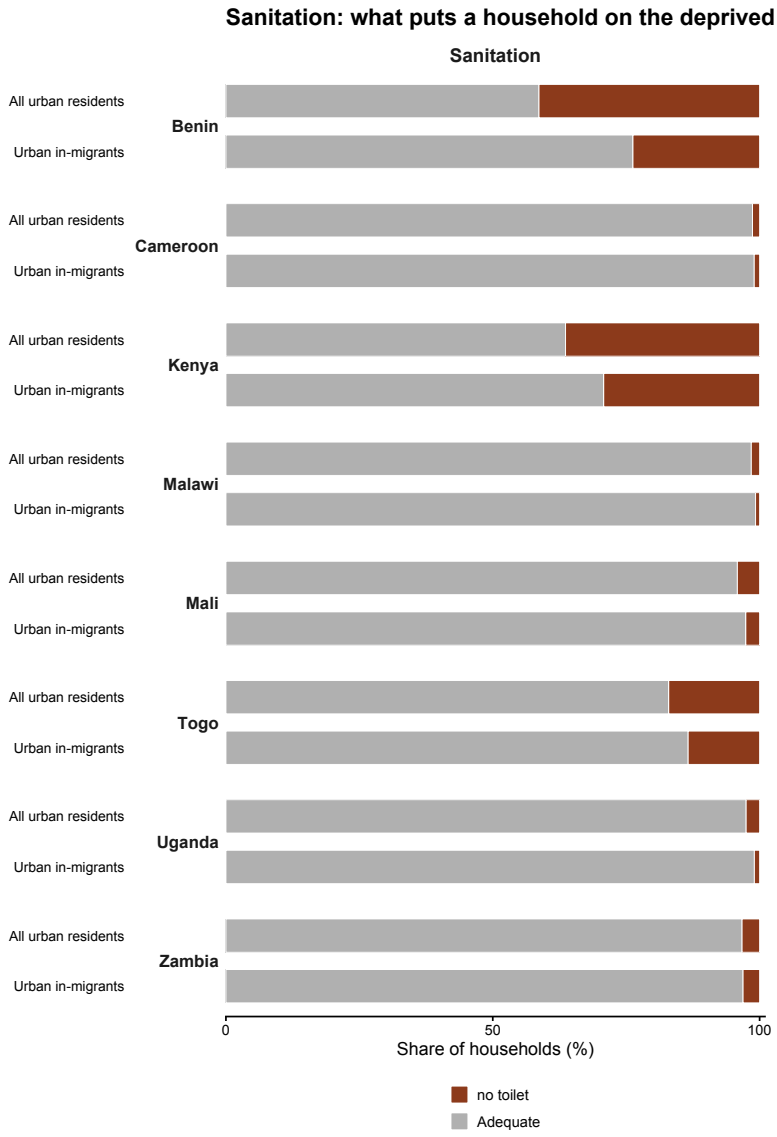
Figure S18: Durability: Materials on the Deprived Side of the Rule




*Notes:* Each bar is the category composition of one dimension for one country and population, over urban households, person-weighted. An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window. Deprived categories are named individually and shown in warm shades when they exceed 5 per cent of a bar, while all adequate categories are pooled into a single grey, the question here being where the deprivation cut falls. Non-permanent wall materials are the unburnt-brick, adobe, mud and plant families; burnt or stabilised brick, cement-stabilised earth, stone, concrete, metal and wood count as permanent, and sheet-metal roofs count as adequate. Water is omitted throughout: deprivation is nil in every sample, these censuses distinguishing only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources.

*Source:* Authors' computations from IPUMS census microdata.

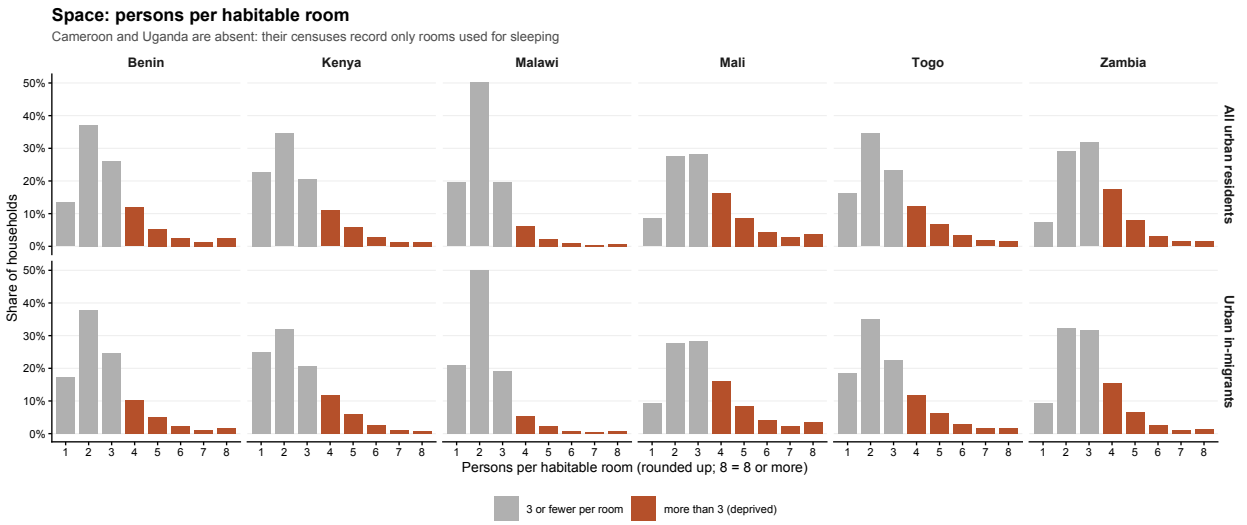
Figure S19: Sanitation: Facilities on the Deprived Side of the Rule




*Notes:* An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window; urban households, person-weighted. Water is omitted throughout, deprivation being nil in every sample: these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. Only the complete absence of a toilet counts as deprivation. Pit latrines, roughly half of the pooled sample, are not counted: IPUMS records them in a single undifferentiated category and UN-Habitat treats an improved pit latrine with a slab as adequate sanitation, so the conservative reading is adopted.

*Source:* Authors' computations from IPUMS census microdata.

Figure S20: Living Space: Persons per Habitable Room




*Notes:* An in-migrant is a person whose unit of origin differs from the unit of enumeration and whose arrival year falls in that census's retrospective window; urban households, person-weighted. Water is omitted throughout, deprivation being nil in every sample: these censuses distinguish only types of piped water while UN-Habitat counts boreholes and protected dug wells as improved sources. A household is deprived when more than three people share a habitable room, the UN-Habitat threshold, habitable rooms excluding kitchens, corridors, verandas and bathrooms. Cameroon 2005 and Uganda 2014 are absent: their censuses record only the number of rooms used for sleeping, which is not comparable to a room count, so crowding is left unmeasured there rather than approximated.

*Source:* Authors' computations from IPUMS census microdata.